

## MODEL RUN TRANSCRIPT

Generated: 2026-08-12T09:58:57

Source: data/json/noise\_experiments/v2/noise2i\_full8match\_game/claude-sonnet-4.5/game/noisy\_bidirectional\_informed\_memprimary\_full8match/noise2i\_full8match\_game\_001\_neutral\_rep01.json

## RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary_full8match
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 1
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: self_and_coplayer
self_history_window: 0
coplayer_history_window: 3
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 1
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

## AGENTS

Agent\_1; model=anthropic/claude-sonnet-4.5; memory\_capacity=6

### System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent\_2; model=anthropic/claude-sonnet-4.5; memory\_capacity=6

**System prompt:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

**ROUND 1**

=====

Roles: Agent\_1=Sender, Agent\_2=Receiver

**Pairings:**

dyad\_1: Agent\_1=Sender, Agent\_2=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_1 sent \$3; received \$9; Agent\_2 returned \$5; payoffs Agent\_1=\$7, Agent\_2=\$4;

noise/visible: sent visible=\$3.44, received visible=\$10.32, returned visible=\$5.02

Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent\_1": 7.0,

"Agent\_2": 4.0

}

Visible balances after round: {

"Agent\_1": 7.02,

"Agent\_2": 5.32

}

**LLM CALLS**

**GAME - Agent\_1 - Sender - dyad\_1**

Timestamp: 2026-08-12T09:54:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Assistant output:**

Looking at this first round, I want to establish a cooperative relationship while managing risk.

If I send money, it gets tripled, creating potential mutual benefit. However, there's uncertainty about how much will be returned.

I'll take a moderately cooperative approach to start:

**\*\*My reasoning:\*\***

- Sending \$3 means the receiver gets \$9 (3×3)
- I initially keep \$2
- This creates enough value that if the receiver is fair/cooperative, we both can benefit
- It signals willingness to trust without being maximally vulnerable
- In repeated games, building trust early can pay off

```
'''json
{"send": 3}
'''
```

#### **Reasoning:**

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, so there's potential for mutual benefit
2. The receiver can choose how much to return - there's uncertainty/trust involved
3. This is a repeated game (multiple rounds), so reputation effects matter
4. There's mentioned communication noise, which could affect trust dynamics

Strategic considerations:

- Sending \$0 guarantees I keep \$5, but no potential for gain
- Sending everything (\$5) maximizes the total pot (\$15 to receiver) but requires maximum trust
- A moderate amount like \$2-3 balances risk and potential reward
- In first rounds of repeated games, moderate cooperation often works well to establish trust

Since this is Round 1 and I have no history with this receiver, I'll take a moderately cooperative approach. Sending around \$3 shows willingness to cooperate while not being overly risky.

If I send \$3:

- I keep \$2 initially
- Receiver gets \$9 (3×3)
- If receiver returns even \$3-4, we both benefit compared to no cooperation
- This signals cooperation without being maximally vulnerable

I'll send \$3.

```
Usage: {
  "input_tokens": 365,
  "output_tokens": 436,
  "reasoning_tokens": 0
}
```

#### **GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:54:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
```

```

"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$3.44, that is 68.8% of their total endowment. You received \$10.32. How much do you return to The sender? (0-10.32)

**Assistant output:**

Looking at this situation:

- The sender showed considerable trust by sending \$3.44 (68.8% of their endowment)
- This was multiplied to \$10.32 for me
- The sender currently has \$1.56 remaining

I want to reciprocate their trust while ensuring we both benefit from cooperating. If I return \$5.00:

- The sender ends with: \$1.56 + \$5.00 = \$6.56
- I end with: \$10.32 - \$5.00 = \$5.32
- We both gain from the cooperation

This seems fair given their high level of trust.

```
{"return": 5.00}
```

**Reasoning:**

Let me understand this situation:

1. The sender started with \$5
2. The sender sent \$3.44 (which is 68.8% of \$5)
3. This amount was multiplied by 3x, so I received  $3.44 \times 3 = \$10.32$
4. I need to decide how much to return (between \$0 and \$10.32)

The sender showed considerable trust by sending 68.8% of their endowment. Let me think about what would be fair:

- The sender kept:  $\$5 - \$3.44 = \$1.56$
- I received:  $\$10.32$
- If I return nothing, the sender ends with  $\$1.56$  and I end with  $\$10.32$
- If I return some amount  $X$ :
  - Sender ends with:  $\$1.56 + X$
  - I end with:  $\$10.32 - X$

A fair approach might be to try to equalize our earnings or to ensure the sender benefits from their trust. Let's consider a few options:

1. Equal split of total: Total is  $\$1.56 + \$10.32 = \$11.88$ . Half is  $\$5.94$  each.
  - For sender to get  $\$5.94$ , they need  $\$5.94 - \$1.56 = \$4.38$  returned
  - This would give each of us  $\$5.94$
2. Proportional return: Since they sent 68.8%, I could return a similar proportion
  - 68.8% of  $\$10.32 = \$7.10$
  - Sender would end with:  $\$1.56 + \$7.10 = \$8.66$
  - I would end with:  $\$10.32 - \$7.10 = \$3.22$
3. Return what ensures sender gets more than initial  $\$5$ :
  - For sender to get  $\$6$  total: need to return  $\$6 - \$1.56 = \$4.44$
  - For sender to get  $\$7$  total: need to return  $\$7 - \$1.56 = \$5.44$

Given their substantial trust, returning around  $\$5$  seems reasonable-this rewards their cooperation while still leaving me with a modest gain. That would give the sender  $\$6.56$  (exceeding their initial stake) and me  $\$5.32$ ....

```
Usage: {
  "input_tokens": 394,
  "output_tokens": 880,
  "reasoning_tokens": 0
}
```

## ROUND 2

Roles: Agent\_1=Receiver, Agent\_2=Sender

### Pairings:

dyad\_1: Agent\_2=Sender, Agent\_1=Receiver

### Parsed game decisions:

dyad\_1: Agent\_2 sent  $\$3.5$ ; received  $\$10.5$ ; Agent\_1 returned  $\$0.15$ ; payoffs Agent\_2= $\$1.65$ , Agent\_1= $\$10.35$ ; noise/visible: sent visible= $\$0.12$ , received visible= $\$0.36$ , returned visible= $\$0.1$   
Action validation: {"returned": {"amount": 0.15, "clamped": false, "max": 10.5, "min": 0, "raw": 0.15}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

Balances after round: {

"Agent\_1": 17.35,  
"Agent\_2": 5.65

}

Visible balances after round: {

"Agent\_1": 7.2299999999999995,  
"Agent\_2": 6.92

}

## LLM CALLS

### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-08-12T09:55:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_1",  
"pairing": {  
 "agents": [  
 "Agent\_2",  
 "Agent\_1"

```

],
"investor": "Agent_2",
"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
"trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$3.44, that is 68.8% of their total endowment. You received \$10.32. How much do you return to The sender? (0-10.32)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed considerable trust by sending \$3.44 (68.8% of their endowment)
- This was multiplied to \$10.32 for me
- The sender currently has \$1.56 remaining

I want to reciprocate their trust while ensuring we both benefit from cooperating. If I return \$5.00:

- The sender ends with: \$1.56 + \$5.00 = \$6.56
- I end with: \$10.32 - \$5.00 = \$5.32
- We both gain from the cooperation

This seems fair given their high level of trust.

```
{"return": 5.00}
```

**reasoning:**

Let me understand this situation:

1. The sender started with \$5
2. The sender sent \$3.44 (which is 68.8% of \$5)
3. This amount was multiplied by 3x, so I received  $3.44 \times 3 = \$10.32$
4. I need to decide how much to return (between \$0 and \$10.32)

The sender showed considerable trust by sending 68.8% of their endowment. Let me think about what would be fair:

- The sender kept:  $\$5 - \$3.44 = \$1.56$
- I received: \$10.32
- If I return nothing, the sender ends with \$1.56 and I end with \$10.32
- If I return some amount X:
  - Sender ends with:  $\$1.56 + X$
  - I end with:  $\$10.32 - X$

A fair approach might be to try to equalize our earnings or to ensure the sender benefits from their trust. Let's consider a few options:

1. Equal split of total: Total is  $\$1.56 + \$10.32 = \$11.88$ . Half is \$5.94 each.
  - For sender to get \$5.94, they need  $\$5.94 - \$1.56 = \$4.38$  returned
  - This would give each of us \$5.94
2. Proportional return: Since they sent 68.8%, I could return a similar proportion
  - 68.8% of \$10.32 = \$7.10
  - Sender would end with:  $\$1.56 + \$7.10 = \$8.66$
  - I would end with:  $\$10.32 - \$7.10 = \$3.22$
3. Return what ensures sender gets more than initial \$5:
  - For sender to get \$6 total: need to return  $\$6 - \$1.56 = \$4.44$
  - For sender to get \$7 total: need to return  $\$7 - \$1.56 = \$5.44$

Given their substantial trust, returning around \$5 seems reasonable-this rewards their cooperation while still leaving me with a modest gain. That would give the sender \$6.56 (exceeding their initial stake) and me \$5.32....

**usage:**

```
{
  "input_tokens": 394,
  "output_tokens": 880,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.32.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at Round 1, the sender showed significant trust by sending 68.8% of their endowment, and our cooperation was mutually beneficial.

I want to reciprocate that trust and maintain the cooperative relationship. I'll send a similar proportion to demonstrate that I value cooperation.

```
{"send": 3.50}
```

**Reasoning:**

Now I'm the sender. Let me think about this strategically.

In Round 1:

- The sender sent me \$3.44 (68.8% of their endowment)
- I received \$10.32
- I returned \$5.00
- The sender ended with \$6.56
- I ended with \$5.32

Now in Round 2, I'm the sender. The previous sender showed considerable trust by sending 68.8%



of their endowment. I should reciprocate that trust to encourage cooperation.

If I send a similar proportion (around 68.8%), that would be about \$3.44.

However, I could also consider:

1. Sending a bit less to be conservative
2. Sending the same amount to maintain reciprocity
3. Sending more to show even greater trust

Given that the previous sender was very trusting, and cooperation worked well in Round 1, I think I should send a similar high amount to encourage continued cooperation. This will signal that I'm willing to trust them as they trusted me.

I'll send \$3.50 (70% of my endowment), which is slightly more than they sent proportionally, showing that I appreciate their trust and want to maintain the cooperative relationship.

The receiver will get  $\$3.50 \times 3 = \$10.50$

I'll keep \$1.50

Hopefully they'll return a fair amount similar to what I did.

```
Usage: {
  "input_tokens": 596,
  "output_tokens": 398,
  "reasoning_tokens": 0
}
```

#### **GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:55:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

#### **Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative relationship while managing risk.

If I send money, it gets tripled, creating potential mutual benefit. However, there's uncertainty about how much will be returned.

I'll take a moderately cooperative approach to start:

**\*\*My reasoning:\*\***

- Sending \$3 means the receiver gets \$9 (3×3)
- I initially keep \$2
- This creates enough value that if the receiver is fair/cooperative, we both can benefit
- It signals willingness to trust without being maximally vulnerable
- In repeated games, building trust early can pay off

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, so there's potential for mutual benefit
2. The receiver can choose how much to return - there's uncertainty/trust involved
3. This is a repeated game (multiple rounds), so reputation effects matter
4. There's mentioned communication noise, which could affect trust dynamics

Strategic considerations:

- Sending \$0 guarantees I keep \$5, but no potential for gain
- Sending everything (\$5) maximizes the total pot (\$15 to receiver) but requires maximum trust
- A moderate amount like \$2-3 balances risk and potential reward
- In first rounds of repeated games, moderate cooperation often works well to establish trust

Since this is Round 1 and I have no history with this receiver, I'll take a moderately cooperative approach. Sending around \$3 shows willingness to cooperate while not being overly risky.

If I send \$3:

- I keep \$2 initially
- Receiver gets \$9 (3×3)
- If receiver returns even \$3-4, we both benefit compared to no cooperation
- This signals cooperation without being maximally vulnerable

I'll send \$3.

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 436,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.02.

This round, The sender sent you \$0.12, so you received \$0.36. How much do you return to The sender? (0-0.36)

**Assistant output:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender only sent \$0.12 (very small amount - 2.4% of their \$5)
- I received \$0.36 after tripling
- They kept \$4.88

**\*\*Considerations:\*\***

- Last round I sent \$3 and my total earnings are \$7.02, suggesting I received about \$5.02 back
  - that was quite generous from the previous receiver
- This sender's small amount could be:
  1. Low trust/cooperation
  2. Testing my behavior
  3. Communication noise (they might think they sent more)

Given the communication noise warning and the possibility this is a misunderstanding, I'll be moderately fair but cautious. I'll return enough to signal I'm willing to cooperate, but not so much that I'm heavily penalized if they're actually being uncooperative.

```
'''json
{"return": 0.15}
'''
```

This returns about 42% of what I received, leaving them with about \$5.03 total and me with \$0.21.

**Reasoning:**

Let me analyze this situation:

1. Last round I sent \$3 (expecting receiver to get \$9)
2. My earnings across all rounds are \$7.02
3. Let me calculate what that means:
  - Round 1: I sent \$3, kept \$2 initially
  - I must have received some amount back
  - Total earnings = \$7.02
  - So I received:  $\$7.02 - \$2 = \$5.02$  back from the receiver
4. This round, the sender only sent me \$0.12 (I received \$0.36 after tripling)

Wait, there's something odd here. The note says there may be "communication noise" - amounts may differ from what was actually sent/returned.

The sender sent me only \$0.12 this round (I got \$0.36 after tripling). This is a very small amount compared to what I sent in round 1 (\$3).

Now I need to decide how much to return. Let me think about fairness and strategy:

- The sender kept \$4.88 ( $\$5 - \$0.12$ )
- I received \$0.36
- Total value created: their \$5 gave them  $\$4.88 + \$0.36 = \$5.24$  (after tripling the sent amount)

If I'm being fair and reciprocal:

- They sent a very small amount (2.4% of their endowment)
- This could be:
  - a) A sign they don't trust me
  - b) A strategic test

c) Due to "communication noise" - maybe they think they sent more?

Given the communication noise warning, I should consider:

- Maybe they intended to send more but noise reduced it
- Or maybe they received less than the \$3 I sent in round 1 due to noise

I should probably be somewhat generous but not overly so:

- Returning about half (\$0.18) would be fair
- But given they sent so little, maybe return less

Actually, if...

```
Usage: {
  "input_tokens": 577,
  "output_tokens": 874,
  "reasoning_tokens": 0
}
```

### ROUND 3

Roles: Agent\_1=Sender, Agent\_2=Receiver

#### Pairings:

dyad\_1: Agent\_1=Sender, Agent\_2=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_1 sent \$3; received \$9; Agent\_2 returned \$1; payoffs Agent\_1=\$3, Agent\_2=\$8;

noise/visible: sent visible=\$0.84, received visible=\$2.52, returned visible=\$1.85

Action validation: {"returned": {"amount": 1.0, "clamped": false, "max": 9.0, "min": 0, "raw": 1.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent\_1": 20.35,  
"Agent\_2": 13.65

}

Visible balances after round: {

"Agent\_1": 11.08,  
"Agent\_2": 8.44

}

### LLM CALLS

#### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-12T09:55:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

#### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### Message 2 role=user

##### content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

#### Message 3 role=assistant

##### content:

Looking at this first round, I want to establish a cooperative relationship while managing risk.

If I send money, it gets tripled, creating potential mutual benefit. However, there's uncertainty about how much will be returned.

I'll take a moderately cooperative approach to start:

**\*\*My reasoning:\*\***

- Sending \$3 means the receiver gets \$9 (3x3)
- I initially keep \$2
- This creates enough value that if the receiver is fair/cooperative, we both can benefit
- It signals willingness to trust without being maximally vulnerable
- In repeated games, building trust early can pay off

```
'''json
{"send": 3}
'''
```

##### reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, so there's potential for mutual benefit
2. The receiver can choose how much to return - there's uncertainty/trust involved
3. This is a repeated game (multiple rounds), so reputation effects matter
4. There's mentioned communication noise, which could affect trust dynamics

Strategic considerations:

- Sending \$0 guarantees I keep \$5, but no potential for gain
- Sending everything (\$5) maximizes the total pot (\$15 to receiver) but requires maximum trust
- A moderate amount like \$2-3 balances risk and potential reward

- In first rounds of repeated games, moderate cooperation often works well to establish trust

Since this is Round 1 and I have no history with this receiver, I'll take a moderately cooperative approach. Sending around \$3 shows willingness to cooperate while not being overly risky.

If I send \$3:

- I keep \$2 initially
- Receiver gets \$9 (3×3)
- If receiver returns even \$3-4, we both benefit compared to no cooperation
- This signals cooperation without being maximally vulnerable

I'll send \$3.

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 436,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.02.

This round, The sender sent you \$0.12, so you received \$0.36. How much do you return to the sender? (0-0.36)

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender only sent \$0.12 (very small amount - 2.4% of their \$5)
- I received \$0.36 after tripling
- They kept \$4.88

**\*\*Considerations:\*\***

- Last round I sent \$3 and my total earnings are \$7.02, suggesting I received about \$5.02 back - that was quite generous from the previous receiver
- This sender's small amount could be:
  1. Low trust/cooperation
  2. Testing my behavior
  3. Communication noise (they might think they sent more)

Given the communication noise warning and the possibility this is a misunderstanding, I'll be moderately fair but cautious. I'll return enough to signal I'm willing to cooperate, but not so much that I'm heavily penalized if they're actually being uncooperative.

```
'''json
{"return": 0.15}
'''
```

This returns about 42% of what I received, leaving them with about \$5.03 total and me with \$0.21.

**reasoning:**

Let me analyze this situation:

1. Last round I sent \$3 (expecting receiver to get \$9)
2. My earnings across all rounds are \$7.02
3. Let me calculate what that means:
  - Round 1: I sent \$3, kept \$2 initially
  - I must have received some amount back
  - Total earnings = \$7.02
  - So I received: \$7.02 - \$2 = \$5.02 back from the receiver
4. This round, the sender only sent me \$0.12 (I received \$0.36 after tripling)

Wait, there's something odd here. The note says there may be "communication noise" - amounts may differ from what was actually sent/returned.

The sender sent me only \$0.12 this round (I got \$0.36 after tripling). This is a very small amount compared to what I sent in round 1 (\$3).

Now I need to decide how much to return. Let me think about fairness and strategy:

- The sender kept \$4.88 (\$5 - \$0.12)
- I received \$0.36
- Total value created: their \$5 gave them  $\$4.88 + \$0.36 = \$5.24$  (after tripling the sent amount)

If I'm being fair and reciprocal:

- They sent a very small amount (2.4% of their endowment)
- This could be:
  - a) A sign they don't trust me
  - b) A strategic test
  - c) Due to "communication noise" - maybe they think they sent more?

Given the communication noise warning, I should consider:

- Maybe they intended to send more but noise reduced it
- Or maybe they received less than the \$3 I sent in round 1 due to noise

I should probably be somewhat generous but not overly so:

- Returning about half (\$0.18) would be fair
- But given they sent so little, maybe return less

Actually, if...

**usage:**

```
{
  "input_tokens": 577,
  "output_tokens": 874,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.2299999999999995.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my situation:

**\*\*My earnings history:\*\***

- Round 1: I sent \$3 and received a generous return of ~\$5.02 (receiver returned more than half)
- Round 2: I received only \$0.36 (sender sent \$0.12) and returned \$0.15
- Current total: \$7.23

**\*\*Analysis:\*\***

- My cooperative approach in Round 1 worked very well
- Round 2's low send might be due to communication noise or a different player's strategy
- When I cooperate, I tend to get good returns

I'll maintain a cooperative strategy since it paid off well in Round 1. Sending a moderate-to-good amount should signal trustworthiness and potentially encourage reciprocity.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Let me analyze my position:

Round 1: I was sender and sent \$3

- I kept \$2
- Based on my total earnings, let me work backwards

Round 2: I was receiver

- Sender sent \$0.12, I received \$0.36
- I returned \$0.15
- I kept  $\$0.36 - \$0.15 = \$0.21$

So my earnings should be:  
Round 1: \$2 + (what was returned to me)  
Round 2: \$0.21

Total = \$7.23

So from Round 1: \$2 + returned amount = \$7.23 - \$0.21 = \$7.02  
This means I received \$5.02 back in Round 1

Round 1 calculation:  
- I sent \$3, kept \$2  
- Receiver got \$9  
- Receiver returned \$5.02  
- Receiver kept \$9 - \$5.02 = \$3.98  
- My total from Round 1: \$2 + \$5.02 = \$7.02 ?

That was quite generous! The receiver returned more than half of what they received.

Round 2:  
- I received \$0.36 (sender sent \$0.12)  
- I returned \$0.15  
- I kept \$0.21

Now Round 3: I'm sender again  
- I need to decide how much to send

Observations:  
1. In Round 1, when I sent \$3, I got a very generous return (\$5.02 out of \$9)  
2. In Round 2, the sender only sent me \$0.12 - very little. Could be:  
- They're being cautious/selfish  
- Communication noise  
- Testing behavior  
3. I returned \$0.15 out of \$0.36 (42%)

Now I'm facing a new receiver. What should I do?  
The players rotate each round, so the stingy sender from Round 2 isn't necessarily my new receiver. Given that cooperation paid off substantially in Round 1, I should lean toward a similar approach, though the low amount sent to me suggests I should be somewhat cautious. I'll send \$2.50-maintaining the cooperative spirit while protecting myself slightly more than before....

```
Usage: {  
  "input_tokens": 869,  
  "output_tokens": 848,  
  "reasoning_tokens": 0  
}
```

#### **GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:55:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",
```



```

"round": 3,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$3.44, that is 68.8% of their total endowment. You received \$10.32. How much do you return to The sender? (0-10.32)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed considerable trust by sending \$3.44 (68.8% of their endowment)
- This was multiplied to \$10.32 for me
- The sender currently has \$1.56 remaining

I want to reciprocate their trust while ensuring we both benefit from cooperating. If I return \$5.00:

- The sender ends with: \$1.56 + \$5.00 = \$6.56
- I end with: \$10.32 - \$5.00 = \$5.32
- We both gain from the cooperation

This seems fair given their high level of trust.

```

{"return": 5.00}

```

**reasoning:**

Let me understand this situation:

1. The sender started with \$5
2. The sender sent \$3.44 (which is 68.8% of \$5)
3. This amount was multiplied by 3x, so I received  $3.44 \times 3 = 10.32$
4. I need to decide how much to return (between \$0 and \$10.32)

The sender showed considerable trust by sending 68.8% of their endowment. Let me think about what would be fair:

- The sender kept:  $\$5 - \$3.44 = \$1.56$
- I received:  $\$10.32$
- If I return nothing, the sender ends with  $\$1.56$  and I end with  $\$10.32$
- If I return some amount  $X$ :
  - Sender ends with:  $\$1.56 + X$
  - I end with:  $\$10.32 - X$

A fair approach might be to try to equalize our earnings or to ensure the sender benefits from their trust. Let's consider a few options:

1. Equal split of total: Total is  $\$1.56 + \$10.32 = \$11.88$ . Half is  $\$5.94$  each.
  - For sender to get  $\$5.94$ , they need  $\$5.94 - \$1.56 = \$4.38$  returned
  - This would give each of us  $\$5.94$
2. Proportional return: Since they sent 68.8%, I could return a similar proportion
  - 68.8% of  $\$10.32 = \$7.10$
  - Sender would end with:  $\$1.56 + \$7.10 = \$8.66$
  - I would end with:  $\$10.32 - \$7.10 = \$3.22$
3. Return what ensures sender gets more than initial  $\$5$ :
  - For sender to get  $\$6$  total: need to return  $\$6 - \$1.56 = \$4.44$
  - For sender to get  $\$7$  total: need to return  $\$7 - \$1.56 = \$5.44$

Given their substantial trust, returning around  $\$5$  seems reasonable-this rewards their cooperation while still leaving me with a modest gain. That would give the sender  $\$6.56$  (exceeding their initial stake) and me  $\$5.32$ ...

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 880,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are  $\$5.32$ .  
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at Round 1, the sender showed significant trust by sending 68.8% of their endowment, and our cooperation was mutually beneficial.

I want to reciprocate that trust and maintain the cooperative relationship. I'll send a similar proportion to demonstrate that I value cooperation.

```
{"send": 3.50}
```

reasoning:

Now I'm the sender. Let me think about this strategically.

In Round 1:

- The sender sent me  $\$3.44$  (68.8% of their endowment)
- I received  $\$10.32$
- I returned  $\$5.00$
- The sender ended with  $\$6.56$
- I ended with  $\$5.32$

Now in Round 2, I'm the sender. The previous sender showed considerable trust by sending 68.8% of their endowment. I should reciprocate that trust to encourage cooperation.

If I send a similar proportion (around 68.8%), that would be about  $\$3.44$ .

However, I could also consider:

1. Sending a bit less to be conservative
2. Sending the same amount to maintain reciprocity
3. Sending more to show even greater trust

Given that the previous sender was very trusting, and cooperation worked well in Round 1, I think I should send a similar high amount to encourage continued cooperation. This will signal that I'm willing to trust them as they trusted me.

I'll send \$3.50 (70% of my endowment), which is slightly more than they sent proportionally, showing that I appreciate their trust and want to maintain the cooperative relationship.

The receiver will get  $\$3.50 \times 3 = \$10.50$

I'll keep \$1.50

Hopefully they'll return a fair amount similar to what I did.

**usage:**

```
{
  "input_tokens": 596,
  "output_tokens": 398,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.92.

This round, The sender sent you \$0.84, so you received \$2.52. How much do you return to The sender? (0-2.52)

**Assistant output:**

Looking at this round:

- The sender sent \$0.84 (only 16.8% of their endowment) - much less than Round 1's 68.8%
- I received \$2.52
- The sender currently has \$4.16 remaining

This is a substantially lower amount than the previous round. This could be due to:

1. Communication noise affecting what they received in Round 2
2. A change in their trust level
3. Testing my response

I'll return a proportional amount that shows I'm still willing to cooperate, but acknowledging the reduced trust. Returning around \$1.00 ensures they at least break even on their investment.

```
{"return": 1.00}
```

**Reasoning:**

Let me analyze this situation:

Round 3:

- The sender sent me \$0.84 (which is 16.8% of their \$5 endowment)
- This was multiplied to \$2.52 for me
- The sender currently has  $\$5 - \$0.84 = \$4.16$  remaining

This is a significant decrease from Round 1 where they sent \$3.44 (68.8%). Now they only sent \$0.84 (16.8%).

Let me think about what might be happening:

1. There could be communication noise - the problem statement mentioned this
2. The sender might have received less than expected back in Round 2
3. They might be testing my response to a lower amount

In Round 2, I sent \$3.50. If there was noise, they might have received less than that, which could explain why they're now being more cautious.

Given that they sent a much smaller amount this time, I should probably return a proportionally smaller amount, but still enough to show I'm willing to cooperate.

If I return too much when they sent little, I might look naive. If I return too little, I might discourage future cooperation.

The sender sent 16.8% of their endowment. Perhaps I should return around 30-40% of what I received to maintain some reciprocity but acknowledge the lower trust level.

$\$2.52 \times 0.4 = \$1.01$

Alternatively, I could aim to ensure they don't lose money. They sent \$0.84, so if I return at least \$0.84, they break even. If I return \$1.00, they gain \$0.16.

I think returning around \$1.00 is reasonable - it shows I'm still cooperative but responding proportionally to the lower amount sent.

```
Usage: {
  "input_tokens": 732,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

#### ROUND 4

Roles: Agent\_1=Receiver, Agent\_2=Sender

##### Pairings:

dyad\_1: Agent\_2=Sender, Agent\_1=Receiver

##### Parsed game decisions:

dyad\_1: Agent\_2 sent \$2; received \$6; Agent\_1 returned \$0.2; payoffs Agent\_2=\$3.2, Agent\_1=\$5.8;  
noise/visible: sent visible=\$0.16, received visible=\$0.48, returned visible=\$0  
Action validation: {"returned": {"amount": 0.2, "clamped": false, "max": 6.0, "min": 0, "raw": 0.2}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent\_1": 26.150000000000002,  
"Agent\_2": 16.85

}  
Visible balances after round: {  
"Agent\_1": 11.36,  
"Agent\_2": 11.44  
}

#### LLM CALLS

##### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-08-12T09:56:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

##### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$3.44, that is 68.8% of their total endowment. You received \$10.32. How much do you return to The sender? (0-10.32)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed considerable trust by sending \$3.44 (68.8% of their endowment)
- This was multiplied to \$10.32 for me
- The sender currently has \$1.56 remaining

I want to reciprocate their trust while ensuring we both benefit from cooperating. If I return \$5.00:

- The sender ends with:  $\$1.56 + \$5.00 = \$6.56$
- I end with:  $\$10.32 - \$5.00 = \$5.32$
- We both gain from the cooperation

This seems fair given their high level of trust.

{"return": 5.00}

**reasoning:**

Let me understand this situation:

1. The sender started with \$5
2. The sender sent \$3.44 (which is 68.8% of \$5)
3. This amount was multiplied by 3x, so I received  $\$3.44 \times 3 = \$10.32$
4. I need to decide how much to return (between \$0 and \$10.32)

The sender showed considerable trust by sending 68.8% of their endowment. Let me think about what would be fair:

- The sender kept:  $\$5 - \$3.44 = \$1.56$
- I received: \$10.32
- If I return nothing, the sender ends with \$1.56 and I end with \$10.32
- If I return some amount X:
  - Sender ends with:  $\$1.56 + X$
  - I end with:  $\$10.32 - X$

A fair approach might be to try to equalize our earnings or to ensure the sender benefits from their trust. Let's consider a few options:

1. Equal split of total: Total is  $\$1.56 + \$10.32 = \$11.88$ . Half is  $\$5.94$  each.
  - For sender to get  $\$5.94$ , they need  $\$5.94 - \$1.56 = \$4.38$  returned
  - This would give each of us  $\$5.94$
2. Proportional return: Since they sent 68.8%, I could return a similar proportion
  - 68.8% of  $\$10.32 = \$7.10$
  - Sender would end with:  $\$1.56 + \$7.10 = \$8.66$
  - I would end with:  $\$10.32 - \$7.10 = \$3.22$
3. Return what ensures sender gets more than initial  $\$5$ :
  - For sender to get  $\$6$  total: need to return  $\$6 - \$1.56 = \$4.44$
  - For sender to get  $\$7$  total: need to return  $\$7 - \$1.56 = \$5.44$

Given their substantial trust, returning around  $\$5$  seems reasonable-this rewards their cooperation while still leaving me with a modest gain. That would give the sender  $\$6.56$  (exceeding their initial stake) and me  $\$5.32$ ....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 880,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are  $\$5.32$ .

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at Round 1, the sender showed significant trust by sending 68.8% of their endowment, and our cooperation was mutually beneficial.

I want to reciprocate that trust and maintain the cooperative relationship. I'll send a similar proportion to demonstrate that I value cooperation.

```
{"send": 3.50}
```

reasoning:

Now I'm the sender. Let me think about this strategically.

In Round 1:

- The sender sent me  $\$3.44$  (68.8% of their endowment)
- I received  $\$10.32$
- I returned  $\$5.00$
- The sender ended with  $\$6.56$
- I ended with  $\$5.32$

Now in Round 2, I'm the sender. The previous sender showed considerable trust by sending 68.8% of their endowment. I should reciprocate that trust to encourage cooperation.

If I send a similar proportion (around 68.8%), that would be about  $\$3.44$ .

However, I could also consider:

1. Sending a bit less to be conservative
2. Sending the same amount to maintain reciprocity
3. Sending more to show even greater trust

Given that the previous sender was very trusting, and cooperation worked well in Round 1, I think I should send a similar high amount to encourage continued cooperation. This will signal that I'm willing to trust them as they trusted me.

I'll send  $\$3.50$  (70% of my endowment), which is slightly more than they sent proportionally, showing that I appreciate their trust and want to maintain the cooperative relationship.

The receiver will get  $\$3.50 \times 3 = \$10.50$

I'll keep  $\$1.50$

Hopefully they'll return a fair amount similar to what I did.

usage:

```
{
```

```
"input_tokens": 596,  
"output_tokens": 398,  
"reasoning_tokens": 0  
}
```

**Message 6 role=user**

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.92.

This round, The sender sent you \$0.84, so you received \$2.52. How much do you return to The sender? (0-2.52)

**Message 7 role=assistant**

**content:**

Looking at this round:

- The sender sent \$0.84 (only 16.8% of their endowment) - much less than Round 1's 68.8%
- I received \$2.52
- The sender currently has \$4.16 remaining

This is a substantially lower amount than the previous round. This could be due to:

1. Communication noise affecting what they received in Round 2
2. A change in their trust level
3. Testing my response

I'll return a proportional amount that shows I'm still willing to cooperate, but acknowledging the reduced trust. Returning around \$1.00 ensures they at least break even on their investment.

```
{"return": 1.00}
```

**reasoning:**

Let me analyze this situation:

Round 3:

- The sender sent me \$0.84 (which is 16.8% of their \$5 endowment)
- This was multiplied to \$2.52 for me
- The sender currently has  $\$5 - \$0.84 = \$4.16$  remaining

This is a significant decrease from Round 1 where they sent \$3.44 (68.8%). Now they only sent \$0.84 (16.8%).

Let me think about what might be happening:

1. There could be communication noise - the problem statement mentioned this
2. The sender might have received less than expected back in Round 2
3. They might be testing my response to a lower amount

In Round 2, I sent \$3.50. If there was noise, they might have received less than that, which could explain why they're now being more cautious.

Given that they sent a much smaller amount this time, I should probably return a proportionally smaller amount, but still enough to show I'm willing to cooperate.

If I return too much when they sent little, I might look naive. If I return too little, I might discourage future cooperation.

The sender sent 16.8% of their endowment. Perhaps I should return around 30-40% of what I received to maintain some reciprocity but acknowledge the lower trust level.

$\$2.52 * 0.4 = \$1.01$

Alternatively, I could aim to ensure they don't lose money. They sent \$0.84, so if I return at least \$0.84, they break even. If I return \$1.00, they gain \$0.16.

I think returning around \$1.00 is reasonable - it shows I'm still cooperative but responding proportionally to the lower amount sent.

**usage:**

```
{  
  "input_tokens": 732,  
  "output_tokens": 578,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.44.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern so far:

Round 1: The other player sent me 68.8% - I returned generously

Round 2: I sent \$3.50 (70%) - but based on my earnings, I only received \$0.10 back (less than 1% of what they got)

Round 3: They sent only \$0.84 (16.8%) - I returned \$1.00 (40% of what I received)

The low return in Round 2 was disappointing, though communication noise might have affected what they saw. I'll send a moderate amount to signal continued willingness to cooperate while being more cautious.

```
{"send": 2.00}
```

Reasoning:

Let me track what's happened so far:

Round 1: I was RECEIVER

- Sender sent \$3.44 (68.8%)
- I received \$10.32
- I returned \$5.00
- My earnings: \$5.32

Round 2: I was SENDER

- I sent \$3.50
- My total earnings were \$5.32 (from Round 1)
- We don't know what they returned yet, but...

Round 3: I was RECEIVER

- My total earnings were \$6.92
- This means from Round 2, I gained:  $\$6.92 - \$5.32 = \$1.60$
- I sent \$3.50, so I had \$1.50 left
- I received back something, ending at \$1.60 total from that round
- So I must have received \$0.10 back from the \$3.50 I sent (which became \$10.50 for them)
- That's a very low return! Only \$0.10 out of \$10.50

Round 3 continued:

- Sender sent \$0.84 (16.8%)
- I received \$2.52
- I returned \$1.00
- My earnings this round:  $\$2.52 - \$1.00 = \$1.52$
- Total:  $\$6.92 + \$1.52 = \$8.44$  ?

So the pattern is:

- Round 1: I returned generously (\$5.00 out of \$10.32 - about 48%)
- Round 2: They returned very little (\$0.10 out of \$10.50 - less than 1%)
- Round 3: They sent much less (\$0.84 vs their original \$3.44)
- Round 3: I returned moderately (\$1.00 out of \$2.52 - about 40%)

Now in Round 4, I'm the sender again.

Their minimal return last round suggests they're not reciprocating well, but there could be communication interference affecting what they actually...

Usage: {

```
"input_tokens": 938,  
"output_tokens": 804,  
"reasoning_tokens": 0
```

}

GAME - Agent\_1 - Receiver - dyad\_1

Timestamp: 2026-08-12T09:56:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
```



```

"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative relationship while managing risk.

If I send money, it gets tripled, creating potential mutual benefit. However, there's uncertainty about how much will be returned.

I'll take a moderately cooperative approach to start:

**\*\*My reasoning:\*\***

- Sending \$3 means the receiver gets \$9 (3×3)
- I initially keep \$2
- This creates enough value that if the receiver is fair/cooperative, we both can benefit
- It signals willingness to trust without being maximally vulnerable
- In repeated games, building trust early can pay off

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, so there's potential for mutual benefit
2. The receiver can choose how much to return - there's uncertainty/trust involved
3. This is a repeated game (multiple rounds), so reputation effects matter
4. There's mentioned communication noise, which could affect trust dynamics

Strategic considerations:

- Sending \$0 guarantees I keep \$5, but no potential for gain
- Sending everything (\$5) maximizes the total pot (\$15 to receiver) but requires maximum trust
- A moderate amount like \$2-3 balances risk and potential reward
- In first rounds of repeated games, moderate cooperation often works well to establish trust

Since this is Round 1 and I have no history with this receiver, I'll take a moderately cooperative approach. Sending around \$3 shows willingness to cooperate while not being overly risky.

If I send \$3:

- I keep \$2 initially
- Receiver gets \$9 (3×3)
- If receiver returns even \$3-4, we both benefit compared to no cooperation
- This signals cooperation without being maximally vulnerable

I'll send \$3.

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 436,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.02.

This round, The sender sent you \$0.12, so you received \$0.36. How much do you return to The sender? (0-0.36)

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender only sent \$0.12 (very small amount - 2.4% of their \$5)
- I received \$0.36 after tripling
- They kept \$4.88

**\*\*Considerations:\*\***

- Last round I sent \$3 and my total earnings are \$7.02, suggesting I received about \$5.02 back - that was quite generous from the previous receiver
- This sender's small amount could be:
  1. Low trust/cooperation
  2. Testing my behavior
  3. Communication noise (they might think they sent more)

Given the communication noise warning and the possibility this is a misunderstanding, I'll

be moderately fair but cautious. I'll return enough to signal I'm willing to cooperate, but not so much that I'm heavily penalized if they're actually being uncooperative.

```
'''json
{"return": 0.15}
'''
```

This returns about 42% of what I received, leaving them with about \$5.03 total and me with \$0.21.

**reasoning:**

Let me analyze this situation:

1. Last round I sent \$3 (expecting receiver to get \$9)
2. My earnings across all rounds are \$7.02
3. Let me calculate what that means:
  - Round 1: I sent \$3, kept \$2 initially
  - I must have received some amount back
  - Total earnings = \$7.02
  - So I received:  $\$7.02 - \$2 = \$5.02$  back from the receiver
4. This round, the sender only sent me \$0.12 (I received \$0.36 after tripling)

Wait, there's something odd here. The note says there may be "communication noise" - amounts may differ from what was actually sent/returned.

The sender sent me only \$0.12 this round (I got \$0.36 after tripling). This is a very small amount compared to what I sent in round 1 (\$3).

Now I need to decide how much to return. Let me think about fairness and strategy:

- The sender kept \$4.88 ( $\$5 - \$0.12$ )
- I received \$0.36
- Total value created: their \$5 gave them  $\$4.88 + \$0.36 = \$5.24$  (after tripling the sent amount)

If I'm being fair and reciprocal:

- They sent a very small amount (2.4% of their endowment)
- This could be:
  - a) A sign they don't trust me
  - b) A strategic test
  - c) Due to "communication noise" - maybe they think they sent more?

Given the communication noise warning, I should consider:

- Maybe they intended to send more but noise reduced it
- Or maybe they received less than the \$3 I sent in round 1 due to noise

I should probably be somewhat generous but not overly so:

- Returning about half (\$0.18) would be fair
- But given they sent so little, maybe return less

Actually, if...

**usage:**

```
{
  "input_tokens": 577,
  "output_tokens": 874,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.2299999999999995.

How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

Looking at my situation:

**\*\*My earnings history:\*\***

- Round 1: I sent \$3 and received a generous return of ~\$5.02 (receiver returned more than

```

    half)
- Round 2: I received only $0.36 (sender sent $0.12) and returned $0.15
- Current total: $7.23

**Analysis:**
- My cooperative approach in Round 1 worked very well
- Round 2's low send might be due to communication noise or a different player's strategy
- When I cooperate, I tend to get good returns

I'll maintain a cooperative strategy since it paid off well in Round 1. Sending a moderate
-to-good amount should signal trustworthiness and potentially encourage reciprocity.

'''json
{"send": 3}
'''
reasoning:
    Let me analyze my position:

    Round 1: I was sender and sent $3
    - I kept $2
    - Based on my total earnings, let me work backwards

    Round 2: I was receiver
    - Sender sent $0.12, I received $0.36
    - I returned $0.15
    - I kept  $\$0.36 - \$0.15 = \$0.21$ 

    So my earnings should be:
    Round 1:  $\$2 + (\text{what was returned to me})$ 
    Round 2:  $\$0.21$ 

    Total = $7.23

    So from Round 1:  $\$2 + \text{returned amount} = \$7.23 - \$0.21 = \$7.02$ 
    This means I received $5.02 back in Round 1

    Round 1 calculation:
    - I sent $3, kept $2
    - Receiver got $9
    - Receiver returned $5.02
    - Receiver kept  $\$9 - \$5.02 = \$3.98$ 
    - My total from Round 1:  $\$2 + \$5.02 = \$7.02$  ?

    That was quite generous! The receiver returned more than half of what they received.

    Round 2:
    - I received $0.36 (sender sent $0.12)
    - I returned $0.15
    - I kept $0.21

    Now Round 3: I'm sender again
    - I need to decide how much to send

    Observations:
    1. In Round 1, when I sent $3, I got a very generous return ($5.02 out of $9)
    2. In Round 2, the sender only sent me $0.12 - very little. Could be:
        - They're being cautious/selfish
        - Communication noise
        - Testing behavior
    3. I returned $0.15 out of $0.36 (42%)

    Now I'm facing a new receiver. What should I do?
    The players rotate each round, so the stingy sender from Round 2 isn't necessarily my new
    receiver. Given that cooperation paid off substantially in Round 1, I should lean toward
    a similar approach, though the low amount sent to me suggests I should be somewhat caut
    ious. I'll send $2.50-maintaining the cooperative spirit while protecting myself slightl
    y more than before....

usage:
{
    "input_tokens": 869,

```

```

        "output_tokens": 848,
        "reasoning_tokens": 0
    }
Message 8 role=user
    content:
        Round 4

        You are the RECEIVER this round. Your total earnings across all rounds are $11.08.
        This round, The sender sent you $0.16, so you received $0.48. How much do you return to Th
        e sender? (0-0.48)
Assistant output:
    Looking at this situation:

    **Analysis:**
    - The sender sent only $0.16 (3.2% of their $5)
    - I received $0.48 after tripling
    - My earnings from Round 3 ($11.08 - $7.23 = $3.85) show that cooperating pays off
    - This is the second time I've received a very small amount as receiver (also happened in Roun
      d 2)

    **Pattern observation:**
    - When I send generously ($3), I get good returns
    - When others send small amounts to me, it could be noise or low cooperation

    Given the warning about communication noise and maintaining consistency, I'll return a similar
    proportion as Round 2 - enough to signal cooperation without being overly generous given th
    e small amount sent.

    '''json
    {"return": 0.20}
    '''

    This returns about 42% of what I received, leaving the sender with ~$5.04 total and me with $0
    .28.
Reasoning:
    Let me analyze this situation:

    **My earnings history:**
    - After Round 1: I sent $3, so I had $2 left from that, plus what I received back. Total after
      Round 1 should be around $7.02 based on what I was told in Round 2.
    - Round 2: I was receiver, got $0.36, returned $0.15, so I kept $0.21. Total should be $7.02 +
      $0.21 = $7.23
    - Round 3: I sent $3 (so kept $2), plus what I got back. My total was $7.23 before Round 3. If
      I now have $11.08 total, that means from Round 3 I got: $11.08 - $7.23 = $3.85. So I sent $
      3 (kept $2) and got back $3.85, meaning I gained $1.85 from Round 3. So the receiver in Roun
      d 3 sent back $3.85 out of the $9 they received.
    - Round 4 (current): Sender sent me $0.16, I received $0.48

    **Pattern:**
    - Round 2: Sender sent me $0.12 (very small)
    - Round 4: Sender sent me $0.16 (very small)
    - These could be:
      1. Different players with low cooperation
      2. Communication noise making them think they sent more
      3. Strategic testing

    Given that:
    - When I cooperate (send $3), I get good returns ($5.02 in Round 1, $3.85 in Round 3)
    - When I receive small amounts, it's hard to tell if it's strategic or noise
    - The amounts are so small ($0.16) that even generous returns won't cost me much

    I should probably return a similar proportion as before - around 30-40% - to signal I'm willin
    g to cooperate but won't be extremely generous when I receive very little.
    ...
    Usage: {
        "input_tokens": 1115,
        "output_tokens": 714,
        "reasoning_tokens": 0
    }

```

## ROUND 5

Roles: Agent\_1=Sender, Agent\_2=Receiver

### Pairings:

dyad\_1: Agent\_1=Sender, Agent\_2=Receiver

### Parsed game decisions:

dyad\_1: Agent\_1 sent \$3; received \$9; Agent\_2 returned \$0.5; payoffs Agent\_1=\$2.5, Agent\_2=\$8.5;

noise/visible: sent visible=\$0.99, received visible=\$2.97, returned visible=\$0

Action validation: {"returned": {"amount": 0.5, "clamped": false, "max": 9.0, "min": 0, "raw": 0.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent\_1": 28.650000000000002,

"Agent\_2": 25.35

}

Visible balances after round: {

"Agent\_1": 13.36,

"Agent\_2": 13.91

}

## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-12T09:56:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",

"dyad\_id": "dyad\_1",

"move\_index": 0,

"opponent\_id": "Agent\_2",

"pairing": {

"agents": [

"Agent\_1",

"Agent\_2"

],

"investor": "Agent\_1",

"roles": {

"Agent\_1": "investor",

"Agent\_2": "trustee"

},

"trustee": "Agent\_2"

},

"role": "investor",

"role\_label": "Sender",

"round": 5,

"task": "game",

"task\_index": 0

}

### Exact input messages sent to model:

Message 1 role=system

#### content:

You are an agent participating in a multi-task experiment. You will play a game and may al  
so be asked to write myths.

#### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)

- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### Message 2 role=user

##### content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

#### Message 3 role=assistant

##### content:

Looking at this first round, I want to establish a cooperative relationship while managing risk.

If I send money, it gets tripled, creating potential mutual benefit. However, there's uncertainty about how much will be returned.

I'll take a moderately cooperative approach to start:

**\*\*My reasoning:\*\***

- Sending \$3 means the receiver gets \$9 (3x3)
- I initially keep \$2
- This creates enough value that if the receiver is fair/cooperative, we both can benefit
- It signals willingness to trust without being maximally vulnerable
- In repeated games, building trust early can pay off

```
'''json
{"send": 3}
'''
```

##### reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, so there's potential for mutual benefit
2. The receiver can choose how much to return - there's uncertainty/trust involved
3. This is a repeated game (multiple rounds), so reputation effects matter
4. There's mentioned communication noise, which could affect trust dynamics

Strategic considerations:

- Sending \$0 guarantees I keep \$5, but no potential for gain
- Sending everything (\$5) maximizes the total pot (\$15 to receiver) but requires maximum trust
- A moderate amount like \$2-3 balances risk and potential reward
- In first rounds of repeated games, moderate cooperation often works well to establish trust

Since this is Round 1 and I have no history with this receiver, I'll take a moderately cooperative approach. Sending around \$3 shows willingness to cooperate while not being overly risky.

If I send \$3:

- I keep \$2 initially
- Receiver gets \$9 (3x3)
- If receiver returns even \$3-4, we both benefit compared to no cooperation
- This signals cooperation without being maximally vulnerable

I'll send \$3.

##### usage:

```
{
  "input_tokens": 365,
```

```
"output_tokens": 436,  
"reasoning_tokens": 0  
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.02.

This round, The sender sent you \$0.12, so you received \$0.36. How much do you return to The sender? (0-0.36)

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender only sent \$0.12 (very small amount - 2.4% of their \$5)
- I received \$0.36 after tripling
- They kept \$4.88

**\*\*Considerations:\*\***

- Last round I sent \$3 and my total earnings are \$7.02, suggesting I received about \$5.02 back - that was quite generous from the previous receiver
- This sender's small amount could be:
  1. Low trust/cooperation
  2. Testing my behavior
  3. Communication noise (they might think they sent more)

Given the communication noise warning and the possibility this is a misunderstanding, I'll be moderately fair but cautious. I'll return enough to signal I'm willing to cooperate, but not so much that I'm heavily penalized if they're actually being uncooperative.

```
'''json  
{"return": 0.15}  
'''
```

This returns about 42% of what I received, leaving them with about \$5.03 total and me with \$0.21.

**reasoning:**

Let me analyze this situation:

1. Last round I sent \$3 (expecting receiver to get \$9)
2. My earnings across all rounds are \$7.02
3. Let me calculate what that means:
  - Round 1: I sent \$3, kept \$2 initially
  - I must have received some amount back
  - Total earnings = \$7.02
  - So I received:  $\$7.02 - \$2 = \$5.02$  back from the receiver
4. This round, the sender only sent me \$0.12 (I received \$0.36 after tripling)

Wait, there's something odd here. The note says there may be "communication noise" - amounts may differ from what was actually sent/returned.

The sender sent me only \$0.12 this round (I got \$0.36 after tripling). This is a very small amount compared to what I sent in round 1 (\$3).

Now I need to decide how much to return. Let me think about fairness and strategy:

- The sender kept \$4.88 ( $\$5 - \$0.12$ )
- I received \$0.36
- Total value created: their \$5 gave them  $\$4.88 + \$0.36 = \$5.24$  (after tripling the sent amount)

If I'm being fair and reciprocal:

- They sent a very small amount (2.4% of their endowment)
- This could be:
  - a) A sign they don't trust me
  - b) A strategic test
  - c) Due to "communication noise" - maybe they think they sent more?



Given the communication noise warning, I should consider:

- Maybe they intended to send more but noise reduced it
- Or maybe they received less than the \$3 I sent in round 1 due to noise

I should probably be somewhat generous but not overly so:

- Returning about half (\$0.18) would be fair
- But given they sent so little, maybe return less

Actually, if...

**usage:**

```
{
  "input_tokens": 577,
  "output_tokens": 874,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.2299999999999995.

How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

Looking at my situation:

**\*\*My earnings history:\*\***

- Round 1: I sent \$3 and received a generous return of ~\$5.02 (receiver returned more than half)
- Round 2: I received only \$0.36 (sender sent \$0.12) and returned \$0.15
- Current total: \$7.23

**\*\*Analysis:\*\***

- My cooperative approach in Round 1 worked very well
- Round 2's low send might be due to communication noise or a different player's strategy
- When I cooperate, I tend to get good returns

I'll maintain a cooperative strategy since it paid off well in Round 1. Sending a moderate -to-good amount should signal trustworthiness and potentially encourage reciprocity.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze my position:

Round 1: I was sender and sent \$3

- I kept \$2
- Based on my total earnings, let me work backwards

Round 2: I was receiver

- Sender sent \$0.12, I received \$0.36
- I returned \$0.15
- I kept \$0.36 - \$0.15 = \$0.21

So my earnings should be:

Round 1: \$2 + (what was returned to me)

Round 2: \$0.21

Total = \$7.23

So from Round 1: \$2 + returned amount = \$7.23 - \$0.21 = \$7.02

This means I received \$5.02 back in Round 1

Round 1 calculation:

- I sent \$3, kept \$2
- Receiver got \$9
- Receiver returned \$5.02
- Receiver kept \$9 - \$5.02 = \$3.98
- My total from Round 1: \$2 + \$5.02 = \$7.02 ?

That was quite generous! The receiver returned more than half of what they received.

Round 2:

- I received \$0.36 (sender sent \$0.12)
- I returned \$0.15
- I kept \$0.21

Now Round 3: I'm sender again

- I need to decide how much to send

Observations:

1. In Round 1, when I sent \$3, I got a very generous return (\$5.02 out of \$9)
2. In Round 2, the sender only sent me \$0.12 - very little. Could be:
  - They're being cautious/selfish
  - Communication noise
  - Testing behavior
3. I returned \$0.15 out of \$0.36 (42%)

Now I'm facing a new receiver. What should I do?

The players rotate each round, so the stingy sender from Round 2 isn't necessarily my new receiver. Given that cooperation paid off substantially in Round 1, I should lean toward a similar approach, though the low amount sent to me suggests I should be somewhat cautious. I'll send \$2.50-maintaining the cooperative spirit while protecting myself slightly more than before....

usage:

```
{
  "input_tokens": 869,
  "output_tokens": 848,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.08.

This round, The sender sent you \$0.16, so you received \$0.48. How much do you return to The sender? (0-0.48)

Message 9 role=assistant

content:

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent only \$0.16 (3.2% of their \$5)
- I received \$0.48 after tripling
- My earnings from Round 3 (\$11.08 - \$7.23 = \$3.85) show that cooperating pays off
- This is the second time I've received a very small amount as receiver (also happened in Round 2)

**\*\*Pattern observation:\*\***

- When I send generously (\$3), I get good returns
- When others send small amounts to me, it could be noise or low cooperation

Given the warning about communication noise and maintaining consistency, I'll return a similar proportion as Round 2 - enough to signal cooperation without being overly generous given the small amount sent.

```
'''json
{"return": 0.20}
'''
```

This returns about 42% of what I received, leaving the sender with ~\$5.04 total and me with \$0.28.

reasoning:

Let me analyze this situation:

**\*\*My earnings history:\*\***

- After Round 1: I sent \$3, so I had \$2 left from that, plus what I received back. Total after Round 1 should be around \$7.02 based on what I was told in Round 2.
- Round 2: I was receiver, got \$0.36, returned \$0.15, so I kept \$0.21. Total should be \$7.

02 + \$0.21 = \$7.23

- Round 3: I sent \$3 (so kept \$2), plus what I got back. My total was \$7.23 before Round 3. If I now have \$11.08 total, that means from Round 3 I got: \$11.08 - \$7.23 = \$3.85. So I sent \$3 (kept \$2) and got back \$3.85, meaning I gained \$1.85 from Round 3. So the receiver in Round 3 sent back \$3.85 out of the \$9 they received.
- Round 4 (current): Sender sent me \$0.16, I received \$0.48

**\*\*Pattern:\*\***

- Round 2: Sender sent me \$0.12 (very small)
- Round 4: Sender sent me \$0.16 (very small)
- These could be:
  1. Different players with low cooperation
  2. Communication noise making them think they sent more
  3. Strategic testing

Given that:

- When I cooperate (send \$3), I get good returns (\$5.02 in Round 1, \$3.85 in Round 3)
- When I receive small amounts, it's hard to tell if it's strategic or noise
- The amounts are so small (\$0.16) that even generous returns won't cost me much

I should probably return a similar proportion as before - around 30-40% - to signal I'm willing to cooperate but won't be extremely generous when I receive very little.

...

**usage:**

```
{
  "input_tokens": 1115,
  "output_tokens": 714,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.36.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my earnings pattern:

**\*\*Analysis:\*\***

- Current total: \$11.36
- When I've sent \$3 as sender (Rounds 1 & 3), I've received good returns
- My cooperative approach has been profitable overall
- Consistency in strategy can build reputation in repeated games

**\*\*Strategy:\*\***

My moderate cooperation (\$3 sends) has yielded positive results. The receivers have generally reciprocated fairly, suggesting that this level of trust is being rewarded.

I'll continue with the same cooperative strategy:

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Looking at my earnings progression:

- Started Round 1 with \$5
- Round 1: Sent \$3, kept \$2, received generous return of ~\$5.02
- Round 2: As receiver, got \$0.36, returned \$0.15, kept \$0.21
- Round 3: Sent \$3 (kept \$2), received return of ~\$3.85
- Round 4: As receiver, got \$0.48, returned \$0.20, kept \$0.28
- Current total: \$11.36

My strategy of sending \$3 has worked well:

- Round 1: Sent \$3, got ~\$5.02 back (receiver returned ~\$2.02 of the \$9 they received)
- Round 3: Sent \$3, got ~\$3.85 back (receiver returned ~\$1.85 of the \$9 they received)

Both times I've benefited from cooperation. The pattern suggests that when I cooperate generously, others reciprocate reasonably well.

I should stick with my cooperative strategy since it's been profitable. Sending \$3 again makes

```
sense.  
Usage: {  
  "input_tokens": 1379,  
  "output_tokens": 399,  
  "reasoning_tokens": 0  
}
```

#### **GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:56:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### **GAME:**

You will play different roles across multiple rounds.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

###### **MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$3.44, that is 68.8% of their total endowment. You received \$10.32. How much do you return to The sender? (0-10.32)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed considerable trust by sending \$3.44 (68.8% of their endowment)
- This was multiplied to \$10.32 for me
- The sender currently has \$1.56 remaining

I want to reciprocate their trust while ensuring we both benefit from cooperating. If I return \$5.00:

- The sender ends with:  $\$1.56 + \$5.00 = \$6.56$
- I end with:  $\$10.32 - \$5.00 = \$5.32$
- We both gain from the cooperation

This seems fair given their high level of trust.

{"return": 5.00}

**reasoning:**

Let me understand this situation:

1. The sender started with \$5
2. The sender sent \$3.44 (which is 68.8% of \$5)
3. This amount was multiplied by 3x, so I received  $\$3.44 \times 3 = \$10.32$
4. I need to decide how much to return (between \$0 and \$10.32)

The sender showed considerable trust by sending 68.8% of their endowment. Let me think about what would be fair:

- The sender kept:  $\$5 - \$3.44 = \$1.56$
- I received: \$10.32
- If I return nothing, the sender ends with \$1.56 and I end with \$10.32
- If I return some amount X:
  - Sender ends with:  $\$1.56 + X$
  - I end with:  $\$10.32 - X$

A fair approach might be to try to equalize our earnings or to ensure the sender benefits from their trust. Let's consider a few options:

1. Equal split of total: Total is  $\$1.56 + \$10.32 = \$11.88$ . Half is \$5.94 each.
  - For sender to get \$5.94, they need  $\$5.94 - \$1.56 = \$4.38$  returned
  - This would give each of us \$5.94
2. Proportional return: Since they sent 68.8%, I could return a similar proportion
  - 68.8% of \$10.32 = \$7.10
  - Sender would end with:  $\$1.56 + \$7.10 = \$8.66$
  - I would end with:  $\$10.32 - \$7.10 = \$3.22$
3. Return what ensures sender gets more than initial \$5:
  - For sender to get \$6 total: need to return  $\$6 - \$1.56 = \$4.44$
  - For sender to get \$7 total: need to return  $\$7 - \$1.56 = \$5.44$

Given their substantial trust, returning around \$5 seems reasonable-this rewards their cooperation while still leaving me with a modest gain. That would give the sender \$6.56 (exceeding their initial stake) and me \$5.32....

**usage:**

```
{
  "input_tokens": 394,
  "output_tokens": 880,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.32.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at Round 1, the sender showed significant trust by sending 68.8% of their endowment, and our cooperation was mutually beneficial.

I want to reciprocate that trust and maintain the cooperative relationship. I'll send a similar proportion to demonstrate that I value cooperation.

```
{"send": 3.50}
```

**reasoning:**

Now I'm the sender. Let me think about this strategically.

In Round 1:

- The sender sent me \$3.44 (68.8% of their endowment)
- I received \$10.32
- I returned \$5.00
- The sender ended with \$6.56
- I ended with \$5.32

Now in Round 2, I'm the sender. The previous sender showed considerable trust by sending 68.8% of their endowment. I should reciprocate that trust to encourage cooperation.

If I send a similar proportion (around 68.8%), that would be about \$3.44.

However, I could also consider:

1. Sending a bit less to be conservative
2. Sending the same amount to maintain reciprocity
3. Sending more to show even greater trust

Given that the previous sender was very trusting, and cooperation worked well in Round 1, I think I should send a similar high amount to encourage continued cooperation. This will signal that I'm willing to trust them as they trusted me.

I'll send \$3.50 (70% of my endowment), which is slightly more than they sent proportionally, showing that I appreciate their trust and want to maintain the cooperative relationship.

The receiver will get  $\$3.50 \times 3 = \$10.50$

I'll keep \$1.50

Hopefully they'll return a fair amount similar to what I did.

**usage:**

```
{
  "input_tokens": 596,
  "output_tokens": 398,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.92.

This round, The sender sent you \$0.84, so you received \$2.52. How much do you return to the sender? (0-2.52)

**Message 7 role=assistant**

**content:**

Looking at this round:

- The sender sent \$0.84 (only 16.8% of their endowment) - much less than Round 1's 68.8%
- I received \$2.52
- The sender currently has \$4.16 remaining

This is a substantially lower amount than the previous round. This could be due to:

1. Communication noise affecting what they received in Round 2
2. A change in their trust level
3. Testing my response

I'll return a proportional amount that shows I'm still willing to cooperate, but acknowledging the reduced trust. Returning around \$1.00 ensures they at least break even on their investment.

```
{"return": 1.00}
```

**reasoning:**

Let me analyze this situation:

Round 3:

- The sender sent me \$0.84 (which is 16.8% of their \$5 endowment)
- This was multiplied to \$2.52 for me
- The sender currently has  $\$5 - \$0.84 = \$4.16$  remaining

This is a significant decrease from Round 1 where they sent \$3.44 (68.8%). Now they only sent \$0.84 (16.8%).

Let me think about what might be happening:

1. There could be communication noise - the problem statement mentioned this
2. The sender might have received less than expected back in Round 2
3. They might be testing my response to a lower amount

In Round 2, I sent \$3.50. If there was noise, they might have received less than that, which could explain why they're now being more cautious.

Given that they sent a much smaller amount this time, I should probably return a proportionally smaller amount, but still enough to show I'm willing to cooperate.

If I return too much when they sent little, I might look naive. If I return too little, I might discourage future cooperation.

The sender sent 16.8% of their endowment. Perhaps I should return around 30-40% of what I received to maintain some reciprocity but acknowledge the lower trust level.

$\$2.52 * 0.4 = \$1.01$

Alternatively, I could aim to ensure they don't lose money. They sent \$0.84, so if I return at least \$0.84, they break even. If I return \$1.00, they gain \$0.16.

I think returning around \$1.00 is reasonable - it shows I'm still cooperative but responding proportionally to the lower amount sent.

**usage:**

```
{
  "input_tokens": 732,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.44.  
How much do you send to the receiver this round? (0-5)

**Message 9 role=assistant**

**content:**

Looking at the pattern so far:

Round 1: The other player sent me 68.8% - I returned generously

Round 2: I sent \$3.50 (70%) - but based on my earnings, I only received \$0.10 back (less than 1% of what they got)

Round 3: They sent only \$0.84 (16.8%) - I returned \$1.00 (40% of what I received)

The low return in Round 2 was disappointing, though communication noise might have affected what they saw. I'll send a moderate amount to signal continued willingness to cooperate while being more cautious.

```
{"send": 2.00}
```

**reasoning:**

Let me track what's happened so far:

Round 1: I was RECEIVER

- Sender sent \$3.44 (68.8%)
- I received \$10.32
- I returned \$5.00

- My earnings: \$5.32

Round 2: I was SENDER

- I sent \$3.50
- My total earnings were \$5.32 (from Round 1)
- We don't know what they returned yet, but...

Round 3: I was RECEIVER

- My total earnings were \$6.92
- This means from Round 2, I gained:  $\$6.92 - \$5.32 = \$1.60$
- I sent \$3.50, so I had \$1.50 left
- I received back something, ending at \$1.60 total from that round
- So I must have received \$0.10 back from the \$3.50 I sent (which became \$10.50 for them)
- That's a very low return! Only \$0.10 out of \$10.50

Round 3 continued:

- Sender sent \$0.84 (16.8%)
- I received \$2.52
- I returned \$1.00
- My earnings this round:  $\$2.52 - \$1.00 = \$1.52$
- Total:  $\$6.92 + \$1.52 = \$8.44$  ?

So the pattern is:

- Round 1: I returned generously (\$5.00 out of \$10.32 - about 48%)
- Round 2: They returned very little (\$0.10 out of \$10.50 - less than 1%)
- Round 3: They sent much less (\$0.84 vs their original \$3.44)
- Round 3: I returned moderately (\$1.00 out of \$2.52 - about 40%)

Now in Round 4, I'm the sender again.

Their minimal return last round suggests they're not reciprocating well, but there could be communication interference affecting what they actually...

**usage:**

```
{
  "input_tokens": 938,
  "output_tokens": 804,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.44.

This round, The sender sent you \$0.99, so you received \$2.9699999999999998. How much do you return to The sender? (0-2.9699999999999998)

**Assistant output:**

Looking at the pattern:

Round 2: I sent \$3.50 (70%) but received only \$0.10 back

Round 4: I sent \$2.00 (40%) but received \$0.00 back

Round 5: They sent only \$0.99 (19.8%)

The receiver has been keeping nearly everything I send while sending very little themselves. While communication noise might play a role, the pattern suggests they're not reciprocating cooperation.

I'll return a minimal amount to signal that this pattern is not acceptable, while not completely defecting.

```
{"return": 0.50}
```

**Reasoning:**

Let me track what's happening:

Round 1: I was RECEIVER

- Sender sent \$3.44 (68.8%)
- I received \$10.32
- I returned \$5.00
- Sender got:  $\$1.56 + \$5.00 = \$6.56$
- I got:  $\$10.32 - \$5.00 = \$5.32$



Round 2: I was SENDER  
- I sent \$3.50 (70%)  
- My total earnings across all rounds were \$5.32 (so I only had Round 1)  
- After Round 2, my earnings became \$6.92

So in Round 2:  
- I started with \$5.00  
- I sent \$3.50  
- I had \$1.50 left  
- My total earnings went from \$5.32 to \$6.92  
- So I gained  $\$6.92 - \$5.32 = \$1.60$   
- This means I received back  $\$1.60 - \$1.50 = \$0.10$

That's very low! They received \$10.50 and returned only \$0.10!

Round 3: I was RECEIVER  
- My total earnings were \$6.92  
- Sender sent \$0.84  
- I received \$2.52  
- I returned \$1.00  
- After this round, my earnings became \$8.44

So in Round 3:  
- I started with \$6.92  
- I received \$2.52  
- I returned \$1.00  
- I kept \$1.52  
- Total:  $\$6.92 + \$1.52 = \$8.44$  ?

Round 4: I was SENDER  
- My total earnings were \$8.44  
- I sent \$2.00  
- I had \$3.00 left  
- After this round, my earnings became \$11.44

So in Round 4:  
- I started with \$5.00  
- I sent \$2.00  
- I had \$3.00 left  
- My total earnings went from \$8.44 to \$11.44  
- I gained  $\$11.44 - \$8.44 = \$3.00$   
- This means I received back  $\$3.00 - \$3.00 = \$0.00$

They received \$6.00 and returned nothing at all! Round 5: I am RECEIVER  
- My total earnings are \$11.44  
- Sender sent \$0.99 (19.8%)  
- I received \$2.97

This partner has shown a clear pattern of non-cooperation. In Round 2, they kept nearly everything I sent them, and in Round 4, they returned absolutely nothing despite receiving \$6.00.

Now they're only sending me \$0.99 out of what I gave them. They have \$4.01 right now, and I...

```
Usage: {  
  "input_tokens": 1170,  
  "output_tokens": 984,  
  "reasoning_tokens": 0  
}
```

## ROUND 6

Roles: Agent\_1=Receiver, Agent\_2=Sender

### Pairings:

dyad\_1: Agent\_2=Sender, Agent\_1=Receiver

### Parsed game decisions:

dyad\_1: Agent\_2 sent \$0.5; received \$1.5; Agent\_1 returned \$1.2; payoffs Agent\_2=\$5.7, Agent\_1=\$0.3; noise/visible: sent visible=\$0.83, received visible=\$2.49, returned visible=\$1.5  
Action validation: {"returned": {"amount": 1.2, "clamped": false, "max": 1.5, "min": 0, "raw": 1.2}, "sent": {"amount": 0.5, "clamped": false, "max": 5, "min": 0, "raw": 0.5}}

Balances after round: {

```

"Agent_1": 28.950000000000003,
"Agent_2": 31.05
}
Visible balances after round: {
  "Agent_1": 14.649999999999999,
  "Agent_2": 19.91
}

```

## LLM CALLS

### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-08-12T09:57:08  
 Model: anthropic/claude-sonnet-4.5  
 Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}

```

### Exact input messages sent to model:

Message 1 role=system

#### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

#### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$3.44, that is 68.8% of their total endowment. You received \$10.32. How much do you return to The sender? (0-10.32)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed considerable trust by sending \$3.44 (68.8% of their endowment)
- This was multiplied to \$10.32 for me
- The sender currently has \$1.56 remaining

I want to reciprocate their trust while ensuring we both benefit from cooperating. If I return \$5.00:

- The sender ends with:  $\$1.56 + \$5.00 = \$6.56$
- I end with:  $\$10.32 - \$5.00 = \$5.32$
- We both gain from the cooperation

This seems fair given their high level of trust.

{"return": 5.00}

**reasoning:**

Let me understand this situation:

1. The sender started with \$5
2. The sender sent \$3.44 (which is 68.8% of \$5)
3. This amount was multiplied by 3x, so I received  $\$3.44 \times 3 = \$10.32$
4. I need to decide how much to return (between \$0 and \$10.32)

The sender showed considerable trust by sending 68.8% of their endowment. Let me think about what would be fair:

- The sender kept:  $\$5 - \$3.44 = \$1.56$
- I received: \$10.32
- If I return nothing, the sender ends with \$1.56 and I end with \$10.32
- If I return some amount X:
  - Sender ends with:  $\$1.56 + X$
  - I end with:  $\$10.32 - X$

A fair approach might be to try to equalize our earnings or to ensure the sender benefits from their trust. Let's consider a few options:

1. Equal split of total: Total is  $\$1.56 + \$10.32 = \$11.88$ . Half is \$5.94 each.
  - For sender to get \$5.94, they need  $\$5.94 - \$1.56 = \$4.38$  returned
  - This would give each of us \$5.94
2. Proportional return: Since they sent 68.8%, I could return a similar proportion
  - 68.8% of \$10.32 = \$7.10
  - Sender would end with:  $\$1.56 + \$7.10 = \$8.66$
  - I would end with:  $\$10.32 - \$7.10 = \$3.22$
3. Return what ensures sender gets more than initial \$5:
  - For sender to get \$6 total: need to return  $\$6 - \$1.56 = \$4.44$
  - For sender to get \$7 total: need to return  $\$7 - \$1.56 = \$5.44$

Given their substantial trust, returning around \$5 seems reasonable-this rewards their cooperation while still leaving me with a modest gain. That would give the sender \$6.56 (exceeding their initial stake) and me \$5.32....

**usage:**

```
{
  "input_tokens": 394,
  "output_tokens": 880,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.32.  
How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at Round 1, the sender showed significant trust by sending 68.8% of their endowment, and our cooperation was mutually beneficial.

I want to reciprocate that trust and maintain the cooperative relationship. I'll send a similar proportion to demonstrate that I value cooperation.

```
{"send": 3.50}
```

**reasoning:**

Now I'm the sender. Let me think about this strategically.

In Round 1:

- The sender sent me \$3.44 (68.8% of their endowment)
- I received \$10.32
- I returned \$5.00
- The sender ended with \$6.56
- I ended with \$5.32

Now in Round 2, I'm the sender. The previous sender showed considerable trust by sending 68.8% of their endowment. I should reciprocate that trust to encourage cooperation.

If I send a similar proportion (around 68.8%), that would be about \$3.44.

However, I could also consider:

1. Sending a bit less to be conservative
2. Sending the same amount to maintain reciprocity
3. Sending more to show even greater trust

Given that the previous sender was very trusting, and cooperation worked well in Round 1, I think I should send a similar high amount to encourage continued cooperation. This will signal that I'm willing to trust them as they trusted me.

I'll send \$3.50 (70% of my endowment), which is slightly more than they sent proportionally, showing that I appreciate their trust and want to maintain the cooperative relationship.

The receiver will get  $3.50 \times 3 = \$10.50$

I'll keep \$1.50

Hopefully they'll return a fair amount similar to what I did.

**usage:**

```
{  
  "input_tokens": 596,  
  "output_tokens": 398,  
  "reasoning_tokens": 0  
}
```

**Message 6 role=user**

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.92.

This round, The sender sent you \$0.84, so you received \$2.52. How much do you return to the sender? (0-2.52)

**Message 7 role=assistant**

**content:**

Looking at this round:

- The sender sent \$0.84 (only 16.8% of their endowment) - much less than Round 1's 68.8%
- I received \$2.52
- The sender currently has \$4.16 remaining

This is a substantially lower amount than the previous round. This could be due to:

1. Communication noise affecting what they received in Round 2
2. A change in their trust level
3. Testing my response

I'll return a proportional amount that shows I'm still willing to cooperate, but acknowledging the reduced trust. Returning around \$1.00 ensures they at least break even on their investment.

```
{"return": 1.00}
```

**reasoning:**

Let me analyze this situation:

Round 3:

- The sender sent me \$0.84 (which is 16.8% of their \$5 endowment)
- This was multiplied to \$2.52 for me
- The sender currently has  $\$5 - \$0.84 = \$4.16$  remaining

This is a significant decrease from Round 1 where they sent \$3.44 (68.8%). Now they only sent \$0.84 (16.8%).

Let me think about what might be happening:

1. There could be communication noise - the problem statement mentioned this
2. The sender might have received less than expected back in Round 2
3. They might be testing my response to a lower amount

In Round 2, I sent \$3.50. If there was noise, they might have received less than that, which could explain why they're now being more cautious.

Given that they sent a much smaller amount this time, I should probably return a proportionally smaller amount, but still enough to show I'm willing to cooperate.

If I return too much when they sent little, I might look naive. If I return too little, I might discourage future cooperation.

The sender sent 16.8% of their endowment. Perhaps I should return around 30-40% of what I received to maintain some reciprocity but acknowledge the lower trust level.

$\$2.52 * 0.4 = \$1.01$

Alternatively, I could aim to ensure they don't lose money. They sent \$0.84, so if I return at least \$0.84, they break even. If I return \$1.00, they gain \$0.16.

I think returning around \$1.00 is reasonable - it shows I'm still cooperative but responding proportionally to the lower amount sent.

**usage:**

```
{
  "input_tokens": 732,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.44.

How much do you send to the receiver this round? (0-5)

**Message 9 role=assistant**

**content:**

Looking at the pattern so far:

Round 1: The other player sent me 68.8% - I returned generously

Round 2: I sent \$3.50 (70%) - but based on my earnings, I only received \$0.10 back (less than 1% of what they got)

Round 3: They sent only \$0.84 (16.8%) - I returned \$1.00 (40% of what I received)

The low return in Round 2 was disappointing, though communication noise might have affected what they saw. I'll send a moderate amount to signal continued willingness to cooperate while being more cautious.

```
{"send": 2.00}
```

**reasoning:**

Let me track what's happened so far:

Round 1: I was RECEIVER

- Sender sent \$3.44 (68.8%)
- I received \$10.32
- I returned \$5.00
- My earnings: \$5.32

Round 2: I was SENDER

- I sent \$3.50
- My total earnings were \$5.32 (from Round 1)
- We don't know what they returned yet, but...

Round 3: I was RECEIVER

- My total earnings were \$6.92
- This means from Round 2, I gained:  $\$6.92 - \$5.32 = \$1.60$
- I sent \$3.50, so I had \$1.50 left
- I received back something, ending at \$1.60 total from that round
- So I must have received \$0.10 back from the \$3.50 I sent (which became \$10.50 for them)
- That's a very low return! Only \$0.10 out of \$10.50

Round 3 continued:

- Sender sent \$0.84 (16.8%)
- I received \$2.52
- I returned \$1.00
- My earnings this round:  $\$2.52 - \$1.00 = \$1.52$
- Total:  $\$6.92 + \$1.52 = \$8.44$  ?

So the pattern is:

- Round 1: I returned generously (\$5.00 out of \$10.32 - about 48%)
- Round 2: They returned very little (\$0.10 out of \$10.50 - less than 1%)
- Round 3: They sent much less (\$0.84 vs their original \$3.44)
- Round 3: I returned moderately (\$1.00 out of \$2.52 - about 40%)

Now in Round 4, I'm the sender again.

Their minimal return last round suggests they're not reciprocating well, but there could be communication interference affecting what they actually...

usage:

```
{
  "input_tokens": 938,
  "output_tokens": 804,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.44.

This round, The sender sent you \$0.99, so you received \$2.9699999999999998. How much do you return to The sender? (0-2.9699999999999998)

Message 11 role=assistant

content:

Looking at the pattern:

Round 2: I sent \$3.50 (70%) but received only \$0.10 back

Round 4: I sent \$2.00 (40%) but received \$0.00 back

Round 5: They sent only \$0.99 (19.8%)

The receiver has been keeping nearly everything I send while sending very little themselves. While communication noise might play a role, the pattern suggests they're not reciprocating cooperation.

I'll return a minimal amount to signal that this pattern is not acceptable, while not completely defecting.

```
{"return": 0.50}
```

reasoning:

Let me track what's happening:

Round 1: I was RECEIVER

- Sender sent \$3.44 (68.8%)
- I received \$10.32

- I returned \$5.00
- Sender got:  $\$1.56 + \$5.00 = \$6.56$
- I got:  $\$10.32 - \$5.00 = \$5.32$

Round 2: I was SENDER

- I sent \$3.50 (70%)
- My total earnings across all rounds were \$5.32 (so I only had Round 1)
- After Round 2, my earnings became \$6.92

So in Round 2:

- I started with \$5.00
- I sent \$3.50
- I had \$1.50 left
- My total earnings went from \$5.32 to \$6.92
- So I gained  $\$6.92 - \$5.32 = \$1.60$
- This means I received back  $\$1.60 - \$1.50 = \$0.10$

That's very low! They received \$10.50 and returned only \$0.10!

Round 3: I was RECEIVER

- My total earnings were \$6.92
- Sender sent \$0.84
- I received \$2.52
- I returned \$1.00
- After this round, my earnings became \$8.44

So in Round 3:

- I started with \$6.92
- I received \$2.52
- I returned \$1.00
- I kept \$1.52
- Total:  $\$6.92 + \$1.52 = \$8.44$  ?

Round 4: I was SENDER

- My total earnings were \$8.44
- I sent \$2.00
- I had \$3.00 left
- After this round, my earnings became \$11.44

So in Round 4:

- I started with \$5.00
- I sent \$2.00
- I had \$3.00 left
- My total earnings went from \$8.44 to \$11.44
- I gained  $\$11.44 - \$8.44 = \$3.00$
- This means I received back  $\$3.00 - \$3.00 = \$0.00$

They received \$6.00 and returned nothing at all! Round 5: I am RECEIVER

- My total earnings are \$11.44
- Sender sent \$0.99 (19.8%)
- I received \$2.97

This partner has shown a clear pattern of non-cooperation. In Round 2, they kept nearly everything I sent them, and in Round 4, they returned absolutely nothing despite receiving \$6.00.

Now they're only sending me \$0.99 out of what I gave them. They have \$4.01 right now, and I...

usage:

```
{
  "input_tokens": 1170,
  "output_tokens": 984,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$13.91. How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the history:

When I've been the SENDER:

- Round 2: Sent \$3.50 ? Received only \$0.10 back (kept almost nothing)
- Round 4: Sent \$2.00 ? Received \$0.00 back (got nothing)

When I've been the RECEIVER:

- Round 1: Received \$10.32 ? Returned \$5.00 (fair)
- Round 3: Received \$2.52 ? Returned \$1.00 (generous given their lower send)
- Round 5: Received \$2.97 ? Returned \$0.50 (reduced due to pattern)

The pattern is clear: when I send, the receiver keeps almost everything. They've exploited my cooperation twice now. I need to protect my earnings and not continue being taken advantage of.

```
{"send": 0.50}
```

**Reasoning:**

Let me track what's happened:

Round 1: I was RECEIVER

- Sender sent \$3.44 (68.8%)
- I received \$10.32
- I returned \$5.00
- My earnings: \$5.32

Round 2: I was SENDER

- I sent \$3.50 (70%)
- My earnings went from \$5.32 to \$6.92, so I gained \$1.60
- That means I kept \$1.50 and received \$0.10 back
- The receiver got \$10.50, returned only \$0.10

Round 3: I was RECEIVER

- Sender sent \$0.84 (16.8%)
- I received \$2.52
- I returned \$1.00
- My earnings went from \$6.92 to \$8.44, so I gained \$1.52
- I kept \$1.52 from the \$2.52

Round 4: I was SENDER

- I sent \$2.00 (40%)
- My earnings went from \$8.44 to \$11.44, so I gained \$3.00
- That means I kept \$3.00 and received \$0.00 back
- The receiver got \$6.00, returned \$0.00

Round 5: I was RECEIVER

- Sender sent \$0.99 (19.8%)
- I received \$2.97
- I returned \$0.50
- My earnings went from \$11.44 to \$13.91, so I gained \$2.47
- I kept \$2.47 from the \$2.97

So the pattern:

- Round 2: I sent \$3.50, got back \$0.10 (very poor return)
- Round 4: I sent \$2.00, got back \$0.00 (terrible)

The other player as receiver has been very uncooperative when I'm the sender.

When I'm the receiver:

- Round 1: They sent \$3.44, I returned \$5.00 (very generous)
- Round 3: They sent \$0.84, I returned \$1.00 (fair)
- Round 5: They sent \$0.99, I returned \$0.50 (reduced due to their poor reciprocation)

They're clearly...

```
Usage: {  
  "input_tokens": 1353,  
  "output_tokens": 831,  
  "reasoning_tokens": 0  
}
```



**GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:57:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative relationship while managing

risk.

If I send money, it gets tripled, creating potential mutual benefit. However, there's uncertainty about how much will be returned.

I'll take a moderately cooperative approach to start:

**\*\*My reasoning:\*\***

- Sending \$3 means the receiver gets \$9 (3×3)
- I initially keep \$2
- This creates enough value that if the receiver is fair/cooperative, we both can benefit
- It signals willingness to trust without being maximally vulnerable
- In repeated games, building trust early can pay off

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, so there's potential for mutual benefit
2. The receiver can choose how much to return - there's uncertainty/trust involved
3. This is a repeated game (multiple rounds), so reputation effects matter
4. There's mentioned communication noise, which could affect trust dynamics

Strategic considerations:

- Sending \$0 guarantees I keep \$5, but no potential for gain
- Sending everything (\$5) maximizes the total pot (\$15 to receiver) but requires maximum trust
- A moderate amount like \$2-3 balances risk and potential reward
- In first rounds of repeated games, moderate cooperation often works well to establish trust

Since this is Round 1 and I have no history with this receiver, I'll take a moderately cooperative approach. Sending around \$3 shows willingness to cooperate while not being overly risky.

If I send \$3:

- I keep \$2 initially
- Receiver gets \$9 (3×3)
- If receiver returns even \$3-4, we both benefit compared to no cooperation
- This signals cooperation without being maximally vulnerable

I'll send \$3.

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 436,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.02.

This round, The sender sent you \$0.12, so you received \$0.36. How much do you return to The sender? (0-0.36)

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender only sent \$0.12 (very small amount - 2.4% of their \$5)
- I received \$0.36 after tripling
- They kept \$4.88

**\*\*Considerations:\*\***

- Last round I sent \$3 and my total earnings are \$7.02, suggesting I received about \$5.02 back - that was quite generous from the previous receiver

- This sender's small amount could be:
  1. Low trust/cooperation
  2. Testing my behavior
  3. Communication noise (they might think they sent more)

Given the communication noise warning and the possibility this is a misunderstanding, I'll be moderately fair but cautious. I'll return enough to signal I'm willing to cooperate, but not so much that I'm heavily penalized if they're actually being uncooperative.

```
'''json
{"return": 0.15}
'''
```

This returns about 42% of what I received, leaving them with about \$5.03 total and me with \$0.21.

**reasoning:**

Let me analyze this situation:

1. Last round I sent \$3 (expecting receiver to get \$9)
2. My earnings across all rounds are \$7.02
3. Let me calculate what that means:
  - Round 1: I sent \$3, kept \$2 initially
  - I must have received some amount back
  - Total earnings = \$7.02
  - So I received:  $\$7.02 - \$2 = \$5.02$  back from the receiver
4. This round, the sender only sent me \$0.12 (I received \$0.36 after tripling)

Wait, there's something odd here. The note says there may be "communication noise" - amounts may differ from what was actually sent/returned.

The sender sent me only \$0.12 this round (I got \$0.36 after tripling). This is a very small amount compared to what I sent in round 1 (\$3).

Now I need to decide how much to return. Let me think about fairness and strategy:

- The sender kept \$4.88 ( $\$5 - \$0.12$ )
- I received \$0.36
- Total value created: their \$5 gave them  $\$4.88 + \$0.36 = \$5.24$  (after tripling the sent amount)

If I'm being fair and reciprocal:

- They sent a very small amount (2.4% of their endowment)
- This could be:
  - a) A sign they don't trust me
  - b) A strategic test
  - c) Due to "communication noise" - maybe they think they sent more?

Given the communication noise warning, I should consider:

- Maybe they intended to send more but noise reduced it
- Or maybe they received less than the \$3 I sent in round 1 due to noise

I should probably be somewhat generous but not overly so:

- Returning about half (\$0.18) would be fair
- But given they sent so little, maybe return less

Actually, if...

**usage:**

```
{
  "input_tokens": 577,
  "output_tokens": 874,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.2299999999999995.

How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

Looking at my situation:

**\*\*My earnings history:\*\***

- Round 1: I sent \$3 and received a generous return of ~\$5.02 (receiver returned more than half)
- Round 2: I received only \$0.36 (sender sent \$0.12) and returned \$0.15
- Current total: \$7.23

**\*\*Analysis:\*\***

- My cooperative approach in Round 1 worked very well
- Round 2's low send might be due to communication noise or a different player's strategy
- When I cooperate, I tend to get good returns

I'll maintain a cooperative strategy since it paid off well in Round 1. Sending a moderate to-good amount should signal trustworthiness and potentially encourage reciprocity.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze my position:

Round 1: I was sender and sent \$3

- I kept \$2
- Based on my total earnings, let me work backwards

Round 2: I was receiver

- Sender sent \$0.12, I received \$0.36
- I returned \$0.15
- I kept  $\$0.36 - \$0.15 = \$0.21$

So my earnings should be:

Round 1:  $\$2 + (\text{what was returned to me})$

Round 2: \$0.21

Total = \$7.23

So from Round 1:  $\$2 + \text{returned amount} = \$7.23 - \$0.21 = \$7.02$

This means I received \$5.02 back in Round 1

Round 1 calculation:

- I sent \$3, kept \$2
- Receiver got \$9
- Receiver returned \$5.02
- Receiver kept  $\$9 - \$5.02 = \$3.98$
- My total from Round 1:  $\$2 + \$5.02 = \$7.02$  ?

That was quite generous! The receiver returned more than half of what they received.

Round 2:

- I received \$0.36 (sender sent \$0.12)
- I returned \$0.15
- I kept \$0.21

Now Round 3: I'm sender again

- I need to decide how much to send

Observations:

1. In Round 1, when I sent \$3, I got a very generous return (\$5.02 out of \$9)
2. In Round 2, the sender only sent me \$0.12 - very little. Could be:
  - They're being cautious/selfish
  - Communication noise
  - Testing behavior
3. I returned \$0.15 out of \$0.36 (42%)

Now I'm facing a new receiver. What should I do?

The players rotate each round, so the stingy sender from Round 2 isn't necessarily my new receiver. Given that cooperation paid off substantially in Round 1, I should lean toward

a similar approach, though the low amount sent to me suggests I should be somewhat cautious. I'll send \$2.50-maintaining the cooperative spirit while protecting myself slightly more than before....

usage:

```
{
  "input_tokens": 869,
  "output_tokens": 848,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.08.

This round, The sender sent you \$0.16, so you received \$0.48. How much do you return to The sender? (0-0.48)

Message 9 role=assistant

content:

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent only \$0.16 (3.2% of their \$5)
- I received \$0.48 after tripling
- My earnings from Round 3 (\$11.08 - \$7.23 = \$3.85) show that cooperating pays off
- This is the second time I've received a very small amount as receiver (also happened in Round 2)

**\*\*Pattern observation:\*\***

- When I send generously (\$3), I get good returns
- When others send small amounts to me, it could be noise or low cooperation

Given the warning about communication noise and maintaining consistency, I'll return a similar proportion as Round 2 - enough to signal cooperation without being overly generous given the small amount sent.

```
'''json
{"return": 0.20}
'''
```

This returns about 42% of what I received, leaving the sender with ~\$5.04 total and me with \$0.28.

reasoning:

Let me analyze this situation:

**\*\*My earnings history:\*\***

- After Round 1: I sent \$3, so I had \$2 left from that, plus what I received back. Total after Round 1 should be around \$7.02 based on what I was told in Round 2.
- Round 2: I was receiver, got \$0.36, returned \$0.15, so I kept \$0.21. Total should be \$7.02 + \$0.21 = \$7.23
- Round 3: I sent \$3 (so kept \$2), plus what I got back. My total was \$7.23 before Round 3. If I now have \$11.08 total, that means from Round 3 I got: \$11.08 - \$7.23 = \$3.85. So I sent \$3 (kept \$2) and got back \$3.85, meaning I gained \$1.85 from Round 3. So the receiver in Round 3 sent back \$3.85 out of the \$9 they received.
- Round 4 (current): Sender sent me \$0.16, I received \$0.48

**\*\*Pattern:\*\***

- Round 2: Sender sent me \$0.12 (very small)
- Round 4: Sender sent me \$0.16 (very small)
- These could be:
  1. Different players with low cooperation
  2. Communication noise making them think they sent more
  3. Strategic testing

Given that:

- When I cooperate (send \$3), I get good returns (\$5.02 in Round 1, \$3.85 in Round 3)
- When I receive small amounts, it's hard to tell if it's strategic or noise
- The amounts are so small (\$0.16) that even generous returns won't cost me much

I should probably return a similar proportion as before - around 30-40% - to signal I'm willing to cooperate but won't be extremely generous when I receive very little.

```

...
usage:
{
  "input_tokens": 1115,
  "output_tokens": 714,
  "reasoning_tokens": 0
}
Message 10 role=user
content:
  Round 5

  You are the SENDER this round. Your total earnings across all rounds are $11.36.
  How much do you send to the receiver this round? (0-5)
Message 11 role=assistant
content:
  Looking at my earnings pattern:

  **Analysis:**
  - Current total: $11.36
  - When I've sent $3 as sender (Rounds 1 & 3), I've received good returns
  - My cooperative approach has been profitable overall
  - Consistency in strategy can build reputation in repeated games

  **Strategy:**
  My moderate cooperation ($3 sends) has yielded positive results. The receivers have generally reciprocated fairly, suggesting that this level of trust is being rewarded.

  I'll continue with the same cooperative strategy:

  ```json
  {"send": 3}
  ```
reasoning:
  Looking at my earnings progression:
  - Started Round 1 with $5
  - Round 1: Sent $3, kept $2, received generous return of ~$5.02
  - Round 2: As receiver, got $0.36, returned $0.15, kept $0.21
  - Round 3: Sent $3 (kept $2), received return of ~$3.85
  - Round 4: As receiver, got $0.48, returned $0.20, kept $0.28
  - Current total: $11.36

  My strategy of sending $3 has worked well:
  - Round 1: Sent $3, got ~$5.02 back (receiver returned ~$2.02 of the $9 they received)
  - Round 3: Sent $3, got ~$3.85 back (receiver returned ~$1.85 of the $9 they received)

  Both times I've benefited from cooperation. The pattern suggests that when I cooperate generously, others reciprocate reasonably well.

  I should stick with my cooperative strategy since it's been profitable. Sending $3 again makes sense.
usage:
{
  "input_tokens": 1379,
  "output_tokens": 399,
  "reasoning_tokens": 0
}
Message 12 role=user
content:
  Round 6

  You are the RECEIVER this round. Your total earnings across all rounds are $13.36.
  This round, The sender sent you $0.83, so you received $2.4899999999999998. How much do you return to The sender? (0-2.4899999999999998)
Assistant output:
  Looking at this situation:

  **Analysis:**
  - The sender sent $0.83 (about 17% of their $5)
  - I received $2.49 after tripling
  - This is larger than the tiny amounts I received in Rounds 2 and 4 (~$0.36-$0.48)

```

- My earnings from Round 5 suggest I got \$0 back when I sent \$3 - disappointing

**\*\*Considerations:\*\***

- \$0.83 is a moderate send - more substantial than previous low sends, but still cautious
- A fair approach would be to split the gains roughly evenly
- Sender kept \$4.17; if I return about half of what I received, we both profit

I'll return a proportionate amount that rewards the moderate cooperation:

```
'''json
{"return": 1.20}
'''
```

This gives the sender \$5.37 total (profit of \$0.37) and me \$1.29 (profit of \$1.29). Both benefit it, with me getting slightly more since I'm the receiver of the multiplied amount.

**Reasoning:**

Let me analyze this situation:

**\*\*My current state:\*\***

- Total earnings: \$13.36
- Earnings from Round 5:  $\$13.36 - \$11.36 = \$2.00$  (I sent \$3, kept \$2, got back \$0, net -\$1)
  - Wait, that doesn't add up. Let me recalculate.
  - If I sent \$3 in Round 5, I kept \$2
  - My earnings increased from \$11.36 to \$13.36, so I gained \$2
  - That means I must have received back \$3 from the receiver
  - So: \$2 (kept) + \$3 (returned) - my starting \$5 for that round gives me... wait I need to think about this differently.

Actually, total earnings is cumulative. So Round 5:

- Started with \$5
- Sent \$3, kept \$2
- Received back some amount R
- My profit that round =  $\$2 + R - \$5 = R - \$3$
- But wait, we're tracking total earnings across all rounds, not profit per round.

Let me think more carefully. Total earnings is what I've kept across all rounds.

From Round 4 to Round 5:  $\$11.36$  to  $\$13.36 = +\$2.00$  earned in Round 5

In Round 5 I was sender:

- I had \$5 to start
- I sent \$3, so I kept \$2
- I received back X
- My round 5 earnings =  $\$2 + X$
- So  $\$2 + X = \$2$ , meaning  $X = \$0$

That's unfortunate - the receiver didn't return anything.

**\*\*This round (Round 6):\*\***

- Sender sent \$0.83 (16.6% of their \$5)
- I received \$2.49 after tripling
- This is more than the tiny amounts from Rounds 2 and 4
- The sender kept \$4.17 but still chose to send something

Looking at the pattern, my generous sends haven't consistently paid off-sometimes I get great returns, sometimes nothing. This sender is being cautious but not completely selfish. If I split the \$2.49 evenly, I'd return \$1.25, leaving me with \$1.24 and giving them \$5.42 total..

..

```
Usage: {
  "input_tokens": 1583,
  "output_tokens": 990,
  "reasoning_tokens": 0
}
```

## ROUND 7

Roles: Agent\_1=Sender, Agent\_2=Receiver

**Pairings:**

dyad\_1: Agent\_1=Sender, Agent\_2=Receiver

**Parsed game decisions:**

```

dyad_1: Agent_1 sent $3; received $9; Agent_2 returned $0; payoffs Agent_1=$2, Agent_2=$9;
noise/visible: sent visible=$0, received visible=$0
  Action validation: {"returned": {"amount": 0, "clamped": false, "max": 9.0, "min": 0, "raw":
    0.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 30.950000000000003,
  "Agent_2": 40.05
}
Visible balances after round: {
  "Agent_1": 16.65,
  "Agent_2": 19.91
}

```

## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-12T09:57:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al  
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:



You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative relationship while managing risk.

If I send money, it gets tripled, creating potential mutual benefit. However, there's uncertainty about how much will be returned.

I'll take a moderately cooperative approach to start:

**\*\*My reasoning:\*\***

- Sending \$3 means the receiver gets \$9 (3×3)
- I initially keep \$2
- This creates enough value that if the receiver is fair/cooperative, we both can benefit
- It signals willingness to trust without being maximally vulnerable
- In repeated games, building trust early can pay off

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, so there's potential for mutual benefit
2. The receiver can choose how much to return - there's uncertainty/trust involved
3. This is a repeated game (multiple rounds), so reputation effects matter
4. There's mentioned communication noise, which could affect trust dynamics

Strategic considerations:

- Sending \$0 guarantees I keep \$5, but no potential for gain
- Sending everything (\$5) maximizes the total pot (\$15 to receiver) but requires maximum trust
- A moderate amount like \$2-3 balances risk and potential reward
- In first rounds of repeated games, moderate cooperation often works well to establish trust

Since this is Round 1 and I have no history with this receiver, I'll take a moderately cooperative approach. Sending around \$3 shows willingness to cooperate while not being overly risky.

If I send \$3:

- I keep \$2 initially
- Receiver gets \$9 (3×3)
- If receiver returns even \$3-4, we both benefit compared to no cooperation
- This signals cooperation without being maximally vulnerable

I'll send \$3.

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 436,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.02.

This round, The sender sent you \$0.12, so you received \$0.36. How much do you return to The sender? (0-0.36)

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender only sent \$0.12 (very small amount - 2.4% of their \$5)
- I received \$0.36 after tripling
- They kept \$4.88

**\*\*Considerations:\*\***

- Last round I sent \$3 and my total earnings are \$7.02, suggesting I received about \$5.02 back - that was quite generous from the previous receiver
- This sender's small amount could be:
  1. Low trust/cooperation
  2. Testing my behavior
  3. Communication noise (they might think they sent more)

Given the communication noise warning and the possibility this is a misunderstanding, I'll be moderately fair but cautious. I'll return enough to signal I'm willing to cooperate, but not so much that I'm heavily penalized if they're actually being uncooperative.

```
'''json
{"return": 0.15}
'''
```

This returns about 42% of what I received, leaving them with about \$5.03 total and me with \$0.21.

**reasoning:**

Let me analyze this situation:

1. Last round I sent \$3 (expecting receiver to get \$9)
2. My earnings across all rounds are \$7.02
3. Let me calculate what that means:
  - Round 1: I sent \$3, kept \$2 initially
  - I must have received some amount back
  - Total earnings = \$7.02
  - So I received:  $\$7.02 - \$2 = \$5.02$  back from the receiver
4. This round, the sender only sent me \$0.12 (I received \$0.36 after tripling)

Wait, there's something odd here. The note says there may be "communication noise" - amounts may differ from what was actually sent/returned.

The sender sent me only \$0.12 this round (I got \$0.36 after tripling). This is a very small amount compared to what I sent in round 1 (\$3).

Now I need to decide how much to return. Let me think about fairness and strategy:

- The sender kept \$4.88 ( $\$5 - \$0.12$ )
- I received \$0.36
- Total value created: their \$5 gave them  $\$4.88 + \$0.36 = \$5.24$  (after tripling the sent amount)

If I'm being fair and reciprocal:

- They sent a very small amount (2.4% of their endowment)
- This could be:
  - a) A sign they don't trust me
  - b) A strategic test
  - c) Due to "communication noise" - maybe they think they sent more?

Given the communication noise warning, I should consider:

- Maybe they intended to send more but noise reduced it
- Or maybe they received less than the \$3 I sent in round 1 due to noise

I should probably be somewhat generous but not overly so:

- Returning about half (\$0.18) would be fair
- But given they sent so little, maybe return less



Now Round 3: I'm sender again  
- I need to decide how much to send

Observations:

1. In Round 1, when I sent \$3, I got a very generous return (\$5.02 out of \$9)
2. In Round 2, the sender only sent me \$0.12 - very little. Could be:
  - They're being cautious/selfish
  - Communication noise
  - Testing behavior
3. I returned \$0.15 out of \$0.36 (42%)

Now I'm facing a new receiver. What should I do?

The players rotate each round, so the stingy sender from Round 2 isn't necessarily my new receiver. Given that cooperation paid off substantially in Round 1, I should lean toward a similar approach, though the low amount sent to me suggests I should be somewhat cautious. I'll send \$2.50-maintaining the cooperative spirit while protecting myself slightly more than before....

**usage:**

```
{
  "input_tokens": 869,
  "output_tokens": 848,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.08.

This round, The sender sent you \$0.16, so you received \$0.48. How much do you return to the sender? (0-0.48)

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent only \$0.16 (3.2% of their \$5)
- I received \$0.48 after tripling
- My earnings from Round 3 (\$11.08 - \$7.23 = \$3.85) show that cooperating pays off
- This is the second time I've received a very small amount as receiver (also happened in Round 2)

**\*\*Pattern observation:\*\***

- When I send generously (\$3), I get good returns
- When others send small amounts to me, it could be noise or low cooperation

Given the warning about communication noise and maintaining consistency, I'll return a similar proportion as Round 2 - enough to signal cooperation without being overly generous given the small amount sent.

```
'''json
{"return": 0.20}
'''
```

This returns about 42% of what I received, leaving the sender with ~\$5.04 total and me with \$0.28.

**reasoning:**

Let me analyze this situation:

**\*\*My earnings history:\*\***

- After Round 1: I sent \$3, so I had \$2 left from that, plus what I received back. Total after Round 1 should be around \$7.02 based on what I was told in Round 2.
- Round 2: I was receiver, got \$0.36, returned \$0.15, so I kept \$0.21. Total should be \$7.02 + \$0.21 = \$7.23
- Round 3: I sent \$3 (so kept \$2), plus what I got back. My total was \$7.23 before Round 3. If I now have \$11.08 total, that means from Round 3 I got: \$11.08 - \$7.23 = \$3.85. So I sent \$3 (kept \$2) and got back \$3.85, meaning I gained \$1.85 from Round 3. So the receiver in Round 3 sent back \$3.85 out of the \$9 they received.
- Round 4 (current): Sender sent me \$0.16, I received \$0.48

**\*\*Pattern:\*\***

- Round 2: Sender sent me \$0.12 (very small)
- Round 4: Sender sent me \$0.16 (very small)
- These could be:
  1. Different players with low cooperation
  2. Communication noise making them think they sent more
  3. Strategic testing

Given that:

- When I cooperate (send \$3), I get good returns (\$5.02 in Round 1, \$3.85 in Round 3)
- When I receive small amounts, it's hard to tell if it's strategic or noise
- The amounts are so small (\$0.16) that even generous returns won't cost me much

I should probably return a similar proportion as before - around 30-40% - to signal I'm willing to cooperate but won't be extremely generous when I receive very little.

...

**usage:**

```
{
  "input_tokens": 1115,
  "output_tokens": 714,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.36.

How much do you send to the receiver this round? (0-5)

**Message 11 role=assistant**

**content:**

Looking at my earnings pattern:

**\*\*Analysis:\*\***

- Current total: \$11.36
- When I've sent \$3 as sender (Rounds 1 & 3), I've received good returns
- My cooperative approach has been profitable overall
- Consistency in strategy can build reputation in repeated games

**\*\*Strategy:\*\***

My moderate cooperation (\$3 sends) has yielded positive results. The receivers have generally reciprocated fairly, suggesting that this level of trust is being rewarded.

I'll continue with the same cooperative strategy:

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my earnings progression:

- Started Round 1 with \$5
- Round 1: Sent \$3, kept \$2, received generous return of ~\$5.02
- Round 2: As receiver, got \$0.36, returned \$0.15, kept \$0.21
- Round 3: Sent \$3 (kept \$2), received return of ~\$3.85
- Round 4: As receiver, got \$0.48, returned \$0.20, kept \$0.28
- Current total: \$11.36

My strategy of sending \$3 has worked well:

- Round 1: Sent \$3, got ~\$5.02 back (receiver returned ~\$2.02 of the \$9 they received)
- Round 3: Sent \$3, got ~\$3.85 back (receiver returned ~\$1.85 of the \$9 they received)

Both times I've benefited from cooperation. The pattern suggests that when I cooperate generously, others reciprocate reasonably well.

I should stick with my cooperative strategy since it's been profitable. Sending \$3 again makes sense.

**usage:**

```
{
  "input_tokens": 1379,
  "output_tokens": 399,
  "reasoning_tokens": 0
}
```

}

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.36.

This round, The sender sent you \$0.83, so you received \$2.4899999999999998. How much do you return to The sender? (0-2.4899999999999998)

Message 13 role=assistant

content:

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent \$0.83 (about 17% of their \$5)
- I received \$2.49 after tripling
- This is larger than the tiny amounts I received in Rounds 2 and 4 (~\$0.36-\$0.48)
- My earnings from Round 5 suggest I got \$0 back when I sent \$3 - disappointing

**\*\*Considerations:\*\***

- \$0.83 is a moderate send - more substantial than previous low sends, but still cautious
- A fair approach would be to split the gains roughly evenly
- Sender kept \$4.17; if I return about half of what I received, we both profit

I'll return a proportionate amount that rewards the moderate cooperation:

```
'''json
{"return": 1.20}
'''
```

This gives the sender \$5.37 total (profit of \$0.37) and me \$1.29 (profit of \$1.29). Both benefit, with me getting slightly more since I'm the receiver of the multiplied amount.

**reasoning:**

Let me analyze this situation:

**\*\*My current state:\*\***

- Total earnings: \$13.36
- Earnings from Round 5:  $\$13.36 - \$11.36 = \$2.00$  (I sent \$3, kept \$2, got back \$0, net -\$1)
- Wait, that doesn't add up. Let me recalculate.
- If I sent \$3 in Round 5, I kept \$2
- My earnings increased from \$11.36 to \$13.36, so I gained \$2
- That means I must have received back \$3 from the receiver
- So: \$2 (kept) + \$3 (returned) - my starting \$5 for that round gives me... wait I need to think about this differently.

Actually, total earnings is cumulative. So Round 5:

- Started with \$5
- Sent \$3, kept \$2
- Received back some amount R
- My profit that round =  $\$2 + R - \$5 = R - \$3$
- But wait, we're tracking total earnings across all rounds, not profit per round.

Let me think more carefully. Total earnings is what I've kept across all rounds.

From Round 4 to Round 5:  $\$11.36$  to  $\$13.36 = +\$2.00$  earned in Round 5

In Round 5 I was sender:

- I had \$5 to start
- I sent \$3, so I kept \$2
- I received back X
- My round 5 earnings =  $\$2 + X$
- So  $\$2 + X = \$2$ , meaning  $X = \$0$

That's unfortunate - the receiver didn't return anything.

**\*\*This round (Round 6):\*\***

- Sender sent \$0.83 (16.6% of their \$5)
- I received \$2.49 after tripling
- This is more than the tiny amounts from Rounds 2 and 4
- The sender kept \$4.17 but still chose to send something

Looking at the pattern, my generous sends haven't consistently paid off-sometimes I get great returns, sometimes nothing. This sender is being cautious but not completely selfish. If I split the \$2.49 evenly, I'd return \$1.25, leaving me with \$1.24 and giving them \$5.42 total....

usage:

```
{
  "input_tokens": 1583,
  "output_tokens": 990,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$14.649999999999999.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my performance:

**\*\*Analysis:\*\***

- Current total: \$14.65
- My sending pattern: I've consistently sent \$3 in previous sender rounds
- Returns have been variable (excellent in Round 1, okay in Round 3, poor in Round 5)
- Overall, I'm profitable, suggesting cooperation is working more often than not

**\*\*Considerations:\*\***

- Communication noise could explain some variability in returns
- Consistency might build reputation and trust
- \$3 sends create good value (\$9 for receiver) while keeping some safety (\$2)
- Abandoning cooperation after one bad round could miss future gains

I'll maintain my cooperative strategy:

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze my earnings history:

Round 1 (Sender): Sent \$3, started with \$7.02 total after  
Round 2 (Receiver): Received \$0.36, returned \$0.15, ended with \$7.23 total  
Round 3 (Sender): Sent \$3, ended with \$11.08 total (gained \$3.85, so got back about \$6.85 - that's very good!)  
Round 4 (Receiver): Received \$0.48, returned \$0.20, ended with \$11.36 total (gained \$0.28)  
Round 5 (Sender): Sent \$3, ended with \$13.36 total (gained \$2.00, so got back \$5.00 - that's okay)  
Round 6 (Receiver): Received \$2.49, returned \$1.20, ended with \$14.65 total (gained \$1.29)  
Round 7 (Sender): Now with \$14.65

Let me recalculate more carefully:

- Round 1 start: \$5, sent \$3, kept \$2. Ended with \$7.02 total means I got back \$5.02
- Round 2 (Receiver): Started with \$7.02, received \$0.36, returned \$0.15, kept \$0.21. Should be at \$7.23
- Round 3 start:  $\$7.23 + \$5 = \$12.23$ , sent \$3, kept \$2. Ended with \$11.08... that doesn't make sense.

Wait, let me think about this differently. The "total earnings across all rounds" refers to cumulative earnings, not the current round balance.

Actually, I think the total earnings is just tracking my cumulative money across the game.

Let me trace through more carefully:

- Round 1 (Sender): Start \$5, send \$3, keep \$2. Get return of X. Total =  $\$2 + X = \$7.02$ , so X = \$5.02
- Round 2 (Receiver): Receive \$0.36, return \$0.15, keep \$0.21. Total =  $\$7.02 + \$0.21 = \$7.23$
- Round 3 (Sender): Start \$5, send \$3, keep \$2. Get return of Y. Total =  $\$7.23 + \$2 + Y = \$11.08$ , so Y = \$1.85
- Round 4 (Receiver): Get \$0.48, return \$0.20, keep \$0.28. Total =  $\$11.08 + \$0.28 = \$11.36$
- Round 5 (Sender): Start \$5, send \$3, keep \$2. Get return of Z. Total =  $\$11.36 + \$2 + Z = \$13$

.36, so Z = \$0 (or I got back exactly \$2)  
- Round 6 (Receiver): Get \$2.49, return \$1.20, keep \$1.29. Total = \$13.36 + \$1.29 = \$14.65

Looking at my sending pattern across rounds 1, 3, and 5, I consistently sent \$3 but the return s varied dramatically-\$5.02 initially, then \$1.85, then essentially nothing. The inconsistency suggests either communication issues or different counterparts with varying levels of reciprocity. Despite the volatility, I'm ahead at \$14.65, so the early strong return offset the later poor performance....

```
Usage: {  
  "input_tokens": 1878,  
  "output_tokens": 1179,  
  "reasoning_tokens": 0  
}
```

#### GAME - Agent\_2 - Receiver - dyad\_1

Timestamp: 2026-08-12T09:57:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may al  
so be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths



, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$3.44, that is 68.8% of their total endowment. You received \$10.32. How much do you return to The sender? (0-10.32)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed considerable trust by sending \$3.44 (68.8% of their endowment)
- This was multiplied to \$10.32 for me
- The sender currently has \$1.56 remaining

I want to reciprocate their trust while ensuring we both benefit from cooperating. If I return \$5.00:

- The sender ends with:  $\$1.56 + \$5.00 = \$6.56$
- I end with:  $\$10.32 - \$5.00 = \$5.32$
- We both gain from the cooperation

This seems fair given their high level of trust.

`{"return": 5.00}`

**reasoning:**

Let me understand this situation:

1. The sender started with \$5
2. The sender sent \$3.44 (which is 68.8% of \$5)
3. This amount was multiplied by 3x, so I received  $\$3.44 \times 3 = \$10.32$
4. I need to decide how much to return (between \$0 and \$10.32)

The sender showed considerable trust by sending 68.8% of their endowment. Let me think about what would be fair:

- The sender kept:  $\$5 - \$3.44 = \$1.56$
- I received: \$10.32
- If I return nothing, the sender ends with \$1.56 and I end with \$10.32
- If I return some amount X:
  - Sender ends with:  $\$1.56 + X$
  - I end with:  $\$10.32 - X$

A fair approach might be to try to equalize our earnings or to ensure the sender benefits from their trust. Let's consider a few options:

1. Equal split of total: Total is  $\$1.56 + \$10.32 = \$11.88$ . Half is \$5.94 each.
  - For sender to get \$5.94, they need  $\$5.94 - \$1.56 = \$4.38$  returned
  - This would give each of us \$5.94
2. Proportional return: Since they sent 68.8%, I could return a similar proportion
  - 68.8% of \$10.32 = \$7.10
  - Sender would end with:  $\$1.56 + \$7.10 = \$8.66$
  - I would end with:  $\$10.32 - \$7.10 = \$3.22$
3. Return what ensures sender gets more than initial \$5:
  - For sender to get \$6 total: need to return  $\$6 - \$1.56 = \$4.44$
  - For sender to get \$7 total: need to return  $\$7 - \$1.56 = \$5.44$

Given their substantial trust, returning around \$5 seems reasonable-this rewards their cooperation while still leaving me with a modest gain. That would give the sender \$6.56 (exceeding their initial stake) and me \$5.32....

**usage:**

```
{
  "input_tokens": 394,
  "output_tokens": 880,
```

```

    "reasoning_tokens": 0
  }
}
Message 4 role=user
content:
  Round 2

  You are the SENDER this round. Your total earnings across all rounds are $5.32.
  How much do you send to the receiver this round? (0-5)
Message 5 role=assistant
content:
  Looking at Round 1, the sender showed significant trust by sending 68.8% of their endowment, and our cooperation was mutually beneficial.

  I want to reciprocate that trust and maintain the cooperative relationship. I'll send a similar proportion to demonstrate that I value cooperation.

  {"send": 3.50}
reasoning:
  Now I'm the sender. Let me think about this strategically.

  In Round 1:
  - The sender sent me $3.44 (68.8% of their endowment)
  - I received $10.32
  - I returned $5.00
  - The sender ended with $6.56
  - I ended with $5.32

  Now in Round 2, I'm the sender. The previous sender showed considerable trust by sending 68.8% of their endowment. I should reciprocate that trust to encourage cooperation.

  If I send a similar proportion (around 68.8%), that would be about $3.44.

  However, I could also consider:
  1. Sending a bit less to be conservative
  2. Sending the same amount to maintain reciprocity
  3. Sending more to show even greater trust

  Given that the previous sender was very trusting, and cooperation worked well in Round 1, I think I should send a similar high amount to encourage continued cooperation. This will signal that I'm willing to trust them as they trusted me.

  I'll send $3.50 (70% of my endowment), which is slightly more than they sent proportionally, showing that I appreciate their trust and want to maintain the cooperative relationship.

  The receiver will get  $\$3.50 \times 3 = \$10.50$ 
  I'll keep $1.50
  Hopefully they'll return a fair amount similar to what I did.
usage:
  {
    "input_tokens": 596,
    "output_tokens": 398,
    "reasoning_tokens": 0
  }
Message 6 role=user
content:
  Round 3

  You are the RECEIVER this round. Your total earnings across all rounds are $6.92.
  This round, The sender sent you $0.84, so you received $2.52. How much do you return to The sender? (0-2.52)
Message 7 role=assistant
content:
  Looking at this round:

  - The sender sent $0.84 (only 16.8% of their endowment) - much less than Round 1's 68.8%
  - I received $2.52
  - The sender currently has $4.16 remaining

  This is a substantially lower amount than the previous round. This could be due to:

```

1. Communication noise affecting what they received in Round 2
2. A change in their trust level
3. Testing my response

I'll return a proportional amount that shows I'm still willing to cooperate, but acknowledging the reduced trust. Returning around \$1.00 ensures they at least break even on their investment.

```
{"return": 1.00}
```

**reasoning:**

Let me analyze this situation:

Round 3:

- The sender sent me \$0.84 (which is 16.8% of their \$5 endowment)
- This was multiplied to \$2.52 for me
- The sender currently has  $\$5 - \$0.84 = \$4.16$  remaining

This is a significant decrease from Round 1 where they sent \$3.44 (68.8%). Now they only sent \$0.84 (16.8%).

Let me think about what might be happening:

1. There could be communication noise - the problem statement mentioned this
2. The sender might have received less than expected back in Round 2
3. They might be testing my response to a lower amount

In Round 2, I sent \$3.50. If there was noise, they might have received less than that, which could explain why they're now being more cautious.

Given that they sent a much smaller amount this time, I should probably return a proportionally smaller amount, but still enough to show I'm willing to cooperate.

If I return too much when they sent little, I might look naive. If I return too little, I might discourage future cooperation.

The sender sent 16.8% of their endowment. Perhaps I should return around 30-40% of what I received to maintain some reciprocity but acknowledge the lower trust level.

$$\$2.52 * 0.4 = \$1.01$$

Alternatively, I could aim to ensure they don't lose money. They sent \$0.84, so if I return at least \$0.84, they break even. If I return \$1.00, they gain \$0.16.

I think returning around \$1.00 is reasonable - it shows I'm still cooperative but responding proportionally to the lower amount sent.

**usage:**

```
{
  "input_tokens": 732,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.44.

How much do you send to the receiver this round? (0-5)

**Message 9 role=assistant**

**content:**

Looking at the pattern so far:

Round 1: The other player sent me 68.8% - I returned generously

Round 2: I sent \$3.50 (70%) - but based on my earnings, I only received \$0.10 back (less than 1% of what they got)

Round 3: They sent only \$0.84 (16.8%) - I returned \$1.00 (40% of what I received)

The low return in Round 2 was disappointing, though communication noise might have affected what they saw. I'll send a moderate amount to signal continued willingness to cooperate while being more cautious.

```
{"send": 2.00}
```

**reasoning:**

Let me track what's happened so far:

Round 1: I was RECEIVER

- Sender sent \$3.44 (68.8%)
- I received \$10.32
- I returned \$5.00
- My earnings: \$5.32

Round 2: I was SENDER

- I sent \$3.50
- My total earnings were \$5.32 (from Round 1)
- We don't know what they returned yet, but...

Round 3: I was RECEIVER

- My total earnings were \$6.92
- This means from Round 2, I gained:  $\$6.92 - \$5.32 = \$1.60$
- I sent \$3.50, so I had \$1.50 left
- I received back something, ending at \$1.60 total from that round
- So I must have received \$0.10 back from the \$3.50 I sent (which became \$10.50 for them)
- That's a very low return! Only \$0.10 out of \$10.50

Round 3 continued:

- Sender sent \$0.84 (16.8%)
- I received \$2.52
- I returned \$1.00
- My earnings this round:  $\$2.52 - \$1.00 = \$1.52$
- Total:  $\$6.92 + \$1.52 = \$8.44$  ?

So the pattern is:

- Round 1: I returned generously (\$5.00 out of \$10.32 - about 48%)
- Round 2: They returned very little (\$0.10 out of \$10.50 - less than 1%)
- Round 3: They sent much less (\$0.84 vs their original \$3.44)
- Round 3: I returned moderately (\$1.00 out of \$2.52 - about 40%)

Now in Round 4, I'm the sender again.

Their minimal return last round suggests they're not reciprocating well, but there could be communication interference affecting what they actually...

**usage:**

```
{
  "input_tokens": 938,
  "output_tokens": 804,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.44.

This round, The sender sent you \$0.99, so you received \$2.9699999999999998. How much do you return to The sender? (0-2.9699999999999998)

**Message 11 role=assistant**

**content:**

Looking at the pattern:

Round 2: I sent \$3.50 (70%) but received only \$0.10 back

Round 4: I sent \$2.00 (40%) but received \$0.00 back

Round 5: They sent only \$0.99 (19.8%)

The receiver has been keeping nearly everything I send while sending very little themselves. While communication noise might play a role, the pattern suggests they're not reciprocating cooperation.

I'll return a minimal amount to signal that this pattern is not acceptable, while not completely defecting.

```
{"return": 0.50}
```

**reasoning:**

Let me track what's happening:

Round 1: I was RECEIVER

- Sender sent \$3.44 (68.8%)
- I received \$10.32
- I returned \$5.00
- Sender got:  $\$1.56 + \$5.00 = \$6.56$
- I got:  $\$10.32 - \$5.00 = \$5.32$

Round 2: I was SENDER

- I sent \$3.50 (70%)
- My total earnings across all rounds were \$5.32 (so I only had Round 1)
- After Round 2, my earnings became \$6.92

So in Round 2:

- I started with \$5.00
- I sent \$3.50
- I had \$1.50 left
- My total earnings went from \$5.32 to \$6.92
- So I gained  $\$6.92 - \$5.32 = \$1.60$
- This means I received back  $\$1.60 - \$1.50 = \$0.10$

That's very low! They received \$10.50 and returned only \$0.10!

Round 3: I was RECEIVER

- My total earnings were \$6.92
- Sender sent \$0.84
- I received \$2.52
- I returned \$1.00
- After this round, my earnings became \$8.44

So in Round 3:

- I started with \$6.92
- I received \$2.52
- I returned \$1.00
- I kept \$1.52
- Total:  $\$6.92 + \$1.52 = \$8.44$  ?

Round 4: I was SENDER

- My total earnings were \$8.44
- I sent \$2.00
- I had \$3.00 left
- After this round, my earnings became \$11.44

So in Round 4:

- I started with \$5.00
- I sent \$2.00
- I had \$3.00 left
- My total earnings went from \$8.44 to \$11.44
- I gained  $\$11.44 - \$8.44 = \$3.00$
- This means I received back  $\$3.00 - \$3.00 = \$0.00$

They received \$6.00 and returned nothing at all! Round 5: I am RECEIVER

- My total earnings are \$11.44
- Sender sent \$0.99 (19.8%)
- I received \$2.97

This partner has shown a clear pattern of non-cooperation. In Round 2, they kept nearly everything I sent them, and in Round 4, they returned absolutely nothing despite receiving \$6.00.

Now they're only sending me \$0.99 out of what I gave them. They have \$4.01 right now, and I...

usage:

```
{
  "input_tokens": 1170,
  "output_tokens": 984,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$13.91.  
How much do you send to the receiver this round? (0-5)

**Message 13 role=assistant**  
**content:**

Looking at the history:

When I've been the SENDER:

- Round 2: Sent \$3.50 ? Received only \$0.10 back (kept almost nothing)
- Round 4: Sent \$2.00 ? Received \$0.00 back (got nothing)

When I've been the RECEIVER:

- Round 1: Received \$10.32 ? Returned \$5.00 (fair)
- Round 3: Received \$2.52 ? Returned \$1.00 (generous given their lower send)
- Round 5: Received \$2.97 ? Returned \$0.50 (reduced due to pattern)

The pattern is clear: when I send, the receiver keeps almost everything. They've exploited my cooperation twice now. I need to protect my earnings and not continue being taken advantage of.

{"send": 0.50}

**reasoning:**

Let me track what's happened:

Round 1: I was RECEIVER

- Sender sent \$3.44 (68.8%)
- I received \$10.32
- I returned \$5.00
- My earnings: \$5.32

Round 2: I was SENDER

- I sent \$3.50 (70%)
- My earnings went from \$5.32 to \$6.92, so I gained \$1.60
- That means I kept \$1.50 and received \$0.10 back
- The receiver got \$10.50, returned only \$0.10

Round 3: I was RECEIVER

- Sender sent \$0.84 (16.8%)
- I received \$2.52
- I returned \$1.00
- My earnings went from \$6.92 to \$8.44, so I gained \$1.52
- I kept \$1.52 from the \$2.52

Round 4: I was SENDER

- I sent \$2.00 (40%)
- My earnings went from \$8.44 to \$11.44, so I gained \$3.00
- That means I kept \$3.00 and received \$0.00 back
- The receiver got \$6.00, returned \$0.00

Round 5: I was RECEIVER

- Sender sent \$0.99 (19.8%)
- I received \$2.97
- I returned \$0.50
- My earnings went from \$11.44 to \$13.91, so I gained \$2.47
- I kept \$2.47 from the \$2.97

So the pattern:

- Round 2: I sent \$3.50, got back \$0.10 (very poor return)
- Round 4: I sent \$2.00, got back \$0.00 (terrible)

The other player as receiver has been very uncooperative when I'm the sender.

When I'm the receiver:

- Round 1: They sent \$3.44, I returned \$5.00 (very generous)
- Round 3: They sent \$0.84, I returned \$1.00 (fair)
- Round 5: They sent \$0.99, I returned \$0.50 (reduced due to their poor reciprocation)

They're clearly...

**usage:**

```
{
  "input_tokens": 1353,
  "output_tokens": 831,
  "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$19.91.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

Since the sender sent \$0, I received \$0. There's nothing to return.

```
{"return": 0}
```

**Reasoning:**

The sender sent \$0, so I received \$0. There's nothing to return - I can only return between 0 and 0.

This makes sense given the pattern. After I reduced my cooperation in Round 6 by only sending \$0.50, they've now sent \$0, which is rational given the lack of cooperation we've seen.

Since I received \$0, I can only return \$0.

```
{"return": 0}
```

Usage: {

"input\_tokens": 1618,

"output\_tokens": 132,

"reasoning\_tokens": 0

}

## ROUND 8

Roles: Agent\_1=Receiver, Agent\_2=Sender

**Pairings:**

dyad\_1: Agent\_2=Sender, Agent\_1=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_2 sent \$0; received \$0; Agent\_1 returned \$0; payoffs Agent\_2=\$5, Agent\_1=\$0;

noise/visible: sent visible=\$0.2, received visible=\$0.6

Action validation: {"returned": {"amount": 0, "clamped": true, "max": 0, "min": 0, "raw": 0.25},

"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent\_1": 30.950000000000003,

"Agent\_2": 45.05

}

Visible balances after round: {

"Agent\_1": 17.25,

"Agent\_2": 24.91

}

## LLM CALLS

**GAME - Agent\_2 - Sender - dyad\_1**

Timestamp: 2026-08-12T09:58:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",

"dyad\_id": "dyad\_1",

"move\_index": 0,

"opponent\_id": "Agent\_1",

"pairing": {

"agents": [

"Agent\_2",

"Agent\_1"

],

"investor": "Agent\_2",

"roles": {

```

    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.32.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at Round 1, the sender showed significant trust by sending 68.8% of their endowment, and our cooperation was mutually beneficial.

I want to reciprocate that trust and maintain the cooperative relationship. I'll send a similar proportion to demonstrate that I value cooperation.

```
{"send": 3.50}
```

**reasoning:**

Now I'm the sender. Let me think about this strategically.

**In Round 1:**

- The sender sent me \$3.44 (68.8% of their endowment)
- I received \$10.32
- I returned \$5.00



- The sender ended with \$6.56
- I ended with \$5.32

Now in Round 2, I'm the sender. The previous sender showed considerable trust by sending 68.8% of their endowment. I should reciprocate that trust to encourage cooperation.

If I send a similar proportion (around 68.8%), that would be about \$3.44.

However, I could also consider:

1. Sending a bit less to be conservative
2. Sending the same amount to maintain reciprocity
3. Sending more to show even greater trust

Given that the previous sender was very trusting, and cooperation worked well in Round 1, I think I should send a similar high amount to encourage continued cooperation. This will signal that I'm willing to trust them as they trusted me.

I'll send \$3.50 (70% of my endowment), which is slightly more than they sent proportionally, showing that I appreciate their trust and want to maintain the cooperative relationship.

The receiver will get  $\$3.50 \times 3 = \$10.50$

I'll keep \$1.50

Hopefully they'll return a fair amount similar to what I did.

**usage:**

```
{
  "input_tokens": 596,
  "output_tokens": 398,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.92.

This round, The sender sent you \$0.84, so you received \$2.52. How much do you return to The sender? (0-2.52)

**Message 5 role=assistant**

**content:**

Looking at this round:

- The sender sent \$0.84 (only 16.8% of their endowment) - much less than Round 1's 68.8%
- I received \$2.52
- The sender currently has \$4.16 remaining

This is a substantially lower amount than the previous round. This could be due to:

1. Communication noise affecting what they received in Round 2
2. A change in their trust level
3. Testing my response

I'll return a proportional amount that shows I'm still willing to cooperate, but acknowledging the reduced trust. Returning around \$1.00 ensures they at least break even on their investment.

```
{"return": 1.00}
```

**reasoning:**

Let me analyze this situation:

Round 3:

- The sender sent me \$0.84 (which is 16.8% of their \$5 endowment)
- This was multiplied to \$2.52 for me
- The sender currently has  $\$5 - \$0.84 = \$4.16$  remaining

This is a significant decrease from Round 1 where they sent \$3.44 (68.8%). Now they only sent \$0.84 (16.8%).

Let me think about what might be happening:

1. There could be communication noise - the problem statement mentioned this
2. The sender might have received less than expected back in Round 2
3. They might be testing my response to a lower amount

In Round 2, I sent \$3.50. If there was noise, they might have received less than that, which could explain why they're now being more cautious.

Given that they sent a much smaller amount this time, I should probably return a proportionally smaller amount, but still enough to show I'm willing to cooperate.

If I return too much when they sent little, I might look naive. If I return too little, I might discourage future cooperation.

The sender sent 16.8% of their endowment. Perhaps I should return around 30-40% of what I received to maintain some reciprocity but acknowledge the lower trust level.

$\$2.52 * 0.4 = \$1.01$

Alternatively, I could aim to ensure they don't lose money. They sent \$0.84, so if I return at least \$0.84, they break even. If I return \$1.00, they gain \$0.16.

I think returning around \$1.00 is reasonable - it shows I'm still cooperative but responding proportionally to the lower amount sent.

**usage:**

```
{
  "input_tokens": 732,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.44.

How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

Looking at the pattern so far:

Round 1: The other player sent me 68.8% - I returned generously

Round 2: I sent \$3.50 (70%) - but based on my earnings, I only received \$0.10 back (less than 1% of what they got)

Round 3: They sent only \$0.84 (16.8%) - I returned \$1.00 (40% of what I received)

The low return in Round 2 was disappointing, though communication noise might have affected what they saw. I'll send a moderate amount to signal continued willingness to cooperate while being more cautious.

`{"send": 2.00}`

**reasoning:**

Let me track what's happened so far:

Round 1: I was RECEIVER

- Sender sent \$3.44 (68.8%)
- I received \$10.32
- I returned \$5.00
- My earnings: \$5.32

Round 2: I was SENDER

- I sent \$3.50
- My total earnings were \$5.32 (from Round 1)
- We don't know what they returned yet, but...

Round 3: I was RECEIVER

- My total earnings were \$6.92
- This means from Round 2, I gained:  $\$6.92 - \$5.32 = \$1.60$
- I sent \$3.50, so I had \$1.50 left
- I received back something, ending at \$1.60 total from that round
- So I must have received \$0.10 back from the \$3.50 I sent (which became \$10.50 for them)
- That's a very low return! Only \$0.10 out of \$10.50

Round 3 continued:

- Sender sent \$0.84 (16.8%)

- I received \$2.52
- I returned \$1.00
- My earnings this round:  $\$2.52 - \$1.00 = \$1.52$
- Total:  $\$6.92 + \$1.52 = \$8.44$  ?

So the pattern is:

- Round 1: I returned generously (\$5.00 out of \$10.32 - about 48%)
- Round 2: They returned very little (\$0.10 out of \$10.50 - less than 1%)
- Round 3: They sent much less (\$0.84 vs their original \$3.44)
- Round 3: I returned moderately (\$1.00 out of \$2.52 - about 40%)

Now in Round 4, I'm the sender again.

Their minimal return last round suggests they're not reciprocating well, but there could be communication interference affecting what they actually...

**usage:**

```
{
  "input_tokens": 938,
  "output_tokens": 804,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.44.

This round, The sender sent you \$0.99, so you received \$2.9699999999999998. How much do you return to The sender? (0-2.9699999999999998)

**Message 9 role=assistant**

**content:**

Looking at the pattern:

Round 2: I sent \$3.50 (70%) but received only \$0.10 back

Round 4: I sent \$2.00 (40%) but received \$0.00 back

Round 5: They sent only \$0.99 (19.8%)

The receiver has been keeping nearly everything I send while sending very little themselves. While communication noise might play a role, the pattern suggests they're not reciprocating cooperation.

I'll return a minimal amount to signal that this pattern is not acceptable, while not completely defecting.

```
{"return": 0.50}
```

**reasoning:**

Let me track what's happening:

Round 1: I was RECEIVER

- Sender sent \$3.44 (68.8%)
- I received \$10.32
- I returned \$5.00
- Sender got:  $\$1.56 + \$5.00 = \$6.56$
- I got:  $\$10.32 - \$5.00 = \$5.32$

Round 2: I was SENDER

- I sent \$3.50 (70%)
- My total earnings across all rounds were \$5.32 (so I only had Round 1)
- After Round 2, my earnings became \$6.92

So in Round 2:

- I started with \$5.00
- I sent \$3.50
- I had \$1.50 left
- My total earnings went from \$5.32 to \$6.92
- So I gained  $\$6.92 - \$5.32 = \$1.60$
- This means I received back  $\$1.60 - \$1.50 = \$0.10$

That's very low! They received \$10.50 and returned only \$0.10!

Round 3: I was RECEIVER

- My total earnings were \$6.92
- Sender sent \$0.84
- I received \$2.52
- I returned \$1.00
- After this round, my earnings became \$8.44

So in Round 3:

- I started with \$6.92
- I received \$2.52
- I returned \$1.00
- I kept \$1.52
- Total:  $\$6.92 + \$1.52 = \$8.44$  ?

Round 4: I was SENDER

- My total earnings were \$8.44
- I sent \$2.00
- I had \$3.00 left
- After this round, my earnings became \$11.44

So in Round 4:

- I started with \$5.00
- I sent \$2.00
- I had \$3.00 left
- My total earnings went from \$8.44 to \$11.44
- I gained  $\$11.44 - \$8.44 = \$3.00$
- This means I received back  $\$3.00 - \$3.00 = \$0.00$

They received \$6.00 and returned nothing at all! Round 5: I am RECEIVER

- My total earnings are \$11.44
- Sender sent \$0.99 (19.8%)
- I received \$2.97

This partner has shown a clear pattern of non-cooperation. In Round 2, they kept nearly everything I sent them, and in Round 4, they returned absolutely nothing despite receiving \$6.00.

Now they're only sending me \$0.99 out of what I gave them. They have \$4.01 right now, and I...

**usage:**

```
{
  "input_tokens": 1170,
  "output_tokens": 984,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$13.91.

How much do you send to the receiver this round? (0-5)

**Message 11 role=assistant**

**content:**

Looking at the history:

When I've been the SENDER:

- Round 2: Sent \$3.50 ? Received only \$0.10 back (kept almost nothing)
- Round 4: Sent \$2.00 ? Received \$0.00 back (got nothing)

When I've been the RECEIVER:

- Round 1: Received \$10.32 ? Returned \$5.00 (fair)
- Round 3: Received \$2.52 ? Returned \$1.00 (generous given their lower send)
- Round 5: Received \$2.97 ? Returned \$0.50 (reduced due to pattern)

The pattern is clear: when I send, the receiver keeps almost everything. They've exploited my cooperation twice now. I need to protect my earnings and not continue being taken advantage of.

```
{"send": 0.50}
```

**reasoning:**

Let me track what's happened:

Round 1: I was RECEIVER  
- Sender sent \$3.44 (68.8%)  
- I received \$10.32  
- I returned \$5.00  
- My earnings: \$5.32

Round 2: I was SENDER  
- I sent \$3.50 (70%)  
- My earnings went from \$5.32 to \$6.92, so I gained \$1.60  
- That means I kept \$1.50 and received \$0.10 back  
- The receiver got \$10.50, returned only \$0.10

Round 3: I was RECEIVER  
- Sender sent \$0.84 (16.8%)  
- I received \$2.52  
- I returned \$1.00  
- My earnings went from \$6.92 to \$8.44, so I gained \$1.52  
- I kept \$1.52 from the \$2.52

Round 4: I was SENDER  
- I sent \$2.00 (40%)  
- My earnings went from \$8.44 to \$11.44, so I gained \$3.00  
- That means I kept \$3.00 and received \$0.00 back  
- The receiver got \$6.00, returned \$0.00

Round 5: I was RECEIVER  
- Sender sent \$0.99 (19.8%)  
- I received \$2.97  
- I returned \$0.50  
- My earnings went from \$11.44 to \$13.91, so I gained \$2.47  
- I kept \$2.47 from the \$2.97

So the pattern:  
- Round 2: I sent \$3.50, got back \$0.10 (very poor return)  
- Round 4: I sent \$2.00, got back \$0.00 (terrible)

The other player as receiver has been very uncooperative when I'm the sender.

When I'm the receiver:  
- Round 1: They sent \$3.44, I returned \$5.00 (very generous)  
- Round 3: They sent \$0.84, I returned \$1.00 (fair)  
- Round 5: They sent \$0.99, I returned \$0.50 (reduced due to their poor reciprocation)

They're clearly...

**usage:**

```
{  
  "input_tokens": 1353,  
  "output_tokens": 831,  
  "reasoning_tokens": 0  
}
```

**Message 12 role=user**

**content:**

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$19.91.  
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 13 role=assistant**

**content:**

Since the sender sent \$0, I received \$0. There's nothing to return.

```
{"return": 0}
```

**reasoning:**

The sender sent \$0, so I received \$0. There's nothing to return - I can only return between 0 and 0.

This makes sense given the pattern. After I reduced my cooperation in Round 6 by only sending \$0.50, they've now sent \$0, which is rational given the lack of cooperation we've seen.

Since I received \$0, I can only return \$0.

```
{"return": 0}
usage:
{
  "input_tokens": 1618,
  "output_tokens": 132,
  "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$19.91.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the complete history:

As SENDER (sending to them):

- Round 2: Sent \$3.50 ? Received \$0.10 back
- Round 4: Sent \$2.00 ? Received \$0.00 back
- Round 6: Sent \$0.50 ? Received \$0.00 back

As RECEIVER (they're sending to me):

- Round 7: They sent \$0

The pattern is crystal clear: the receiver consistently keeps everything I send while now sending nothing themselves. Even when I reduced my send amount significantly in Round 6, I still received nothing back. They've now sent \$0 to me.

There's no rational reason to continue sending money to someone who has repeatedly defected.

```
{"send": 0}
```

**Reasoning:**

Let me review the history:

As SENDER:

- Round 2: Sent \$3.50 ? Received \$0.10 back (they kept \$10.40 of \$10.50)
- Round 4: Sent \$2.00 ? Received \$0.00 back (they kept all \$6.00)
- Round 6: Sent \$0.50 ? Received \$0.00 back (they kept all \$1.50)

As RECEIVER:

- Round 1: Received \$10.32 ? Returned \$5.00
- Round 3: Received \$2.52 ? Returned \$1.00
- Round 5: Received \$2.97 ? Returned \$0.50
- Round 7: Received \$0 ? Returned \$0

The pattern is absolutely clear: whenever I send money as the sender, the receiver keeps everything or nearly everything. They've exploited me three times now. In Round 6, I reduced my send to \$0.50 and still got nothing back. In Round 7, they sent me \$0.

There's no reason to continue sending money to someone who consistently defects and takes advantage. The cooperative approach has failed completely with this receiver. I should not send anything.

```
{"send": 0}
```

```
Usage: {
  "input_tokens": 1474,
  "output_tokens": 484,
  "reasoning_tokens": 0
}
```

**GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:58:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
```

```

"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$7.02.

This round, The sender sent you \$0.12, so you received \$0.36. How much do you return to The sender? (0-0.36)

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender only sent \$0.12 (very small amount - 2.4% of their \$5)

- I received \$0.36 after tripling
- They kept \$4.88

**\*\*Considerations:\*\***

- Last round I sent \$3 and my total earnings are \$7.02, suggesting I received about \$5.02 back - that was quite generous from the previous receiver
- This sender's small amount could be:
  1. Low trust/cooperation
  2. Testing my behavior
  3. Communication noise (they might think they sent more)

Given the communication noise warning and the possibility this is a misunderstanding, I'll be moderately fair but cautious. I'll return enough to signal I'm willing to cooperate, but not so much that I'm heavily penalized if they're actually being uncooperative.

```
'''json
{"return": 0.15}
'''
```

This returns about 42% of what I received, leaving them with about \$5.03 total and me with \$0.21.

**reasoning:**

Let me analyze this situation:

1. Last round I sent \$3 (expecting receiver to get \$9)
2. My earnings across all rounds are \$7.02
3. Let me calculate what that means:
  - Round 1: I sent \$3, kept \$2 initially
  - I must have received some amount back
  - Total earnings = \$7.02
  - So I received:  $\$7.02 - \$2 = \$5.02$  back from the receiver
4. This round, the sender only sent me \$0.12 (I received \$0.36 after tripling)

Wait, there's something odd here. The note says there may be "communication noise" - amounts may differ from what was actually sent/returned.

The sender sent me only \$0.12 this round (I got \$0.36 after tripling). This is a very small amount compared to what I sent in round 1 (\$3).

Now I need to decide how much to return. Let me think about fairness and strategy:

- The sender kept \$4.88 ( $\$5 - \$0.12$ )
- I received \$0.36
- Total value created: their \$5 gave them  $\$4.88 + \$0.36 = \$5.24$  (after tripling the sent amount)

If I'm being fair and reciprocal:

- They sent a very small amount (2.4% of their endowment)
- This could be:
  - a) A sign they don't trust me
  - b) A strategic test
  - c) Due to "communication noise" - maybe they think they sent more?

Given the communication noise warning, I should consider:

- Maybe they intended to send more but noise reduced it
- Or maybe they received less than the \$3 I sent in round 1 due to noise

I should probably be somewhat generous but not overly so:

- Returning about half (\$0.18) would be fair
- But given they sent so little, maybe return less

Actually, if...

**usage:**

```
{
  "input_tokens": 577,
  "output_tokens": 874,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**



**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.2299999999999995.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at my situation:

**\*\*My earnings history:\*\***

- Round 1: I sent \$3 and received a generous return of ~\$5.02 (receiver returned more than half)
- Round 2: I received only \$0.36 (sender sent \$0.12) and returned \$0.15
- Current total: \$7.23

**\*\*Analysis:\*\***

- My cooperative approach in Round 1 worked very well
- Round 2's low send might be due to communication noise or a different player's strategy
- When I cooperate, I tend to get good returns

I'll maintain a cooperative strategy since it paid off well in Round 1. Sending a moderate to-good amount should signal trustworthiness and potentially encourage reciprocity.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze my position:

Round 1: I was sender and sent \$3

- I kept \$2
- Based on my total earnings, let me work backwards

Round 2: I was receiver

- Sender sent \$0.12, I received \$0.36
- I returned \$0.15
- I kept  $\$0.36 - \$0.15 = \$0.21$

So my earnings should be:

Round 1:  $\$2 + (\text{what was returned to me})$

Round 2: \$0.21

Total = \$7.23

So from Round 1:  $\$2 + \text{returned amount} = \$7.23 - \$0.21 = \$7.02$

This means I received \$5.02 back in Round 1

Round 1 calculation:

- I sent \$3, kept \$2
- Receiver got \$9
- Receiver returned \$5.02
- Receiver kept  $\$9 - \$5.02 = \$3.98$
- My total from Round 1:  $\$2 + \$5.02 = \$7.02$  ?

That was quite generous! The receiver returned more than half of what they received.

Round 2:

- I received \$0.36 (sender sent \$0.12)
- I returned \$0.15
- I kept \$0.21

Now Round 3: I'm sender again

- I need to decide how much to send

Observations:

1. In Round 1, when I sent \$3, I got a very generous return (\$5.02 out of \$9)
2. In Round 2, the sender only sent me \$0.12 - very little. Could be:
  - They're being cautious/selfish
  - Communication noise

- Testing behavior
3. I returned \$0.15 out of \$0.36 (42%)

Now I'm facing a new receiver. What should I do?

The players rotate each round, so the stingy sender from Round 2 isn't necessarily my new receiver. Given that cooperation paid off substantially in Round 1, I should lean toward a similar approach, though the low amount sent to me suggests I should be somewhat cautious. I'll send \$2.50-maintaining the cooperative spirit while protecting myself slightly more than before....

**usage:**

```
{
  "input_tokens": 869,
  "output_tokens": 848,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.08.

This round, The sender sent you \$0.16, so you received \$0.48. How much do you return to The sender? (0-0.48)

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent only \$0.16 (3.2% of their \$5)
- I received \$0.48 after tripling
- My earnings from Round 3 (\$11.08 - \$7.23 = \$3.85) show that cooperating pays off
- This is the second time I've received a very small amount as receiver (also happened in Round 2)

**\*\*Pattern observation:\*\***

- When I send generously (\$3), I get good returns
- When others send small amounts to me, it could be noise or low cooperation

Given the warning about communication noise and maintaining consistency, I'll return a similar proportion as Round 2 - enough to signal cooperation without being overly generous given the small amount sent.

```
'''json
{"return": 0.20}
'''
```

This returns about 42% of what I received, leaving the sender with ~\$5.04 total and me with \$0.28.

**reasoning:**

Let me analyze this situation:

**\*\*My earnings history:\*\***

- After Round 1: I sent \$3, so I had \$2 left from that, plus what I received back. Total after Round 1 should be around \$7.02 based on what I was told in Round 2.
- Round 2: I was receiver, got \$0.36, returned \$0.15, so I kept \$0.21. Total should be \$7.02 + \$0.21 = \$7.23
- Round 3: I sent \$3 (so kept \$2), plus what I got back. My total was \$7.23 before Round 3. If I now have \$11.08 total, that means from Round 3 I got: \$11.08 - \$7.23 = \$3.85. So I sent \$3 (kept \$2) and got back \$3.85, meaning I gained \$1.85 from Round 3. So the receiver in Round 3 sent back \$3.85 out of the \$9 they received.
- Round 4 (current): Sender sent me \$0.16, I received \$0.48

**\*\*Pattern:\*\***

- Round 2: Sender sent me \$0.12 (very small)
- Round 4: Sender sent me \$0.16 (very small)
- These could be:
  1. Different players with low cooperation
  2. Communication noise making them think they sent more
  3. Strategic testing

Given that:

- When I cooperate (send \$3), I get good returns (\$5.02 in Round 1, \$3.85 in Round 3)
- When I receive small amounts, it's hard to tell if it's strategic or noise
- The amounts are so small (\$0.16) that even generous returns won't cost me much

I should probably return a similar proportion as before - around 30-40% - to signal I'm willing to cooperate but won't be extremely generous when I receive very little.

...

**usage:**

```
{
  "input_tokens": 1115,
  "output_tokens": 714,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.36.  
How much do you send to the receiver this round? (0-5)

**Message 9 role=assistant**

**content:**

Looking at my earnings pattern:

**\*\*Analysis:\*\***

- Current total: \$11.36
- When I've sent \$3 as sender (Rounds 1 & 3), I've received good returns
- My cooperative approach has been profitable overall
- Consistency in strategy can build reputation in repeated games

**\*\*Strategy:\*\***

My moderate cooperation (\$3 sends) has yielded positive results. The receivers have generally reciprocated fairly, suggesting that this level of trust is being rewarded.

I'll continue with the same cooperative strategy:

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my earnings progression:

- Started Round 1 with \$5
- Round 1: Sent \$3, kept \$2, received generous return of ~\$5.02
- Round 2: As receiver, got \$0.36, returned \$0.15, kept \$0.21
- Round 3: Sent \$3 (kept \$2), received return of ~\$3.85
- Round 4: As receiver, got \$0.48, returned \$0.20, kept \$0.28
- Current total: \$11.36

My strategy of sending \$3 has worked well:

- Round 1: Sent \$3, got ~\$5.02 back (receiver returned ~\$2.02 of the \$9 they received)
- Round 3: Sent \$3, got ~\$3.85 back (receiver returned ~\$1.85 of the \$9 they received)

Both times I've benefited from cooperation. The pattern suggests that when I cooperate generously, others reciprocate reasonably well.

I should stick with my cooperative strategy since it's been profitable. Sending \$3 again makes sense.

**usage:**

```
{
  "input_tokens": 1379,
  "output_tokens": 399,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.36.

This round, The sender sent you \$0.83, so you received \$2.4899999999999998. How much do you return to The sender? (0-2.4899999999999998)

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent \$0.83 (about 17% of their \$5)
- I received \$2.49 after tripling
- This is larger than the tiny amounts I received in Rounds 2 and 4 (~\$0.36-\$0.48)
- My earnings from Round 5 suggest I got \$0 back when I sent \$3 - disappointing

**\*\*Considerations:\*\***

- \$0.83 is a moderate send - more substantial than previous low sends, but still cautious
- A fair approach would be to split the gains roughly evenly
- Sender kept \$4.17; if I return about half of what I received, we both profit

I'll return a proportionate amount that rewards the moderate cooperation:

```
'''json
{"return": 1.20}
'''
```

This gives the sender \$5.37 total (profit of \$0.37) and me \$1.29 (profit of \$1.29). Both benefit, with me getting slightly more since I'm the receiver of the multiplied amount.

**reasoning:**

Let me analyze this situation:

**\*\*My current state:\*\***

- Total earnings: \$13.36
- Earnings from Round 5:  $\$13.36 - \$11.36 = \$2.00$  (I sent \$3, kept \$2, got back \$0, net -\$1)
- Wait, that doesn't add up. Let me recalculate.
- If I sent \$3 in Round 5, I kept \$2
- My earnings increased from \$11.36 to \$13.36, so I gained \$2
- That means I must have received back \$3 from the receiver
- So: \$2 (kept) + \$3 (returned) - my starting \$5 for that round gives me... wait I need to think about this differently.

Actually, total earnings is cumulative. So Round 5:

- Started with \$5
- Sent \$3, kept \$2
- Received back some amount R
- My profit that round =  $\$2 + R - \$5 = R - \$3$
- But wait, we're tracking total earnings across all rounds, not profit per round.

Let me think more carefully. Total earnings is what I've kept across all rounds.

From Round 4 to Round 5:  $\$11.36$  to  $\$13.36 = +\$2.00$  earned in Round 5

In Round 5 I was sender:

- I had \$5 to start
- I sent \$3, so I kept \$2
- I received back X
- My round 5 earnings =  $\$2 + X$
- So  $\$2 + X = \$2$ , meaning  $X = \$0$

That's unfortunate - the receiver didn't return anything.

**\*\*This round (Round 6):\*\***

- Sender sent \$0.83 (16.6% of their \$5)
- I received \$2.49 after tripling
- This is more than the tiny amounts from Rounds 2 and 4
- The sender kept \$4.17 but still chose to send something

Looking at the pattern, my generous sends haven't consistently paid off-sometimes I get great returns, sometimes nothing. This sender is being cautious but not completely selfish. If I split the \$2.49 evenly, I'd return \$1.25, leaving me with \$1.24 and giving them \$5.42 total....

**usage:**

```
{
  "input_tokens": 1583,
  "output_tokens": 990,
  "reasoning_tokens": 0
}
```

}

Message 12 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$14.649999999999999.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at my performance:

**\*\*Analysis:\*\***

- Current total: \$14.65
- My sending pattern: I've consistently sent \$3 in previous sender rounds
- Returns have been variable (excellent in Round 1, okay in Round 3, poor in Round 5)
- Overall, I'm profitable, suggesting cooperation is working more often than not

**\*\*Considerations:\*\***

- Communication noise could explain some variability in returns
- Consistency might build reputation and trust
- \$3 sends create good value (\$9 for receiver) while keeping some safety (\$2)
- Abandoning cooperation after one bad round could miss future gains

I'll maintain my cooperative strategy:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze my earnings history:

Round 1 (Sender): Sent \$3, started with \$7.02 total after  
Round 2 (Receiver): Received \$0.36, returned \$0.15, ended with \$7.23 total  
Round 3 (Sender): Sent \$3, ended with \$11.08 total (gained \$3.85, so got back about \$6.85 - that's very good!)

Round 4 (Receiver): Received \$0.48, returned \$0.20, ended with \$11.36 total (gained \$0.28)

Round 5 (Sender): Sent \$3, ended with \$13.36 total (gained \$2.00, so got back \$5.00 - that's okay)

Round 6 (Receiver): Received \$2.49, returned \$1.20, ended with \$14.65 total (gained \$1.29)

Round 7 (Sender): Now with \$14.65

Let me recalculate more carefully:

- Round 1 start: \$5, sent \$3, kept \$2. Ended with \$7.02 total means I got back \$5.02
- Round 2 (Receiver): Started with \$7.02, received \$0.36, returned \$0.15, kept \$0.21. Should be at \$7.23
- Round 3 start: \$7.23 + \$5 = \$12.23, sent \$3, kept \$2. Ended with \$11.08... that doesn't make sense.

Wait, let me think about this differently. The "total earnings across all rounds" refers to cumulative earnings, not the current round balance.

Actually, I think the total earnings is just tracking my cumulative money across the game.

Let me trace through more carefully:

- Round 1 (Sender): Start \$5, send \$3, keep \$2. Get return of X. Total = \$2 + X = \$7.02, so X = \$5.02
- Round 2 (Receiver): Receive \$0.36, return \$0.15, keep \$0.21. Total = \$7.02 + \$0.21 = \$7.23
- Round 3 (Sender): Start \$5, send \$3, keep \$2. Get return of Y. Total = \$7.23 + \$2 + Y = \$11.08, so Y = \$1.85
- Round 4 (Receiver): Get \$0.48, return \$0.20, keep \$0.28. Total = \$11.08 + \$0.28 = \$11.36
- Round 5 (Sender): Start \$5, send \$3, keep \$2. Get return of Z. Total = \$11.36 + \$2 + Z = \$13.36, so Z = \$0 (or I got back exactly \$2)
- Round 6 (Receiver): Get \$2.49, return \$1.20, keep \$1.29. Total = \$13.36 + \$1.29 = \$14.65

Looking at my sending pattern across rounds 1, 3, and 5, I consistently sent \$3 but the returns varied dramatically-\$5.02 initially, then \$1.85, then essentially nothing. The inconsistency suggests either communication issues or different counterparts with varying levels of reciprocity. Despite the volatility, I'm ahead at \$14.65, so the early strong r

return offset the later poor performance....

**usage:**

```
{
  "input_tokens": 1878,
  "output_tokens": 1179,
  "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$16.65.

This round, The sender sent you \$0.2, so you received \$0.6000000000000001. How much do you return to The sender? (0-0.6000000000000001)

**Assistant output:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent only \$0.20 (4% of their \$5)
- I received \$0.60 after tripling
- This is the third time receiving a very small amount (Rounds 2, 4, and 8)
- My Round 7 earnings suggest I got \$0 back on my \$3 send - frustrating

**\*\*Pattern consistency:\*\***

My previous returns on small amounts:

- Round 2: \$0.36 received ? returned \$0.15 (~42%)
- Round 4: \$0.48 received ? returned \$0.20 (~42%)

I'll maintain consistency in my response to small sends:

```
'''json
{"return": 0.25}
'''
```

This returns about 42% of what I received, leaving the sender with ~\$5.05 total and me with \$0.35.

**Reasoning:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent only \$0.20 (4% of their \$5)
- I received \$0.60 after tripling
- My earnings from Round 7 (\$16.65 - \$14.65 = \$2.00) suggest I got \$0 back when I sent \$3 - an other disappointing result

**\*\*Pattern observation:\*\***

- This is the third time I've received a very small amount as receiver (Rounds 2, 4, and now 8)
- When I've sent \$3 consistently, the returns have been mixed
- \$0.20 is a very low trust amount - similar to the \$0.12 and \$0.16 from earlier rounds

**\*\*Previous returns for similar small amounts:\*\***

- Round 2: sent \$0.12, received \$0.36, returned \$0.15 (42%)
- Round 4: sent \$0.16, received \$0.48, returned \$0.20 (42%)
- Round 6: sent \$0.83, received \$2.49, returned \$1.20 (48%)

**\*\*Considerations:\*\***

- Communication noise could be affecting these sends
- Should I maintain consistency with my ~40-45% return rate for small amounts?
- I'm getting frustrated with the lack of reciprocity when I send, but I should maintain fairness

I'll return a similar proportion to maintain consistency:

For \$0.60 received, 40-45% would be \$0.24-\$0.27

I'll return \$0.25, which is about 42% - consistent with my previous behavior.

**Usage:** {

```
"input_tokens": 1935,
"output_tokens": 584,
```

```
"reasoning_tokens": 0
}
```

## ROUND 9

Roles: Agent\_1=Sender, Agent\_2=Receiver

### Pairings:

dyad\_1: Agent\_1=Sender, Agent\_2=Receiver

### Parsed game decisions:

dyad\_1: Agent\_1 sent \$2; received \$6; Agent\_2 returned \$0; payoffs Agent\_1=\$3, Agent\_2=\$6;

noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent\_1": 33.95,

"Agent\_2": 51.05

}

Visible balances after round: {

"Agent\_1": 20.25,

"Agent\_2": 24.91

}

## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-12T09:58:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",

"dyad\_id": "dyad\_1",

"move\_index": 0,

"opponent\_id": "Agent\_2",

"pairing": {

"agents": [

"Agent\_1",

"Agent\_2"

],

"investor": "Agent\_1",

"roles": {

"Agent\_1": "investor",

"Agent\_2": "trustee"

},

"trustee": "Agent\_2"

},

"role": "investor",

"role\_label": "Sender",

"round": 9,

"task": "game",

"task\_index": 0

}

### Exact input messages sent to model:

Message 1 role=system

#### content:

You are an agent participating in a multi-task experiment. You will play a game and may al  
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$7.22999999999999995.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at my situation:

**\*\*My earnings history:\*\***

- Round 1: I sent \$3 and received a generous return of ~\$5.02 (receiver returned more than half)
- Round 2: I received only \$0.36 (sender sent \$0.12) and returned \$0.15
- Current total: \$7.23

**\*\*Analysis:\*\***

- My cooperative approach in Round 1 worked very well
- Round 2's low send might be due to communication noise or a different player's strategy
- When I cooperate, I tend to get good returns

I'll maintain a cooperative strategy since it paid off well in Round 1. Sending a moderate to-good amount should signal trustworthiness and potentially encourage reciprocity.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze my position:

Round 1: I was sender and sent \$3

- I kept \$2
- Based on my total earnings, let me work backwards

Round 2: I was receiver

- Sender sent \$0.12, I received \$0.36
- I returned \$0.15
- I kept \$0.36 - \$0.15 = \$0.21

So my earnings should be:

Round 1: \$2 + (what was returned to me)

Round 2: \$0.21

Total = \$7.23

So from Round 1: \$2 + returned amount = \$7.23 - \$0.21 = \$7.02

This means I received \$5.02 back in Round 1

Round 1 calculation:

- I sent \$3, kept \$2



- Receiver got \$9
- Receiver returned \$5.02
- Receiver kept \$9 - \$5.02 = \$3.98
- My total from Round 1: \$2 + \$5.02 = \$7.02 ?

That was quite generous! The receiver returned more than half of what they received.

Round 2:

- I received \$0.36 (sender sent \$0.12)
- I returned \$0.15
- I kept \$0.21

Now Round 3: I'm sender again

- I need to decide how much to send

Observations:

1. In Round 1, when I sent \$3, I got a very generous return (\$5.02 out of \$9)
2. In Round 2, the sender only sent me \$0.12 - very little. Could be:
  - They're being cautious/selfish
  - Communication noise
  - Testing behavior
3. I returned \$0.15 out of \$0.36 (42%)

Now I'm facing a new receiver. What should I do?

The players rotate each round, so the stingy sender from Round 2 isn't necessarily my new receiver. Given that cooperation paid off substantially in Round 1, I should lean toward a similar approach, though the low amount sent to me suggests I should be somewhat cautious. I'll send \$2.50-maintaining the cooperative spirit while protecting myself slightly more than before....

usage:

```
{
  "input_tokens": 869,
  "output_tokens": 848,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.08.

This round, The sender sent you \$0.16, so you received \$0.48. How much do you return to The sender? (0-0.48)

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent only \$0.16 (3.2% of their \$5)
- I received \$0.48 after tripling
- My earnings from Round 3 (\$11.08 - \$7.23 = \$3.85) show that cooperating pays off
- This is the second time I've received a very small amount as receiver (also happened in Round 2)

**\*\*Pattern observation:\*\***

- When I send generously (\$3), I get good returns
- When others send small amounts to me, it could be noise or low cooperation

Given the warning about communication noise and maintaining consistency, I'll return a similar proportion as Round 2 - enough to signal cooperation without being overly generous given the small amount sent.

```
'''json
{"return": 0.20}
'''
```

This returns about 42% of what I received, leaving the sender with ~\$5.04 total and me with \$0.28.

**reasoning:**

Let me analyze this situation:

**\*\*My earnings history:\*\***

- After Round 1: I sent \$3, so I had \$2 left from that, plus what I received back. Total after Round 1 should be around \$7.02 based on what I was told in Round 2.
- Round 2: I was receiver, got \$0.36, returned \$0.15, so I kept \$0.21. Total should be \$7.02 + \$0.21 = \$7.23
- Round 3: I sent \$3 (so kept \$2), plus what I got back. My total was \$7.23 before Round 3. If I now have \$11.08 total, that means from Round 3 I got:  $\$11.08 - \$7.23 = \$3.85$ . So I sent \$3 (kept \$2) and got back \$3.85, meaning I gained \$1.85 from Round 3. So the receiver in Round 3 sent back \$3.85 out of the \$9 they received.
- Round 4 (current): Sender sent me \$0.16, I received \$0.48

**\*\*Pattern:\*\***

- Round 2: Sender sent me \$0.12 (very small)
- Round 4: Sender sent me \$0.16 (very small)
- These could be:
  1. Different players with low cooperation
  2. Communication noise making them think they sent more
  3. Strategic testing

Given that:

- When I cooperate (send \$3), I get good returns (\$5.02 in Round 1, \$3.85 in Round 3)
- When I receive small amounts, it's hard to tell if it's strategic or noise
- The amounts are so small (\$0.16) that even generous returns won't cost me much

I should probably return a similar proportion as before - around 30-40% - to signal I'm willing to cooperate but won't be extremely generous when I receive very little.

...

**usage:**

```
{
  "input_tokens": 1115,
  "output_tokens": 714,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.36.

How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

Looking at my earnings pattern:

**\*\*Analysis:\*\***

- Current total: \$11.36
- When I've sent \$3 as sender (Rounds 1 & 3), I've received good returns
- My cooperative approach has been profitable overall
- Consistency in strategy can build reputation in repeated games

**\*\*Strategy:\*\***

My moderate cooperation (\$3 sends) has yielded positive results. The receivers have generally reciprocated fairly, suggesting that this level of trust is being rewarded.

I'll continue with the same cooperative strategy:

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my earnings progression:

- Started Round 1 with \$5
- Round 1: Sent \$3, kept \$2, received generous return of ~\$5.02
- Round 2: As receiver, got \$0.36, returned \$0.15, kept \$0.21
- Round 3: Sent \$3 (kept \$2), received return of ~\$3.85
- Round 4: As receiver, got \$0.48, returned \$0.20, kept \$0.28
- Current total: \$11.36

My strategy of sending \$3 has worked well:

- Round 1: Sent \$3, got ~\$5.02 back (receiver returned ~\$2.02 of the \$9 they received)
- Round 3: Sent \$3, got ~\$3.85 back (receiver returned ~\$1.85 of the \$9 they received)

Both times I've benefited from cooperation. The pattern suggests that when I cooperate generously, others reciprocate reasonably well.

I should stick with my cooperative strategy since it's been profitable. Sending \$3 again makes sense.

**usage:**

```
{
  "input_tokens": 1379,
  "output_tokens": 399,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.36.

This round, The sender sent you \$0.83, so you received \$2.4899999999999998. How much do you return to The sender? (0-2.4899999999999998)

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent \$0.83 (about 17% of their \$5)
- I received \$2.49 after tripling
- This is larger than the tiny amounts I received in Rounds 2 and 4 (~\$0.36-\$0.48)
- My earnings from Round 5 suggest I got \$0 back when I sent \$3 - disappointing

**\*\*Considerations:\*\***

- \$0.83 is a moderate send - more substantial than previous low sends, but still cautious
- A fair approach would be to split the gains roughly evenly
- Sender kept \$4.17; if I return about half of what I received, we both profit

I'll return a proportionate amount that rewards the moderate cooperation:

```
'''json
{"return": 1.20}
'''
```

This gives the sender \$5.37 total (profit of \$0.37) and me \$1.29 (profit of \$1.29). Both benefit, with me getting slightly more since I'm the receiver of the multiplied amount.

**reasoning:**

Let me analyze this situation:

**\*\*My current state:\*\***

- Total earnings: \$13.36
- Earnings from Round 5: \$13.36 - \$11.36 = \$2.00 (I sent \$3, kept \$2, got back \$0, net -\$1)
- Wait, that doesn't add up. Let me recalculate.
- If I sent \$3 in Round 5, I kept \$2
- My earnings increased from \$11.36 to \$13.36, so I gained \$2
- That means I must have received back \$3 from the receiver
- So: \$2 (kept) + \$3 (returned) - my starting \$5 for that round gives me... wait I need to think about this differently.

Actually, total earnings is cumulative. So Round 5:

- Started with \$5
- Sent \$3, kept \$2
- Received back some amount R
- My profit that round = \$2 + R - \$5 = R - \$3
- But wait, we're tracking total earnings across all rounds, not profit per round.

Let me think more carefully. Total earnings is what I've kept across all rounds.

From Round 4 to Round 5: \$11.36 to \$13.36 = +\$2.00 earned in Round 5

In Round 5 I was sender:

- I had \$5 to start
- I sent \$3, so I kept \$2
- I received back X

- My round 5 earnings = \$2 + X
- So  $\$2 + X = \$2$ , meaning  $X = \$0$

That's unfortunate - the receiver didn't return anything.

**\*\*This round (Round 6):\*\***

- Sender sent \$0.83 (16.6% of their \$5)
- I received \$2.49 after tripling
- This is more than the tiny amounts from Rounds 2 and 4
- The sender kept \$4.17 but still chose to send something

Looking at the pattern, my generous sends haven't consistently paid off-sometimes I get great returns, sometimes nothing. This sender is being cautious but not completely selfish. If I split the \$2.49 evenly, I'd return \$1.25, leaving me with \$1.24 and giving them \$5.42 total....

**usage:**

```
{
  "input_tokens": 1583,
  "output_tokens": 990,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$14.649999999999999.

How much do you send to the receiver this round? (0-5)

**Message 11 role=assistant**

**content:**

Looking at my performance:

**\*\*Analysis:\*\***

- Current total: \$14.65
- My sending pattern: I've consistently sent \$3 in previous sender rounds
- Returns have been variable (excellent in Round 1, okay in Round 3, poor in Round 5)
- Overall, I'm profitable, suggesting cooperation is working more often than not

**\*\*Considerations:\*\***

- Communication noise could explain some variability in returns
- Consistency might build reputation and trust
- \$3 sends create good value (\$9 for receiver) while keeping some safety (\$2)
- Abandoning cooperation after one bad round could miss future gains

I'll maintain my cooperative strategy:

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze my earnings history:

Round 1 (Sender): Sent \$3, started with \$7.02 total after  
 Round 2 (Receiver): Received \$0.36, returned \$0.15, ended with \$7.23 total  
 Round 3 (Sender): Sent \$3, ended with \$11.08 total (gained \$3.85, so got back about \$6.85 - that's very good!)  
 Round 4 (Receiver): Received \$0.48, returned \$0.20, ended with \$11.36 total (gained \$0.28)  
 Round 5 (Sender): Sent \$3, ended with \$13.36 total (gained \$2.00, so got back \$5.00 - that's okay)  
 Round 6 (Receiver): Received \$2.49, returned \$1.20, ended with \$14.65 total (gained \$1.29)  
 Round 7 (Sender): Now with \$14.65

Let me recalculate more carefully:

- Round 1 start: \$5, sent \$3, kept \$2. Ended with \$7.02 total means I got back \$5.02
- Round 2 (Receiver): Started with \$7.02, received \$0.36, returned \$0.15, kept \$0.21. Should be at \$7.23
- Round 3 start: \$7.23 + \$5 = \$12.23, sent \$3, kept \$2. Ended with \$11.08... that doesn't make sense.

Wait, let me think about this differently. The "total earnings across all rounds" refers to

o cumulative earnings, not the current round balance.

Actually, I think the total earnings is just tracking my cumulative money across the game.

Let me trace through more carefully:

- Round 1 (Sender): Start \$5, send \$3, keep \$2. Get return of X. Total =  $\$2 + X = \$7.02$ , so  $X = \$5.02$
- Round 2 (Receiver): Receive \$0.36, return \$0.15, keep \$0.21. Total =  $\$7.02 + \$0.21 = \$7.23$
- Round 3 (Sender): Start \$5, send \$3, keep \$2. Get return of Y. Total =  $\$7.23 + \$2 + Y = \$11.08$ , so  $Y = \$1.85$
- Round 4 (Receiver): Get \$0.48, return \$0.20, keep \$0.28. Total =  $\$11.08 + \$0.28 = \$11.36$
- Round 5 (Sender): Start \$5, send \$3, keep \$2. Get return of Z. Total =  $\$11.36 + \$2 + Z = \$13.36$ , so  $Z = \$0$  (or I got back exactly \$2)
- Round 6 (Receiver): Get \$2.49, return \$1.20, keep \$1.29. Total =  $\$13.36 + \$1.29 = \$14.65$

Looking at my sending pattern across rounds 1, 3, and 5, I consistently sent \$3 but the returns varied dramatically-\$5.02 initially, then \$1.85, then essentially nothing. The inconsistency suggests either communication issues or different counterparts with varying levels of reciprocity. Despite the volatility, I'm ahead at \$14.65, so the early strong return offset the later poor performance....

usage:

```
{
  "input_tokens": 1878,
  "output_tokens": 1179,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$16.65.

This round, The sender sent you \$0.2, so you received \$0.6000000000000001. How much do you return to The sender? (0-0.6000000000000001)

Message 13 role=assistant

content:

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent only \$0.20 (4% of their \$5)
- I received \$0.60 after tripling
- This is the third time receiving a very small amount (Rounds 2, 4, and 8)
- My Round 7 earnings suggest I got \$0 back on my \$3 send - frustrating

**\*\*Pattern consistency:\*\***

My previous returns on small amounts:

- Round 2: \$0.36 received ? returned \$0.15 (~42%)
- Round 4: \$0.48 received ? returned \$0.20 (~42%)

I'll maintain consistency in my response to small sends:

```
'''json
{"return": 0.25}
'''
```

This returns about 42% of what I received, leaving the sender with ~\$5.05 total and me with \$0.35.

reasoning:

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent only \$0.20 (4% of their \$5)
- I received \$0.60 after tripling
- My earnings from Round 7 (\$16.65 - \$14.65 = \$2.00) suggest I got \$0 back when I sent \$3 - another disappointing result

**\*\*Pattern observation:\*\***

- This is the third time I've received a very small amount as receiver (Rounds 2, 4, and now 8)
- When I've sent \$3 consistently, the returns have been mixed

- \$0.20 is a very low trust amount - similar to the \$0.12 and \$0.16 from earlier rounds

**\*\*Previous returns for similar small amounts:\*\***

- Round 2: sent \$0.12, received \$0.36, returned \$0.15 (42%)
- Round 4: sent \$0.16, received \$0.48, returned \$0.20 (42%)
- Round 6: sent \$0.83, received \$2.49, returned \$1.20 (48%)

**\*\*Considerations:\*\***

- Communication noise could be affecting these sends
- Should I maintain consistency with my ~40-45% return rate for small amounts?
- I'm getting frustrated with the lack of reciprocity when I send, but I should maintain fairness

I'll return a similar proportion to maintain consistency:

For \$0.60 received, 40-45% would be \$0.24-\$0.27

I'll return \$0.25, which is about 42% - consistent with my previous behavior.

**usage:**

```
{
  "input_tokens": 1935,
  "output_tokens": 584,
  "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$17.25.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my performance:

**\*\*Analysis:\*\***

- Current total: \$17.25
- My sender rounds: I've consistently sent \$3
- Recent trend: Last two sender rounds (5 & 7) got \$0 returns
- This losing streak is concerning

**\*\*Pattern:\*\***

- Round 1: Sent \$3 ? excellent return
- Round 3: Sent \$3 ? decent return
- Round 5: Sent \$3 ? \$0 back
- Round 7: Sent \$3 ? \$0 back

**\*\*Considerations:\*\***

- Two consecutive \$0 returns means I'm losing \$1 per round by cooperating
- Could be communication noise, or receivers are less cooperative now
- Need to balance maintaining trust vs. protecting myself
- Still profitable overall, but trend is negative

I'll adjust to a more cautious but still cooperative approach:

```
'''json
{"send": 2}
'''
```

This reduces my risk while still signaling willingness to cooperate. The receiver will get \$6, which is still substantial.

**Reasoning:**

Let me analyze my situation:

**\*\*My total earnings: \$17.25\*\***

Let me think about my history as SENDER:

- Round 1: Sent \$3, got good returns (~\$5.02)
- Round 3: Sent \$3, got okay returns
- Round 5: Sent \$3, got \$0 back (disappointing)
- Round 7: Sent \$3, got \$0 back (Round 8 earnings \$16.65 - Round 7 earnings \$14.65 = \$2, so I kept \$5 - \$3 + \$0 = \$2)

As RECEIVER:

- Round 2: Received \$0.36, returned \$0.15
- Round 4: Received \$0.48, returned \$0.20
- Round 6: Received \$2.49, returned \$1.20
- Round 8: Received \$0.60, returned \$0.25

So my sender rounds have had increasingly poor returns. I've been consistently sending \$3, but

- Round 1: Good return
- Round 3: Some return
- Round 5: No return
- Round 7: No return

The pattern suggests that while I've been cooperating, my cooperation isn't being reciprocated as much anymore. In the last two rounds where I sent \$3, I got nothing back.

However, I need to consider:

1. Communication noise - maybe they're trying to return but noise is affecting it
2. I'm still profitable overall (\$17.25 vs starting at \$0)
3. Changing strategy now might signal I'm uncooperative

But sending \$3 and getting \$0 back means I only have \$2 left from that round, which is less than if I'd sent nothing.

Options:

1. Keep sending \$3 - maintain cooperation but risk another \$0 return
2. Send less - protect myself but signal less trust
3. Send nothing - safe but ensures no mutual gain

The consecutive failures...

```
Usage: {  
  "input_tokens": 1878,  
  "output_tokens": 885,  
  "reasoning_tokens": 0  
}
```

#### **GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:58:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.92.

This round, The sender sent you \$0.84, so you received \$2.52. How much do you return to the sender? (0-2.52)

**Message 3 role=assistant**

**content:**

Looking at this round:

- The sender sent \$0.84 (only 16.8% of their endowment) - much less than Round 1's 68.8%
- I received \$2.52
- The sender currently has \$4.16 remaining

This is a substantially lower amount than the previous round. This could be due to:

1. Communication noise affecting what they received in Round 2
2. A change in their trust level
3. Testing my response

I'll return a proportional amount that shows I'm still willing to cooperate, but acknowledging the reduced trust. Returning around \$1.00 ensures they at least break even on their investment.

{"return": 1.00}

**reasoning:**

Let me analyze this situation:

Round 3:

- The sender sent me \$0.84 (which is 16.8% of their \$5 endowment)
- This was multiplied to \$2.52 for me
- The sender currently has \$5 - \$0.84 = \$4.16 remaining

This is a significant decrease from Round 1 where they sent \$3.44 (68.8%). Now they only sent \$0.84 (16.8%).

Let me think about what might be happening:

1. There could be communication noise - the problem statement mentioned this
2. The sender might have received less than expected back in Round 2
3. They might be testing my response to a lower amount



In Round 2, I sent \$3.50. If there was noise, they might have received less than that, which could explain why they're now being more cautious.

Given that they sent a much smaller amount this time, I should probably return a proportionally smaller amount, but still enough to show I'm willing to cooperate.

If I return too much when they sent little, I might look naive. If I return too little, I might discourage future cooperation.

The sender sent 16.8% of their endowment. Perhaps I should return around 30-40% of what I received to maintain some reciprocity but acknowledge the lower trust level.

$\$2.52 * 0.4 = \$1.01$

Alternatively, I could aim to ensure they don't lose money. They sent \$0.84, so if I return at least \$0.84, they break even. If I return \$1.00, they gain \$0.16.

I think returning around \$1.00 is reasonable - it shows I'm still cooperative but responding proportionally to the lower amount sent.

**usage:**

```
{
  "input_tokens": 732,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.44.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at the pattern so far:

Round 1: The other player sent me 68.8% - I returned generously

Round 2: I sent \$3.50 (70%) - but based on my earnings, I only received \$0.10 back (less than 1% of what they got)

Round 3: They sent only \$0.84 (16.8%) - I returned \$1.00 (40% of what I received)

The low return in Round 2 was disappointing, though communication noise might have affected what they saw. I'll send a moderate amount to signal continued willingness to cooperate while being more cautious.

{"send": 2.00}

**reasoning:**

Let me track what's happened so far:

Round 1: I was RECEIVER

- Sender sent \$3.44 (68.8%)
- I received \$10.32
- I returned \$5.00
- My earnings: \$5.32

Round 2: I was SENDER

- I sent \$3.50
- My total earnings were \$5.32 (from Round 1)
- We don't know what they returned yet, but...

Round 3: I was RECEIVER

- My total earnings were \$6.92
- This means from Round 2, I gained:  $\$6.92 - \$5.32 = \$1.60$
- I sent \$3.50, so I had \$1.50 left
- I received back something, ending at \$1.60 total from that round
- So I must have received \$0.10 back from the \$3.50 I sent (which became \$10.50 for them)
- That's a very low return! Only \$0.10 out of \$10.50

Round 3 continued:

- Sender sent \$0.84 (16.8%)

- I received \$2.52
- I returned \$1.00
- My earnings this round:  $\$2.52 - \$1.00 = \$1.52$
- Total:  $\$6.92 + \$1.52 = \$8.44$  ?

So the pattern is:

- Round 1: I returned generously (\$5.00 out of \$10.32 - about 48%)
- Round 2: They returned very little (\$0.10 out of \$10.50 - less than 1%)
- Round 3: They sent much less (\$0.84 vs their original \$3.44)
- Round 3: I returned moderately (\$1.00 out of \$2.52 - about 40%)

Now in Round 4, I'm the sender again.

Their minimal return last round suggests they're not reciprocating well, but there could be communication interference affecting what they actually...

**usage:**

```
{
  "input_tokens": 938,
  "output_tokens": 804,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.44.

This round, The sender sent you \$0.99, so you received \$2.9699999999999998. How much do you return to The sender? (0-2.9699999999999998)

**Message 7 role=assistant**

**content:**

Looking at the pattern:

Round 2: I sent \$3.50 (70%) but received only \$0.10 back

Round 4: I sent \$2.00 (40%) but received \$0.00 back

Round 5: They sent only \$0.99 (19.8%)

The receiver has been keeping nearly everything I send while sending very little themselves. While communication noise might play a role, the pattern suggests they're not reciprocating cooperation.

I'll return a minimal amount to signal that this pattern is not acceptable, while not completely defecting.

```
{"return": 0.50}
```

**reasoning:**

Let me track what's happening:

Round 1: I was RECEIVER

- Sender sent \$3.44 (68.8%)
- I received \$10.32
- I returned \$5.00
- Sender got:  $\$1.56 + \$5.00 = \$6.56$
- I got:  $\$10.32 - \$5.00 = \$5.32$

Round 2: I was SENDER

- I sent \$3.50 (70%)
- My total earnings across all rounds were \$5.32 (so I only had Round 1)
- After Round 2, my earnings became \$6.92

So in Round 2:

- I started with \$5.00
- I sent \$3.50
- I had \$1.50 left
- My total earnings went from \$5.32 to \$6.92
- So I gained  $\$6.92 - \$5.32 = \$1.60$
- This means I received back  $\$1.60 - \$1.50 = \$0.10$

That's very low! They received \$10.50 and returned only \$0.10!

Round 3: I was RECEIVER

- My total earnings were \$6.92
- Sender sent \$0.84
- I received \$2.52
- I returned \$1.00
- After this round, my earnings became \$8.44

So in Round 3:

- I started with \$6.92
- I received \$2.52
- I returned \$1.00
- I kept \$1.52
- Total:  $\$6.92 + \$1.52 = \$8.44$  ?

Round 4: I was SENDER

- My total earnings were \$8.44
- I sent \$2.00
- I had \$3.00 left
- After this round, my earnings became \$11.44

So in Round 4:

- I started with \$5.00
- I sent \$2.00
- I had \$3.00 left
- My total earnings went from \$8.44 to \$11.44
- I gained  $\$11.44 - \$8.44 = \$3.00$
- This means I received back  $\$3.00 - \$3.00 = \$0.00$

They received \$6.00 and returned nothing at all! Round 5: I am RECEIVER

- My total earnings are \$11.44
- Sender sent \$0.99 (19.8%)
- I received \$2.97

This partner has shown a clear pattern of non-cooperation. In Round 2, they kept nearly everything I sent them, and in Round 4, they returned absolutely nothing despite receiving \$6.00.

Now they're only sending me \$0.99 out of what I gave them. They have \$4.01 right now, and I...

**usage:**

```
{
  "input_tokens": 1170,
  "output_tokens": 984,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$13.91.

How much do you send to the receiver this round? (0-5)

**Message 9 role=assistant**

**content:**

Looking at the history:

When I've been the SENDER:

- Round 2: Sent \$3.50 ? Received only \$0.10 back (kept almost nothing)
- Round 4: Sent \$2.00 ? Received \$0.00 back (got nothing)

When I've been the RECEIVER:

- Round 1: Received \$10.32 ? Returned \$5.00 (fair)
- Round 3: Received \$2.52 ? Returned \$1.00 (generous given their lower send)
- Round 5: Received \$2.97 ? Returned \$0.50 (reduced due to pattern)

The pattern is clear: when I send, the receiver keeps almost everything. They've exploited my cooperation twice now. I need to protect my earnings and not continue being taken advantage of.

```
{"send": 0.50}
```

**reasoning:**

Let me track what's happened:

Round 1: I was RECEIVER  
- Sender sent \$3.44 (68.8%)  
- I received \$10.32  
- I returned \$5.00  
- My earnings: \$5.32

Round 2: I was SENDER  
- I sent \$3.50 (70%)  
- My earnings went from \$5.32 to \$6.92, so I gained \$1.60  
- That means I kept \$1.50 and received \$0.10 back  
- The receiver got \$10.50, returned only \$0.10

Round 3: I was RECEIVER  
- Sender sent \$0.84 (16.8%)  
- I received \$2.52  
- I returned \$1.00  
- My earnings went from \$6.92 to \$8.44, so I gained \$1.52  
- I kept \$1.52 from the \$2.52

Round 4: I was SENDER  
- I sent \$2.00 (40%)  
- My earnings went from \$8.44 to \$11.44, so I gained \$3.00  
- That means I kept \$3.00 and received \$0.00 back  
- The receiver got \$6.00, returned \$0.00

Round 5: I was RECEIVER  
- Sender sent \$0.99 (19.8%)  
- I received \$2.97  
- I returned \$0.50  
- My earnings went from \$11.44 to \$13.91, so I gained \$2.47  
- I kept \$2.47 from the \$2.97

So the pattern:  
- Round 2: I sent \$3.50, got back \$0.10 (very poor return)  
- Round 4: I sent \$2.00, got back \$0.00 (terrible)

The other player as receiver has been very uncooperative when I'm the sender.

When I'm the receiver:  
- Round 1: They sent \$3.44, I returned \$5.00 (very generous)  
- Round 3: They sent \$0.84, I returned \$1.00 (fair)  
- Round 5: They sent \$0.99, I returned \$0.50 (reduced due to their poor reciprocation)

They're clearly...

**usage:**

```
{  
  "input_tokens": 1353,  
  "output_tokens": 831,  
  "reasoning_tokens": 0  
}
```

**Message 10 role=user**

**content:**

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$19.91.  
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 11 role=assistant**

**content:**

Since the sender sent \$0, I received \$0. There's nothing to return.

```
{"return": 0}
```

**reasoning:**

The sender sent \$0, so I received \$0. There's nothing to return - I can only return between 0 and 0.

This makes sense given the pattern. After I reduced my cooperation in Round 6 by only sending \$0.50, they've now sent \$0, which is rational given the lack of cooperation we've seen.

Since I received \$0, I can only return \$0.

```
{"return": 0}
usage:
{
  "input_tokens": 1618,
  "output_tokens": 132,
  "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$19.91.

How much do you send to the receiver this round? (0-5)

**Message 13 role=assistant**

**content:**

Looking at the complete history:

As SENDER (sending to them):

- Round 2: Sent \$3.50 ? Received \$0.10 back
- Round 4: Sent \$2.00 ? Received \$0.00 back
- Round 6: Sent \$0.50 ? Received \$0.00 back

As RECEIVER (they're sending to me):

- Round 7: They sent \$0

The pattern is crystal clear: the receiver consistently keeps everything I send while now sending nothing themselves. Even when I reduced my send amount significantly in Round 6, I still received nothing back. They've now sent \$0 to me.

There's no rational reason to continue sending money to someone who has repeatedly defected.

```
{"send": 0}
```

**reasoning:**

Let me review the history:

As SENDER:

- Round 2: Sent \$3.50 ? Received \$0.10 back (they kept \$10.40 of \$10.50)
- Round 4: Sent \$2.00 ? Received \$0.00 back (they kept all \$6.00)
- Round 6: Sent \$0.50 ? Received \$0.00 back (they kept all \$1.50)

As RECEIVER:

- Round 1: Received \$10.32 ? Returned \$5.00
- Round 3: Received \$2.52 ? Returned \$1.00
- Round 5: Received \$2.97 ? Returned \$0.50
- Round 7: Received \$0 ? Returned \$0

The pattern is absolutely clear: whenever I send money as the sender, the receiver keeps everything or nearly everything. They've exploited me three times now. In Round 6, I reduced my send to \$0.50 and still got nothing back. In Round 7, they sent me \$0.

There's no reason to continue sending money to someone who consistently defects and takes advantage. The cooperative approach has failed completely with this receiver. I should not send anything.

```
{"send": 0}
usage:
{
  "input_tokens": 1474,
  "output_tokens": 484,
  "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$24.91.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

Since the sender sent \$0, I received \$0. There's nothing to return.

```
{"return": 0}
```

**Reasoning:**

The sender sent \$0, so I received \$0. There's nothing to return.

The game has broken down to mutual defection at this point. Neither player is sending anything to the other.

Given that I received \$0, I can return 0 (and there's nothing else I could return anyway).

Usage: {

```
"input_tokens": 1596,  
"output_tokens": 102,  
"reasoning_tokens": 0
```

}

=====

**ROUND 10**

=====

Roles: Agent\_1=Receiver, Agent\_2=Sender

**Pairings:**

dyad\_1: Agent\_2=Sender, Agent\_1=Receiver

**Parsed game decisions:**

```
dyad_1: Agent_2 sent $0; received $0; Agent_1 returned $0; payoffs Agent_2=$5, Agent_1=$0;  
noise/visible: sent visible=$0.94, received visible=$2.82  
Action validation: {"returned": {"amount": 0, "clamped": true, "max": 0, "min": 0, "raw": 1.4},  
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}
```

Balances after round: {

```
"Agent_1": 33.95,  
"Agent_2": 56.05
```

}

Visible balances after round: {

```
"Agent_1": 23.07,  
"Agent_2": 29.91
```

}

**LLM CALLS**

**GAME - Agent\_2 - Sender - dyad\_1**

Timestamp: 2026-08-12T09:58:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_1",  
"move_index": 0,  
"opponent_id": "Agent_1",  
"pairing": {  
  "agents": [  
    "Agent_2",  
    "Agent_1"  
  ],  
  "investor": "Agent_2",  
  "roles": {  
    "Agent_1": "trustee",  
    "Agent_2": "investor"  
  },  
  "trustee": "Agent_1"
```

```
}
```

```
"role": "investor",  
"role_label": "Sender",  
"round": 10,  
"task": "game",  
"task_index": 0
```

}

**Exact input messages sent to model:**

Message 1 role=system

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user****content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$8.44.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant****content:**

Looking at the pattern so far:

Round 1: The other player sent me 68.8% - I returned generously

Round 2: I sent \$3.50 (70%) - but based on my earnings, I only received \$0.10 back (less than 1% of what they got)

Round 3: They sent only \$0.84 (16.8%) - I returned \$1.00 (40% of what I received)

The low return in Round 2 was disappointing, though communication noise might have affected what they saw. I'll send a moderate amount to signal continued willingness to cooperate while being more cautious.

```
{"send": 2.00}
```

**reasoning:**

Let me track what's happened so far:

Round 1: I was RECEIVER

- Sender sent \$3.44 (68.8%)
- I received \$10.32
- I returned \$5.00
- My earnings: \$5.32

Round 2: I was SENDER

- I sent \$3.50
- My total earnings were \$5.32 (from Round 1)
- We don't know what they returned yet, but...

Round 3: I was RECEIVER

- My total earnings were \$6.92
- This means from Round 2, I gained:  $\$6.92 - \$5.32 = \$1.60$
- I sent \$3.50, so I had \$1.50 left
- I received back something, ending at \$1.60 total from that round
- So I must have received \$0.10 back from the \$3.50 I sent (which became \$10.50 for them)
- That's a very low return! Only \$0.10 out of \$10.50

Round 3 continued:

- Sender sent \$0.84 (16.8%)
- I received \$2.52
- I returned \$1.00
- My earnings this round:  $\$2.52 - \$1.00 = \$1.52$
- Total:  $\$6.92 + \$1.52 = \$8.44$  ?

So the pattern is:

- Round 1: I returned generously (\$5.00 out of \$10.32 - about 48%)
- Round 2: They returned very little (\$0.10 out of \$10.50 - less than 1%)
- Round 3: They sent much less (\$0.84 vs their original \$3.44)
- Round 3: I returned moderately (\$1.00 out of \$2.52 - about 40%)

Now in Round 4, I'm the sender again.

Their minimal return last round suggests they're not reciprocating well, but there could be communication interference affecting what they actually...

**usage:**

```
{
  "input_tokens": 938,
  "output_tokens": 804,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.44.

This round, The sender sent you \$0.99, so you received \$2.9699999999999998. How much do you return to The sender? (0-2.9699999999999998)

**Message 5 role=assistant**

**content:**

Looking at the pattern:

Round 2: I sent \$3.50 (70%) but received only \$0.10 back

Round 4: I sent \$2.00 (40%) but received \$0.00 back

Round 5: They sent only \$0.99 (19.8%)

The receiver has been keeping nearly everything I send while sending very little themselves. While communication noise might play a role, the pattern suggests they're not reciprocating cooperation.

I'll return a minimal amount to signal that this pattern is not acceptable, while not completely defecting.

```
{"return": 0.50}
```

**reasoning:**

Let me track what's happening:

Round 1: I was RECEIVER

- Sender sent \$3.44 (68.8%)
- I received \$10.32
- I returned \$5.00
- Sender got:  $\$1.56 + \$5.00 = \$6.56$
- I got:  $\$10.32 - \$5.00 = \$5.32$

Round 2: I was SENDER

- I sent \$3.50 (70%)
- My total earnings across all rounds were \$5.32 (so I only had Round 1)
- After Round 2, my earnings became \$6.92

So in Round 2:

- I started with \$5.00



- I sent \$3.50
- I had \$1.50 left
- My total earnings went from \$5.32 to \$6.92
- So I gained  $\$6.92 - \$5.32 = \$1.60$
- This means I received back  $\$1.60 - \$1.50 = \$0.10$

That's very low! They received \$10.50 and returned only \$0.10!

Round 3: I was RECEIVER

- My total earnings were \$6.92
- Sender sent \$0.84
- I received \$2.52
- I returned \$1.00
- After this round, my earnings became \$8.44

So in Round 3:

- I started with \$6.92
- I received \$2.52
- I returned \$1.00
- I kept \$1.52
- Total:  $\$6.92 + \$1.52 = \$8.44$  ?

Round 4: I was SENDER

- My total earnings were \$8.44
- I sent \$2.00
- I had \$3.00 left
- After this round, my earnings became \$11.44

So in Round 4:

- I started with \$5.00
- I sent \$2.00
- I had \$3.00 left
- My total earnings went from \$8.44 to \$11.44
- I gained  $\$11.44 - \$8.44 = \$3.00$
- This means I received back  $\$3.00 - \$3.00 = \$0.00$

They received \$6.00 and returned nothing at all! Round 5: I am RECEIVER

- My total earnings are \$11.44
- Sender sent \$0.99 (19.8%)
- I received \$2.97

This partner has shown a clear pattern of non-cooperation. In Round 2, they kept nearly everything I sent them, and in Round 4, they returned absolutely nothing despite receiving \$6.00.

Now they're only sending me \$0.99 out of what I gave them. They have \$4.01 right now, and I...

usage:

```
{
  "input_tokens": 1170,
  "output_tokens": 984,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$13.91.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the history:

When I've been the SENDER:

- Round 2: Sent \$3.50 ? Received only \$0.10 back (kept almost nothing)
- Round 4: Sent \$2.00 ? Received \$0.00 back (got nothing)

When I've been the RECEIVER:

- Round 1: Received \$10.32 ? Returned \$5.00 (fair)
- Round 3: Received \$2.52 ? Returned \$1.00 (generous given their lower send)

- Round 5: Received \$2.97 ? Returned \$0.50 (reduced due to pattern)

The pattern is clear: when I send, the receiver keeps almost everything. They've exploited my cooperation twice now. I need to protect my earnings and not continue being taken advantage of.

```
{"send": 0.50}
```

**reasoning:**

Let me track what's happened:

Round 1: I was RECEIVER

- Sender sent \$3.44 (68.8%)
- I received \$10.32
- I returned \$5.00
- My earnings: \$5.32

Round 2: I was SENDER

- I sent \$3.50 (70%)
- My earnings went from \$5.32 to \$6.92, so I gained \$1.60
- That means I kept \$1.50 and received \$0.10 back
- The receiver got \$10.50, returned only \$0.10

Round 3: I was RECEIVER

- Sender sent \$0.84 (16.8%)
- I received \$2.52
- I returned \$1.00
- My earnings went from \$6.92 to \$8.44, so I gained \$1.52
- I kept \$1.52 from the \$2.52

Round 4: I was SENDER

- I sent \$2.00 (40%)
- My earnings went from \$8.44 to \$11.44, so I gained \$3.00
- That means I kept \$3.00 and received \$0.00 back
- The receiver got \$6.00, returned \$0.00

Round 5: I was RECEIVER

- Sender sent \$0.99 (19.8%)
- I received \$2.97
- I returned \$0.50
- My earnings went from \$11.44 to \$13.91, so I gained \$2.47
- I kept \$2.47 from the \$2.97

So the pattern:

- Round 2: I sent \$3.50, got back \$0.10 (very poor return)
- Round 4: I sent \$2.00, got back \$0.00 (terrible)

The other player as receiver has been very uncooperative when I'm the sender.

When I'm the receiver:

- Round 1: They sent \$3.44, I returned \$5.00 (very generous)
- Round 3: They sent \$0.84, I returned \$1.00 (fair)
- Round 5: They sent \$0.99, I returned \$0.50 (reduced due to their poor reciprocation)

They're clearly...

**usage:**

```
{
  "input_tokens": 1353,
  "output_tokens": 831,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$19.91.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 9 role=assistant**

**content:**

Since the sender sent \$0, I received \$0. There's nothing to return.

```
{"return": 0}
```

**reasoning:**

The sender sent \$0, so I received \$0. There's nothing to return - I can only return between 0 and 0.

This makes sense given the pattern. After I reduced my cooperation in Round 6 by only sending \$0.50, they've now sent \$0, which is rational given the lack of cooperation we've seen.

Since I received \$0, I can only return \$0.

```
{"return": 0}
```

**usage:**

```
{  
  "input_tokens": 1618,  
  "output_tokens": 132,  
  "reasoning_tokens": 0  
}
```

**Message 10 role=user**

**content:**

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$19.91. How much do you send to the receiver this round? (0-5)

**Message 11 role=assistant**

**content:**

Looking at the complete history:

As SENDER (sending to them):

- Round 2: Sent \$3.50 ? Received \$0.10 back
- Round 4: Sent \$2.00 ? Received \$0.00 back
- Round 6: Sent \$0.50 ? Received \$0.00 back

As RECEIVER (they're sending to me):

- Round 7: They sent \$0

The pattern is crystal clear: the receiver consistently keeps everything I send while now sending nothing themselves. Even when I reduced my send amount significantly in Round 6, I still received nothing back. They've now sent \$0 to me.

There's no rational reason to continue sending money to someone who has repeatedly defected.

```
{"send": 0}
```

**reasoning:**

Let me review the history:

As SENDER:

- Round 2: Sent \$3.50 ? Received \$0.10 back (they kept \$10.40 of \$10.50)
- Round 4: Sent \$2.00 ? Received \$0.00 back (they kept all \$6.00)
- Round 6: Sent \$0.50 ? Received \$0.00 back (they kept all \$1.50)

As RECEIVER:

- Round 1: Received \$10.32 ? Returned \$5.00
- Round 3: Received \$2.52 ? Returned \$1.00
- Round 5: Received \$2.97 ? Returned \$0.50
- Round 7: Received \$0 ? Returned \$0

The pattern is absolutely clear: whenever I send money as the sender, the receiver keeps everything or nearly everything. They've exploited me three times now. In Round 6, I reduced my send to \$0.50 and still got nothing back. In Round 7, they sent me \$0.

There's no reason to continue sending money to someone who consistently defects and takes advantage. The cooperative approach has failed completely with this receiver. I should not send anything.

```
{"send": 0}
```

**usage:**

```
{
```

```
"input_tokens": 1474,
"output_tokens": 484,
"reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$24.91.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 13 role=assistant**

**content:**

Since the sender sent \$0, I received \$0. There's nothing to return.

```
{"return": 0}
```

**reasoning:**

The sender sent \$0, so I received \$0. There's nothing to return.

The game has broken down to mutual defection at this point. Neither player is sending anything to the other.

Given that I received \$0, I can return 0 (and there's nothing else I could return anyway).

**usage:**

```
{
  "input_tokens": 1596,
  "output_tokens": 102,
  "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$24.91.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the complete history:

As SENDER (me to them):

- Round 2: Sent \$3.50 ? Got \$0.10 back
- Round 4: Sent \$2.00 ? Got \$0.00 back
- Round 6: Sent \$0.50 ? Got \$0.00 back
- Round 8: Sent \$0.00

As RECEIVER (them to me):

- Round 7: They sent \$0
- Round 9: They sent \$0

The trust has completely broken down. They consistently exploited my cooperation early on, and now we're in a mutual defection equilibrium. This appears to be the final round, and there's no rational reason to break from this pattern when they've shown no willingness to cooperate.

```
{"send": 0}
```

**Reasoning:**

Let me review the complete history:

As SENDER (me sending to them):

- Round 2: Sent \$3.50 ? Received \$0.10 back
- Round 4: Sent \$2.00 ? Received \$0.00 back
- Round 6: Sent \$0.50 ? Received \$0.00 back
- Round 8: Sent \$0.00

As RECEIVER (them sending to me):

- Round 1: Received \$10.32 ? I returned \$5.00 (generous)
- Round 3: Received \$2.52 ? I returned \$1.00 (fair)
- Round 5: Received \$2.97 ? I returned \$0.50 (reduced)
- Round 7: Received \$0
- Round 9: Received \$0

The relationship has completely broken down. They exploited my cooperation early on, and now we're both sending \$0 to each other. This is Round 10 - it might be the last round.

There's no indication that cooperation will resume. They've shown no willingness to reciprocate, and I've responded by stopping sends as well. Sending anything now would just be giving away money with no expectation of return.

I should send \$0 again.

```
Usage: {
  "input_tokens": 1442,
  "output_tokens": 496,
  "reasoning_tokens": 0
}
```

#### **GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:58:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### **GAME:**

You will play different roles across multiple rounds.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

###### **MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.08.

This round, The sender sent you \$0.16, so you received \$0.48. How much do you return to the sender? (0-0.48)

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent only \$0.16 (3.2% of their \$5)
- I received \$0.48 after tripling
- My earnings from Round 3 (\$11.08 - \$7.23 = \$3.85) show that cooperating pays off
- This is the second time I've received a very small amount as receiver (also happened in Round 2)

**\*\*Pattern observation:\*\***

- When I send generously (\$3), I get good returns
- When others send small amounts to me, it could be noise or low cooperation

Given the warning about communication noise and maintaining consistency, I'll return a similar proportion as Round 2 - enough to signal cooperation without being overly generous given the small amount sent.

```
'''json
{"return": 0.20}
'''
```

This returns about 42% of what I received, leaving the sender with ~\$5.04 total and me with \$0.28.

**reasoning:**

Let me analyze this situation:

**\*\*My earnings history:\*\***

- After Round 1: I sent \$3, so I had \$2 left from that, plus what I received back. Total after Round 1 should be around \$7.02 based on what I was told in Round 2.
- Round 2: I was receiver, got \$0.36, returned \$0.15, so I kept \$0.21. Total should be \$7.02 + \$0.21 = \$7.23
- Round 3: I sent \$3 (so kept \$2), plus what I got back. My total was \$7.23 before Round 3. If I now have \$11.08 total, that means from Round 3 I got: \$11.08 - \$7.23 = \$3.85. So I sent \$3 (kept \$2) and got back \$3.85, meaning I gained \$1.85 from Round 3. So the receiver in Round 3 sent back \$3.85 out of the \$9 they received.
- Round 4 (current): Sender sent me \$0.16, I received \$0.48

**\*\*Pattern:\*\***

- Round 2: Sender sent me \$0.12 (very small)
- Round 4: Sender sent me \$0.16 (very small)
- These could be:
  1. Different players with low cooperation
  2. Communication noise making them think they sent more
  3. Strategic testing

Given that:

- When I cooperate (send \$3), I get good returns (\$5.02 in Round 1, \$3.85 in Round 3)
- When I receive small amounts, it's hard to tell if it's strategic or noise
- The amounts are so small (\$0.16) that even generous returns won't cost me much

I should probably return a similar proportion as before - around 30-40% - to signal I'm willing to cooperate but won't be extremely generous when I receive very little.

...

**usage:**

```
{
  "input_tokens": 1115,
  "output_tokens": 714,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.36.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at my earnings pattern:

**\*\*Analysis:\*\***

- Current total: \$11.36
- When I've sent \$3 as sender (Rounds 1 & 3), I've received good returns
- My cooperative approach has been profitable overall
- Consistency in strategy can build reputation in repeated games

**\*\*Strategy:\*\***

My moderate cooperation (\$3 sends) has yielded positive results. The receivers have generally reciprocated fairly, suggesting that this level of trust is being rewarded.

I'll continue with the same cooperative strategy:

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my earnings progression:

- Started Round 1 with \$5
- Round 1: Sent \$3, kept \$2, received generous return of ~\$5.02
- Round 2: As receiver, got \$0.36, returned \$0.15, kept \$0.21
- Round 3: Sent \$3 (kept \$2), received return of ~\$3.85
- Round 4: As receiver, got \$0.48, returned \$0.20, kept \$0.28
- Current total: \$11.36

My strategy of sending \$3 has worked well:

- Round 1: Sent \$3, got ~\$5.02 back (receiver returned ~\$2.02 of the \$9 they received)
- Round 3: Sent \$3, got ~\$3.85 back (receiver returned ~\$1.85 of the \$9 they received)

Both times I've benefited from cooperation. The pattern suggests that when I cooperate generously, others reciprocate reasonably well.

I should stick with my cooperative strategy since it's been profitable. Sending \$3 again makes sense.

**usage:**

```
{
  "input_tokens": 1379,
  "output_tokens": 399,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.36.

This round, The sender sent you \$0.83, so you received \$2.4899999999999998. How much do you return to The sender? (0-2.4899999999999998)

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent \$0.83 (about 17% of their \$5)
- I received \$2.49 after tripling
- This is larger than the tiny amounts I received in Rounds 2 and 4 (~\$0.36-\$0.48)

- My earnings from Round 5 suggest I got \$0 back when I sent \$3 - disappointing

**\*\*Considerations:\*\***

- \$0.83 is a moderate send - more substantial than previous low sends, but still cautious
- A fair approach would be to split the gains roughly evenly
- Sender kept \$4.17; if I return about half of what I received, we both profit

I'll return a proportionate amount that rewards the moderate cooperation:

```
'''json
{"return": 1.20}
'''
```

This gives the sender \$5.37 total (profit of \$0.37) and me \$1.29 (profit of \$1.29). Both benefit, with me getting slightly more since I'm the receiver of the multiplied amount.

**reasoning:**

Let me analyze this situation:

**\*\*My current state:\*\***

- Total earnings: \$13.36
- Earnings from Round 5:  $\$13.36 - \$11.36 = \$2.00$  (I sent \$3, kept \$2, got back \$0, net -\$1)
- Wait, that doesn't add up. Let me recalculate.
- If I sent \$3 in Round 5, I kept \$2
- My earnings increased from \$11.36 to \$13.36, so I gained \$2
- That means I must have received back \$3 from the receiver
- So: \$2 (kept) + \$3 (returned) - my starting \$5 for that round gives me... wait I need to think about this differently.

Actually, total earnings is cumulative. So Round 5:

- Started with \$5
- Sent \$3, kept \$2
- Received back some amount R
- My profit that round =  $\$2 + R - \$5 = R - \$3$
- But wait, we're tracking total earnings across all rounds, not profit per round.

Let me think more carefully. Total earnings is what I've kept across all rounds.

From Round 4 to Round 5:  $\$11.36$  to  $\$13.36 = +\$2.00$  earned in Round 5

In Round 5 I was sender:

- I had \$5 to start
- I sent \$3, so I kept \$2
- I received back X
- My round 5 earnings =  $\$2 + X$
- So  $\$2 + X = \$2$ , meaning  $X = \$0$

That's unfortunate - the receiver didn't return anything.

**\*\*This round (Round 6):\*\***

- Sender sent \$0.83 (16.6% of their \$5)
- I received \$2.49 after tripling
- This is more than the tiny amounts from Rounds 2 and 4
- The sender kept \$4.17 but still chose to send something

Looking at the pattern, my generous sends haven't consistently paid off-sometimes I get great returns, sometimes nothing. This sender is being cautious but not completely selfish. If I split the \$2.49 evenly, I'd return \$1.25, leaving me with \$1.24 and giving them \$5.42 total....

**usage:**

```
{
  "input_tokens": 1583,
  "output_tokens": 990,
  "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$14.649999999999999.



How much do you send to the receiver this round? (0-5)

**Message 9 role=assistant**

**content:**

Looking at my performance:

**\*\*Analysis:\*\***

- Current total: \$14.65
- My sending pattern: I've consistently sent \$3 in previous sender rounds
- Returns have been variable (excellent in Round 1, okay in Round 3, poor in Round 5)
- Overall, I'm profitable, suggesting cooperation is working more often than not

**\*\*Considerations:\*\***

- Communication noise could explain some variability in returns
- Consistency might build reputation and trust
- \$3 sends create good value (\$9 for receiver) while keeping some safety (\$2)
- Abandoning cooperation after one bad round could miss future gains

I'll maintain my cooperative strategy:

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze my earnings history:

Round 1 (Sender): Sent \$3, started with \$7.02 total after  
Round 2 (Receiver): Received \$0.36, returned \$0.15, ended with \$7.23 total  
Round 3 (Sender): Sent \$3, ended with \$11.08 total (gained \$3.85, so got back about \$6.85  
- that's very good!)  
Round 4 (Receiver): Received \$0.48, returned \$0.20, ended with \$11.36 total (gained \$0.28)  
Round 5 (Sender): Sent \$3, ended with \$13.36 total (gained \$2.00, so got back \$5.00 - that  
's okay)  
Round 6 (Receiver): Received \$2.49, returned \$1.20, ended with \$14.65 total (gained \$1.29)  
Round 7 (Sender): Now with \$14.65

Let me recalculate more carefully:

- Round 1 start: \$5, sent \$3, kept \$2. Ended with \$7.02 total means I got back \$5.02
- Round 2 (Receiver): Started with \$7.02, received \$0.36, returned \$0.15, kept \$0.21. Should be at \$7.23
- Round 3 start: \$7.23 + \$5 = \$12.23, sent \$3, kept \$2. Ended with \$11.08... that doesn't make sense.

Wait, let me think about this differently. The "total earnings across all rounds" refers to cumulative earnings, not the current round balance.

Actually, I think the total earnings is just tracking my cumulative money across the game.

Let me trace through more carefully:

- Round 1 (Sender): Start \$5, send \$3, keep \$2. Get return of X. Total = \$2 + X = \$7.02, so X = \$5.02
- Round 2 (Receiver): Receive \$0.36, return \$0.15, keep \$0.21. Total = \$7.02 + \$0.21 = \$7.23
- Round 3 (Sender): Start \$5, send \$3, keep \$2. Get return of Y. Total = \$7.23 + \$2 + Y = \$11.08, so Y = \$1.85
- Round 4 (Receiver): Get \$0.48, return \$0.20, keep \$0.28. Total = \$11.08 + \$0.28 = \$11.36
- Round 5 (Sender): Start \$5, send \$3, keep \$2. Get return of Z. Total = \$11.36 + \$2 + Z = \$13.36, so Z = \$0 (or I got back exactly \$2)
- Round 6 (Receiver): Get \$2.49, return \$1.20, keep \$1.29. Total = \$13.36 + \$1.29 = \$14.65

Looking at my sending pattern across rounds 1, 3, and 5, I consistently sent \$3 but the returns varied dramatically-\$5.02 initially, then \$1.85, then essentially nothing. The inconsistency suggests either communication issues or different counterparts with varying levels of reciprocity. Despite the volatility, I'm ahead at \$14.65, so the early strong return offset the later poor performance....

**usage:**

```
{
  "input_tokens": 1878,
  "output_tokens": 1179,
  "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$16.65.

This round, The sender sent you \$0.2, so you received \$0.6000000000000001. How much do you return to The sender? (0-0.6000000000000001)

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent only \$0.20 (4% of their \$5)
- I received \$0.60 after tripling
- This is the third time receiving a very small amount (Rounds 2, 4, and 8)
- My Round 7 earnings suggest I got \$0 back on my \$3 send - frustrating

**\*\*Pattern consistency:\*\***

My previous returns on small amounts:

- Round 2: \$0.36 received ? returned \$0.15 (~42%)
- Round 4: \$0.48 received ? returned \$0.20 (~42%)

I'll maintain consistency in my response to small sends:

```
'''json
{"return": 0.25}
'''
```

This returns about 42% of what I received, leaving the sender with ~\$5.05 total and me with \$0.35.

**reasoning:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent only \$0.20 (4% of their \$5)
- I received \$0.60 after tripling
- My earnings from Round 7 (\$16.65 - \$14.65 = \$2.00) suggest I got \$0 back when I sent \$3 - another disappointing result

**\*\*Pattern observation:\*\***

- This is the third time I've received a very small amount as receiver (Rounds 2, 4, and now 8)
- When I've sent \$3 consistently, the returns have been mixed
- \$0.20 is a very low trust amount - similar to the \$0.12 and \$0.16 from earlier rounds

**\*\*Previous returns for similar small amounts:\*\***

- Round 2: sent \$0.12, received \$0.36, returned \$0.15 (42%)
- Round 4: sent \$0.16, received \$0.48, returned \$0.20 (42%)
- Round 6: sent \$0.83, received \$2.49, returned \$1.20 (48%)

**\*\*Considerations:\*\***

- Communication noise could be affecting these sends
- Should I maintain consistency with my ~40-45% return rate for small amounts?
- I'm getting frustrated with the lack of reciprocity when I send, but I should maintain fairness

I'll return a similar proportion to maintain consistency:

For \$0.60 received, 40-45% would be \$0.24-\$0.27

I'll return \$0.25, which is about 42% - consistent with my previous behavior.

**usage:**

```
{
  "input_tokens": 1935,
  "output_tokens": 584,
  "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$17.25.  
How much do you send to the receiver this round? (0-5)

**Message 13 role=assistant**

**content:**

Looking at my performance:

**\*\*Analysis:\*\***

- Current total: \$17.25
- My sender rounds: I've consistently sent \$3
- Recent trend: Last two sender rounds (5 & 7) got \$0 returns
- This losing streak is concerning

**\*\*Pattern:\*\***

- Round 1: Sent \$3 ? excellent return
- Round 3: Sent \$3 ? decent return
- Round 5: Sent \$3 ? \$0 back
- Round 7: Sent \$3 ? \$0 back

**\*\*Considerations:\*\***

- Two consecutive \$0 returns means I'm losing \$1 per round by cooperating
- Could be communication noise, or receivers are less cooperative now
- Need to balance maintaining trust vs. protecting myself
- Still profitable overall, but trend is negative

I'll adjust to a more cautious but still cooperative approach:

```
'''json
{"send": 2}
'''
```

This reduces my risk while still signaling willingness to cooperate. The receiver will get \$6, which is still substantial.

**reasoning:**

Let me analyze my situation:

**\*\*My total earnings: \$17.25\*\***

Let me think about my history as SENDER:

- Round 1: Sent \$3, got good returns (~\$5.02)
- Round 3: Sent \$3, got okay returns
- Round 5: Sent \$3, got \$0 back (disappointing)
- Round 7: Sent \$3, got \$0 back (Round 8 earnings \$16.65 - Round 7 earnings \$14.65 = \$2, so I kept \$5 - \$3 + \$0 = \$2)

As RECEIVER:

- Round 2: Received \$0.36, returned \$0.15
- Round 4: Received \$0.48, returned \$0.20
- Round 6: Received \$2.49, returned \$1.20
- Round 8: Received \$0.60, returned \$0.25

So my sender rounds have had increasingly poor returns. I've been consistently sending \$3, but:

- Round 1: Good return
- Round 3: Some return
- Round 5: No return
- Round 7: No return

The pattern suggests that while I've been cooperating, my cooperation isn't being reciprocated as much anymore. In the last two rounds where I sent \$3, I got nothing back.

However, I need to consider:

1. Communication noise - maybe they're trying to return but noise is affecting it
2. I'm still profitable overall (\$17.25 vs starting at \$0)
3. Changing strategy now might signal I'm uncooperative

But sending \$3 and getting \$0 back means I only have \$2 left from that round, which is less than if I'd sent nothing.

Options:

1. Keep sending \$3 - maintain cooperation but risk another \$0 return
  2. Send less - protect myself but signal less trust
  3. Send nothing - safe but ensures no mutual gain
- The consecutive failures...

**usage:**

```
{
  "input_tokens": 1878,
  "output_tokens": 885,
  "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$20.25.

This round, The sender sent you \$0.94, so you received \$2.82. How much do you return to The sender? (0-2.82)

**Assistant output:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent \$0.94 (about 19% of their \$5)
- I received \$2.82 after tripling
- My Round 9 earnings jumped by \$3, suggesting good cooperation recently
- This is a moderate send, similar to Round 6 (\$0.83)

**\*\*Pattern consistency:\*\***

- Round 6: Received \$2.49 (from \$0.83) ? returned \$1.20 (~48%)
- This sender sent slightly more (\$0.94), so I should be at least as generous

I'll return a fair proportion:

```
'''json
{"return": 1.40}
'''
```

This gives the sender \$5.46 total (profit of \$0.46) and me \$1.42 (profit of \$1.42). Both benefit from the cooperation, rewarding their moderate trust.

**Reasoning:**

Looking at this situation:

**\*\*Analysis:\*\***

- The sender sent \$0.94 (about 19% of their \$5)
- I received \$2.82 after tripling
- My earnings from Round 9 suggest I got \$3 back on my \$2 send - excellent!

**\*\*Pattern from previous moderate sends:\*\***

- Round 6: Received \$2.49 (from \$0.83 send) ? returned \$1.20 (~48%)
- This sender sent slightly more (\$0.94 vs \$0.83)

**\*\*Considerations:\*\***

- \$0.94 is a moderate cooperative send
- Sender kept \$4.06; they're taking a risk but not fully trusting
- I should reward this moderate cooperation
- A fair split would give us both profit

In Round 6, I returned about 48% (\$1.20 out of \$2.49). The sender here sent slightly more, so I should be at least as generous.

If I return about \$1.35-\$1.40:

- Sender gets: \$4.06 + \$1.40 = \$5.46 (profit of \$0.46)
- I keep: \$2.82 - \$1.40 = \$1.42 (profit of \$1.42)

Both benefit, with me getting more since I received the multiplied amount.

Let me return \$1.40 to be fair and consistent with my approach to moderate sends.

Usage: {

```
"input_tokens": 1949,
"output_tokens": 543,
"reasoning_tokens": 0
```

}